

Patient Robot Advocacy

A Phase 1, First-in-Human, PDAC Clinical Trial Protocol of a LLM-Directed
Robotic Whipple with Daraxonrasib (RMC-6236)

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30 figures

21 documented concerns answered

CEO Kevin Kawchak , ChemicalQDevice | kevink@chemicalqdevice.com
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Who is in control?

A named human surgeon approves every motion. The model has no path to the arms. Stopping is bounded at 3 ms across the arms and 500 ms system-wide. §7, Figures 18 and 19.

What could go wrong?

Twenty-one documented concerns, each wired to the clause, limit, or number that answers it, including the five that are answered only by governance. §3, Figures 5, 6, and 7.

What is this costing me?

Roughly 27 days of contact over 24 months, 18 visits, one inpatient stay. The sponsor pays study-only items; two items remain unsettled. §9 and §12, Figures 24 and 30.

Independent research paper and practical adoption guide. It is not medical or regulatory advice and is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. All figures derive from the author's repository sources and are illustrative unless tied to a cited reference.

Disclaimer: This work is independent and is not endorsed or sponsored by any trial sponsor; CRO, site, IRB, regulator, or medical society; and was adapted using Claude Code Opus 5.

San Diego
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The paper (v1.0) is at <https://doi.org/10.5281/zenodo.xxxxxxxx> and the repository (v1.0.0) is at <https://github.com/kevinkawchak/robotic-surgeries/tree/main/patient-robot-advocacy>. The parent clinical protocol this paper advocates for is 10.5281/zenodo.20780121 [1]; bill citations use H. R. 9510 v5, 10.5281/zenodo.20619762 [2].

Keywords Patient advocacy; Informed consent; Pancreatic Ductal Adenocarcinoma; Robotic Whipple; Daraxonrasib; Physical AI; On-premises LLM; Surgical autonomy; Patient concerns; Phase 1 Trial Protocol; H. R. 9510

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1 Statement of Patient Commitment

The parent clinical protocol opens with a Statement of Compliance addressed to the Food and Drug Administration, the Institutional Review Board, and the sponsor [1]. That statement is necessary, it is correct, and it is not addressed to the person whose abdomen will be opened. It tells the regulator which parts of Title 21 the study satisfies. It does not tell the participant what the study owes them.

This paper opens instead with the seven things the same protocol commits to the participant. Each one is tied to a clause that makes it enforceable rather than aspirational, and each one names something the participant can check. The distinction matters, because the surveyed literature on patient attitudes to robotic and artificial-intelligence-assisted surgery reports again and again that reassurance without a checkable referent is heard as marketing [3, 4].

The premise is taken from the author’s earlier analysis of proposed United States legislation for Physical AI oncology trials: the cancer patient, not the doctor, not the nurse, not the trial sponsor, not the Institutional Review Board, and not the regulator, is the priority participant [5, 2]. That premise is not a sentiment. It has a structural consequence, which the whole of this paper works out: if the participant is the priority, then every safeguard written as a sponsor obligation can also be written as a participant capability, and the two readings should agree. Where they do not agree, this paper says so.

This is an advocacy document assembled from the protocol’s own record. It is not the consent form and it is not a substitute for one. It is designed to be read before the consent conversation, so that the conversation can be spent on the questions this document cannot answer.

1.1 The seven commitments

One. Nothing starts without you. No study procedure occurs before informed consent is documented under 21 CFR §50.20 [6]. That includes the staging imaging repeated for study purposes, the tissue re-review, and every laboratory draw the protocol adds beyond routine care. The check is simple: the consent form carries a date, and no study record may predate it.

Two. The robot is proven before you. A Phase 0 simulation-validation gate precedes any patient-facing robotic activity: at least 1000 simulated procedures across at least two independent simulation frameworks, reproducing motion within a 2 mm positional and 0.5 N force tolerance, with a documented Unified Safety Level of at least 7.0 [7, 8]. The check is that the gate sign-off is a dated document, and the participant is shown it at the baseline visit.

Three. You are never the untested dose. Daraxonrasib is escalated by a standard 3+3 rule with staggered sentinel enrollment. A dose level opens only after the level below has been cleared, and the second participant at a new level is not dosed until the first participant’s safety window has closed. The check is the dose level printed on the participant’s own assignment, and the cohort record that shows what happened at the level below.

Four. A human approves every motion. The system is classified as a Class II collaborative device under continuous human oversight. The on-premises language model proposes; a deterministic safety gate tests the proposal; the operating surgeon approves, modifies, or declines. No message travels from the model to the arms. The check is the audit trail, in which every executed motion carries a surgeon signature and no motion carries none.

Five. Stopping is faster than harm. The emergency-stop budget is at most 3 ms across arms and at most 500 ms system-wide, on a 10 kHz heartbeat broadcast bus [9, 10]. A human blink takes roughly 100 to 150 ms. Every arm is held before a blink completes. The check is that the latency is measured on every procedure and reported, not assumed.

Six. Your record cannot be edited. Every model advisory, every gate decision, every surgeon approval, and every safety-limit event is written to a hash-chained audit trail bound to a deterministic seed and a specific repository commit, under 21 CFR part 11 [11]. The check is that the participant may request their own extract, and the chain either verifies or it does not.

Seven. Leaving costs you nothing. Withdrawal is available at any visit, without a reason, and standard-of-care oncology continues unchanged. The check is in §8: four routes out are enumerated, and the data consequence of each is stated, including the two consequences that are unwelcome.

#	Commitment	The clause that makes it enforceable
1	Nothing starts without you	Consent under 21 CFR §50.20 precedes every study procedure
2	The robot is proven before you	Phase 0 gate: at least 1000 simulations across at least two independent frameworks, Unified Safety Level at least 7.0
3	You are never the untested dose	3+3 escalation with sentinel staggering; a dose level opens only after the level below is cleared
4	A human approves every motion	Class II collaborative device under continuous human oversight; no model-to-actuator path exists
5	Stopping is faster than harm	Emergency stop at most 3 ms across arms and at most 500 ms system-wide
6	Your record cannot be edited	Hash-chained audit trail under 21 CFR part 11, bound to a deterministic seed and a repository commit
7	Leaving costs you nothing	Withdrawal at any visit without reason; standard care continues unchanged

1.2 How to check a commitment, without taking anyone's word

A commitment that cannot be checked is a slogan. Each of the seven above names an artefact, and an artefact can be asked for. The list below is what to ask for, and who holds it.

#	Ask to see	Who holds it, and what a satisfactory answer looks like
1	The dated consent form, and the date of the first study-specific record	The site study coordinator. The consent date is on or before every study record date. A record that predates consent is a deviation and is reportable
2	The Phase 0 gate sign-off sheet	The Sponsor-Investigator, through the site. It carries a date, the number of simulated procedures, the two framework names, and the Unified Safety Level as a number
3	The dose level you are assigned to, and its clearance record	The site Principal Investigator. Your level is named in your consent; the level below it has a written clearance and closed sentinel windows
4	The device classification and the oversight statement	The site. The device is Class II under continuous human oversight; the correct answer names the operating surgeon, not the software

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#	Ask to see	Who holds it, and what a satisfactory answer looks like
5	The most recent emergency-stop latency check for your room	The site biomedical engineering record. Two numbers, cross-arm and system-wide, each with a date and each under budget
6	Your own audit extract	The Sponsor-Investigator, on request under §11.5. It arrives as a hash-chained record with a seed and a commit identifier, not as a summary
7	The withdrawal paragraph of your consent form	The site. It states that standard care continues unchanged and that no reason is required

None of those requests is unusual and none of them requires a lawyer. A site that cannot produce the artefact behind a commitment has not met the commitment, whatever else it has done, and §11.6 names the escalation route that applies in that case.

1.3 What this paper is, and what it is not

It is an independent research paper and a practical adoption guide, built entirely from the author's repository sources. It is not medical advice, not regulatory advice, and not endorsed by the FDA, the NIH, HHS, an Institutional Review Board, the International Council for Harmonisation, or any sponsor, contract research organisation, site, regulator, or medical society.

It is also not the informed consent form. A consent form has a required content list, a required reading level, and a required signature block, and the National Cancer Institute publishes a plain-language guide to what that document must contain [12]. This paper does none of that work. What it does is prior: it sets out the twenty-one concerns people actually raise about surgical robots, and it says, for each one, which clause of this specific protocol answers it and how completely. A participant who has read this paper arrives at the consent conversation with better questions.

Every figure in this paper derives from the author's repository sources and is illustrative unless it is tied to a cited reference. Where a number is simulated rather than measured in humans, §10 says so on the same line as the number, and Figure 28 traces it to the source that produced it.

1.4 How to read the thirty figures

The paper carries thirty figures across five diagram types, and the type is not decoration. Each type answers a different shape of question, and the choice of type for a given figure is itself an argument about what kind of question is being asked.

Diagram type	Count	The question it answers
Mermaid-type	9	What happens, in what order, and who decides? Decisions in time
D2-type	7	How much of this is there, and how does it group? Containment and true grids
PlantUML-type	5	What exactly do you guarantee? Formal notation with a defined semantics

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Diagram type	Count	The question it answers
Graphviz-type	5	What depends on what, and can you prove it? Pure graph structure
Diagrams (Python)-type	4	Where does this physically live? Rooms, cabinets, cables

The monospace tag at the top left of every frame names the native construct the figure reproduces, for example `flowchart TD` or `digraph`, so a reader can carry any figure back to the platform it came from. Every figure is vector line art drawn in the document itself. There is no raster image anywhere in this paper.

1.5 Accountability, stated once, at the front

The single most common accountability question in the surveyed literature is the plainest one: if something goes wrong, who is to blame [3, 4, 13]. When a human surgeon errs, the chain of liability is familiar. When a semi-autonomous movement or a model-generated suggestion contributes to harm, participants report being told that the surgeon, the hospital, the software developer, and the device manufacturer are all involved, which is not an answer.

This protocol answers it with a rule rather than a list: exactly one accountable party per failure class, named in advance. Section 11 sets out the full responsibility matrix, and Figure 29 renders it with a final column giving the person the participant should actually call. The rule is checkable, because a matrix with two accountable parties in one row would fail it, and the matrix is published.

Figure 2 states the same structure as a graph. The chart is rooted at the participant rather than at the sponsor, and it is acyclic, which means that no body in it reviews itself. Three edges point back at the root, and only three, because three is what the protocol actually owes the participant as an affirmative obligation to inform.

The graph is acyclic by construction, and the property is load-bearing rather than aesthetic: a cycle would mean that some body in the chart reviews its own work, which is precisely the failure the accountability concern describes. The three blue edges returning to the root are the affirmative obligations to inform. They are not courtesies, and §11.6 names the escalation route that applies if one of them is not met.

1.6 What this paper does not promise

Advocacy that promises too much is worth less than advocacy that draws its own boundary, so the boundary is drawn here rather than buried in §10.

This paper does not promise that the operation will succeed. It is a Phase 1 study, and the primary questions are feasibility and safety, not cure. A participant who joins expecting a better oncologic outcome than a conventional resection is joining for a reason the design cannot support, and §4.1 states the reason the design cannot support it.

It does not promise that the platform is better than a human-only Whipple. No comparison of that kind is possible at this sample size, and §10.5 gives the arithmetic that shows why: with at most 18 participants and no randomised control, the study can detect a gross safety signal and nothing finer. Where this paper cites a comparator number, the comparator is a published series and is labelled C in the provenance table, never a result of this study.

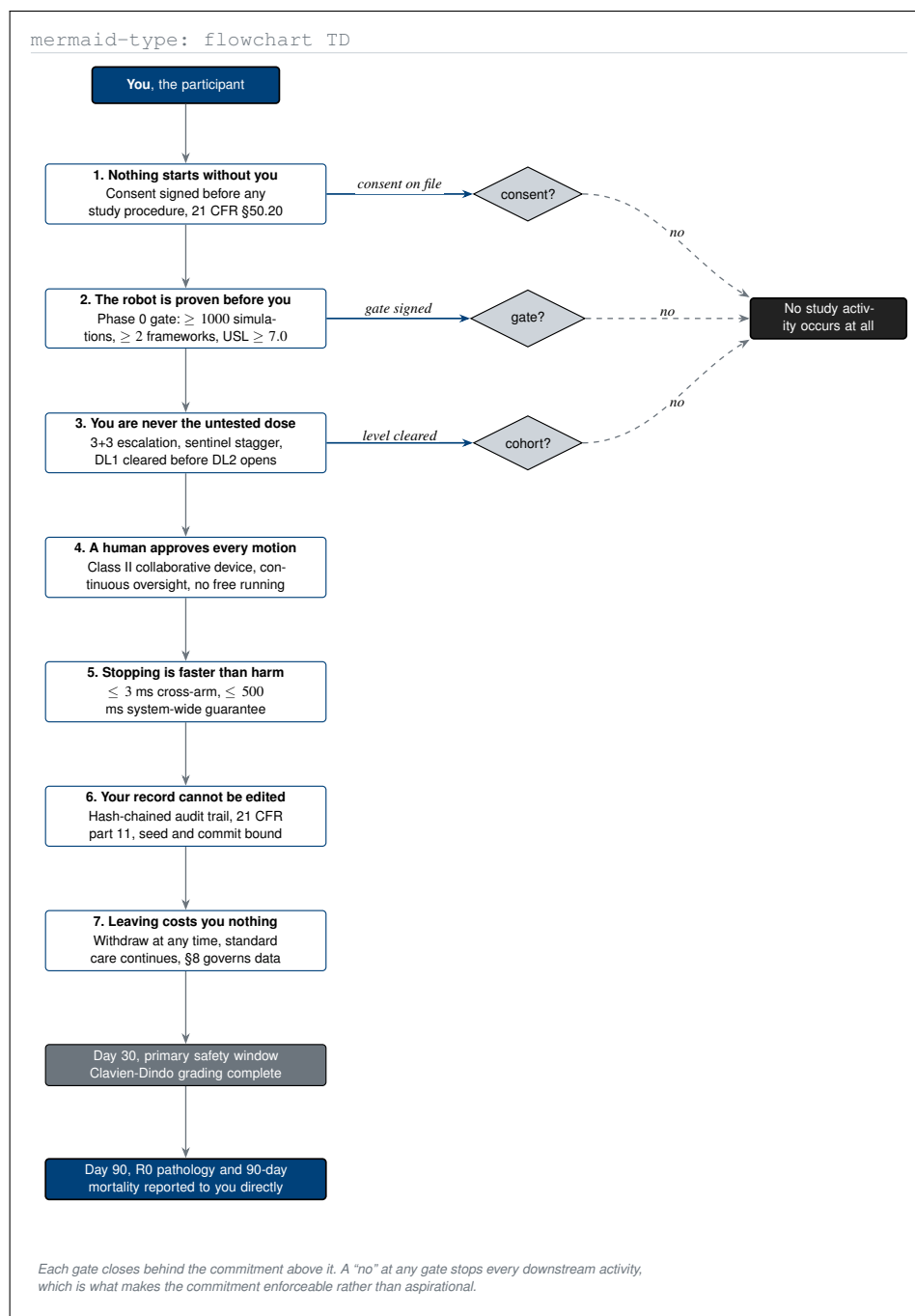


Figure 1. The seven commitments made to the participant, from the first conversation through consent, the Phase 0 gate, and dose assignment, to the day-30 safety window and the day-90 pathology result that is reported back to them directly.

It does not promise that nothing will hurt. A Whipple is one of the largest operations in general surgery under any technique. The relevant claim is narrower and is the one the protocol can keep: that the additional risk introduced by the platform, over and above the operation itself, is bounded by limits that are enforced in hardware and measured on every case.

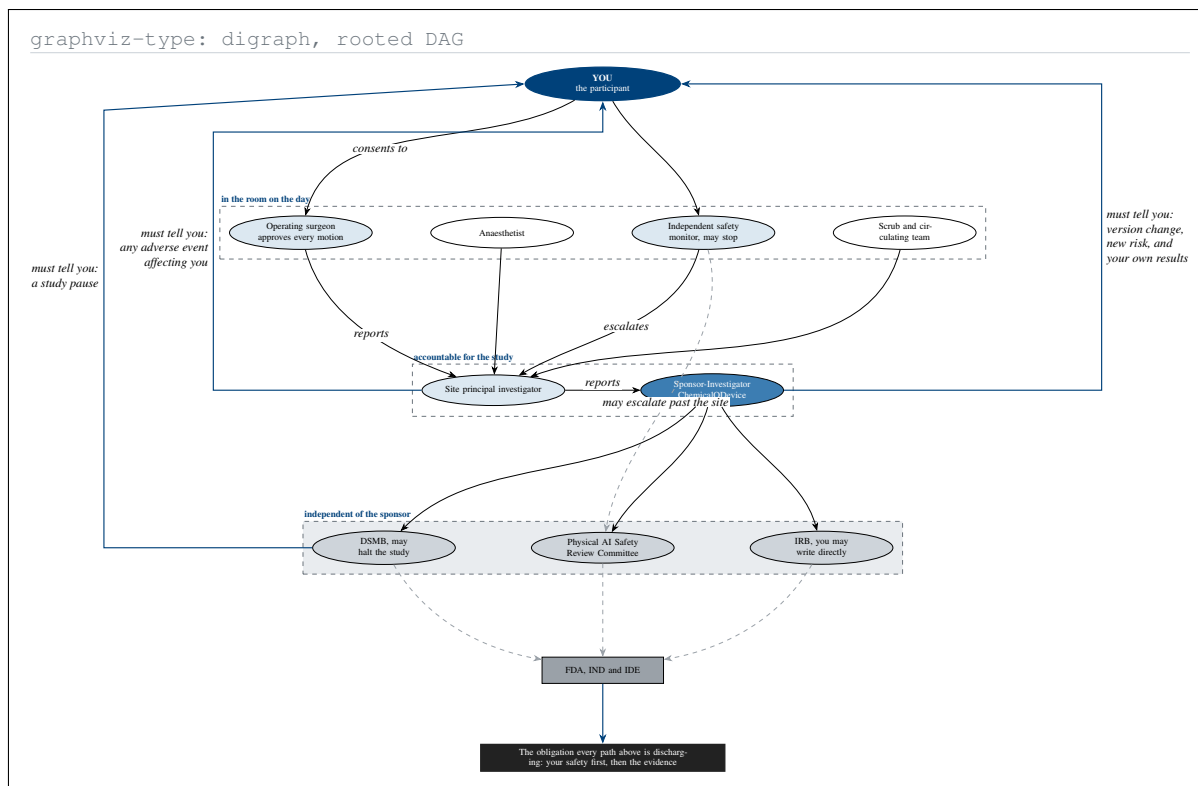


Figure 2. The accountability chain drawn as a rooted directed acyclic graph with the participant at the root, showing the people in the room, the bodies accountable for the study, the independent reviewers, and the three obligations that point back.

It does not promise that every concern is answered. Five of the twenty-one are answered by governance or disclosure alone, and §3.8 names each of those five as such rather than dressing it as a guarantee. Two cost items in §12.3 remain unsettled at the time of writing, and they are printed unsettled.

Finally, it does not promise that the participant's judgement should defer to the protocol's. The purpose of the document is the opposite: to put the participant in a position to disagree with it in specific terms.

2 Plain-Language Protocol Summary

2.1 The study in one paragraph

This is a first-in-human study of two new things used together: a drug called daraxonrasib, given before and after surgery, and an eight-arm robotic system that performs the surgery under the continuous control of a human surgeon, with an on-premises computer model advising that surgeon. The operation is a pancreaticoduodenectomy, usually called a Whipple, which removes the head of the pancreas along with the first part of the small intestine, part of the bile duct, and the gallbladder, and then rebuilds three connections. The disease is pancreatic ductal adenocarcinoma, PDAC, carrying a KRAS G12 mutation. PDAC is the third leading cause of cancer death in the United States, and five-year survival across all stages is about 13 percent [14]. The primary question this study asks is not whether the combination cures the disease. It is whether the combination is safe enough, and practical enough, to be tested in a study that could answer that. Up to 18 participants will take part.

2.2 The words you will hear, defined once

The consent conversation moves faster if the vocabulary is already settled. The terms below are the ones this protocol cannot avoid using. Each is defined here in the sense the protocol uses it, which is occasionally narrower than the sense the same word carries in general oncology.

Term	What it means in this protocol
PDAC	Pancreatic ductal adenocarcinoma, the most common form of pancreatic cancer and the only one this study enrolls
Whipple	Pancreaticoduodenectomy: removal of the head of the pancreas, the duodenum, the gallbladder and bile duct, and part of the stomach, followed by three reconnections
Resectable	The tumour can be removed with a clear margin without reconstructing a major artery. Borderline and locally advanced disease are not eligible
KRAS G12	A family of mutations in the KRAS gene. Daraxonrasib targets this family, so the mutation must be confirmed on tissue before enrollment
Daraxonrasib (RMC-6236)	The investigational oral drug in this study, a multi-selective inhibitor of the RAS pathway
Physical AI	Software that proposes actions for a machine that touches the patient. Here it proposes; it never actuates
On-premises	The language model runs on hardware inside the hospital. No patient data leaves the building during a procedure
Advisory	A proposed motion or step produced by the model. An advisory is not an instruction until a surgeon signs it
Safety gate	Deterministic software that tests every advisory against the numeric limits before a surgeon ever sees it
No-fly corridor	A 4 mm exclusion volume around the superior mesenteric and portal veins that no instrument tip may enter
Heartbeat bus	A 10 kHz signal every arm must hear. Silence for one interval triggers an automatic halt

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Term	What it means in this protocol
Emergency stop	The surgeon-initiated halt: at most 3 ms across arms, at most 500 ms system-wide
Phase 0 gate	The simulation-validation milestone that must be signed before any patient-facing robotic activity
USL	Unified Safety Level, the composite score the Phase 0 gate must reach, at least 7.0
3+3 escalation	The dose-finding rule: three participants per level, expandable to six, with a level opening only when the level below is cleared
Sentinel window	The observation period after the first participant at a dose level, which must close before the second is dosed
DLT	Dose-limiting toxicity, a prespecified severe drug reaction that counts against the 3+3 rule
ISGPS grade	The international grading of a pancreatic fistula, the most common serious complication of a Whipple
R0	A resection margin with no tumour at the inked edge, assessed on the day-90 pathology report
PFS and OS	Progression-free survival and overall survival, followed here to 24 months and reported as descriptive, not comparative

2.3 What is being tested, in three parts

The drug. Daraxonrasib, also called RMC-6236, is a RAS(ON) multi-selective inhibitor: it binds the active form of the RAS protein that a KRAS G12 mutation leaves switched on. It has been studied in previously treated metastatic pancreatic cancer in a randomised Phase 3 programme [15, 16]. In this study it is used differently, in the perioperative setting, with a planned pause before the operation and an advised restart at 7, 14, or 21 days afterwards, keyed to how the pancreatic join is healing and to the drug level in the blood.

The operation. A robotic Whipple is an established operation at high-volume centres. A nationwide series of 1000 robotic pancreatoduodenectomies shows that the rate of a serious pancreatic leak and the rate of death within 90 days both fall substantially as an institution's experience grows [17]. What is new here is not the robot: it is the eight-arm configuration, the hard force and geometry limits enforced in hardware, and the advisory layer above them [1].

The software. An on-premises large language model observes the sensor stream and proposes the next operative step. It cannot move anything. A systematic review of autonomy levels in FDA-cleared surgical robots found that no cleared system exceeds task-level autonomy under supervision [18], and this system does not either. Section 7.3 sets out exactly where the model sits in the chain, and Figure 19 draws it message by message.

2.4 Your journey, start to finish

Figure 3 draws the whole journey in five phases: screening of up to 28 days, a baseline day, the operation on day 0, an acute window of days 1 to 7, and follow-up through day 30, day 90, and every 12 weeks to the 24-month overall survival endpoint. The figure is drawn so that every node the participant controls is

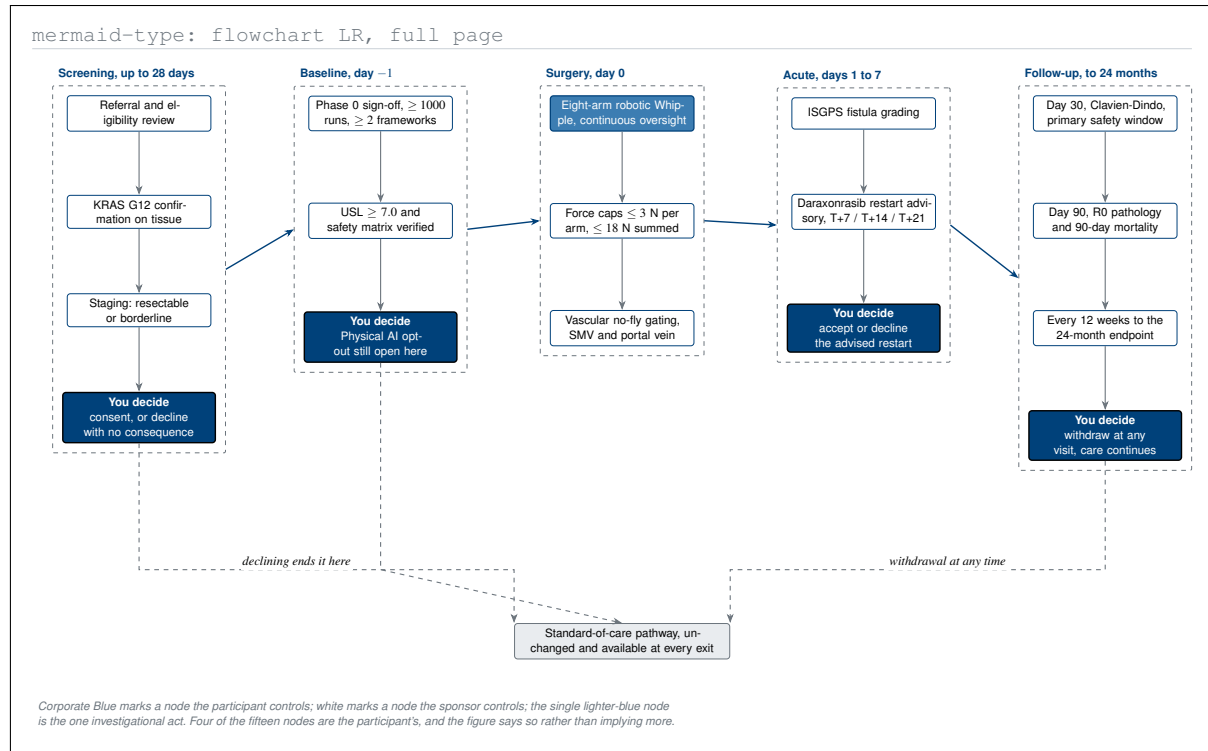


Figure 3. The participant journey across five phases, screening through the 24-month endpoint, with every node the participant controls filled Corporate Blue and every node the sponsor controls left white, so that the balance of control is visible.

filled Corporate Blue and every node the sponsor controls is left white. Four of the fifteen nodes are the participant's. The figure does not pretend that is a majority; it makes the balance visible so that it can be argued about rather than assumed.

The structure of the journey is adapted from the author's simulation of a fully autonomous, single-patient journey through a regulated Physical AI oncology trial [19]. That simulation was non-small cell lung cancer, and four differences matter to a reader with PDAC. The operation is a pancreaticoduodenectomy with three anastomoses rather than a lobectomy with one bronchial closure. The drug is a RAS(ON) inhibitor rather than a checkpoint inhibitor. The dominant early complication is a postoperative pancreatic fistula rather than a prolonged air leak. And the survival baseline is far lower, which changes what a participant is weighing when they decide.

2.5 Your schedule, and what it costs you in time

The parent protocol carries a Schedule of Activities: a matrix of activities against visits, with a cross in each cell where something is collected. It is the right document for a study coordinator and the wrong one for a participant, because it answers "what is collected" and not "what do I do, and how long does it take". Figure 4 redraws the same content as a grid whose cells state what the participant does, with a totals row giving the approximate hours each visit consumes. Both forms are given, because a participant may want to hand the conventional one to their own oncologist [20].

2.6 Schedule of Activities, in the conventional form

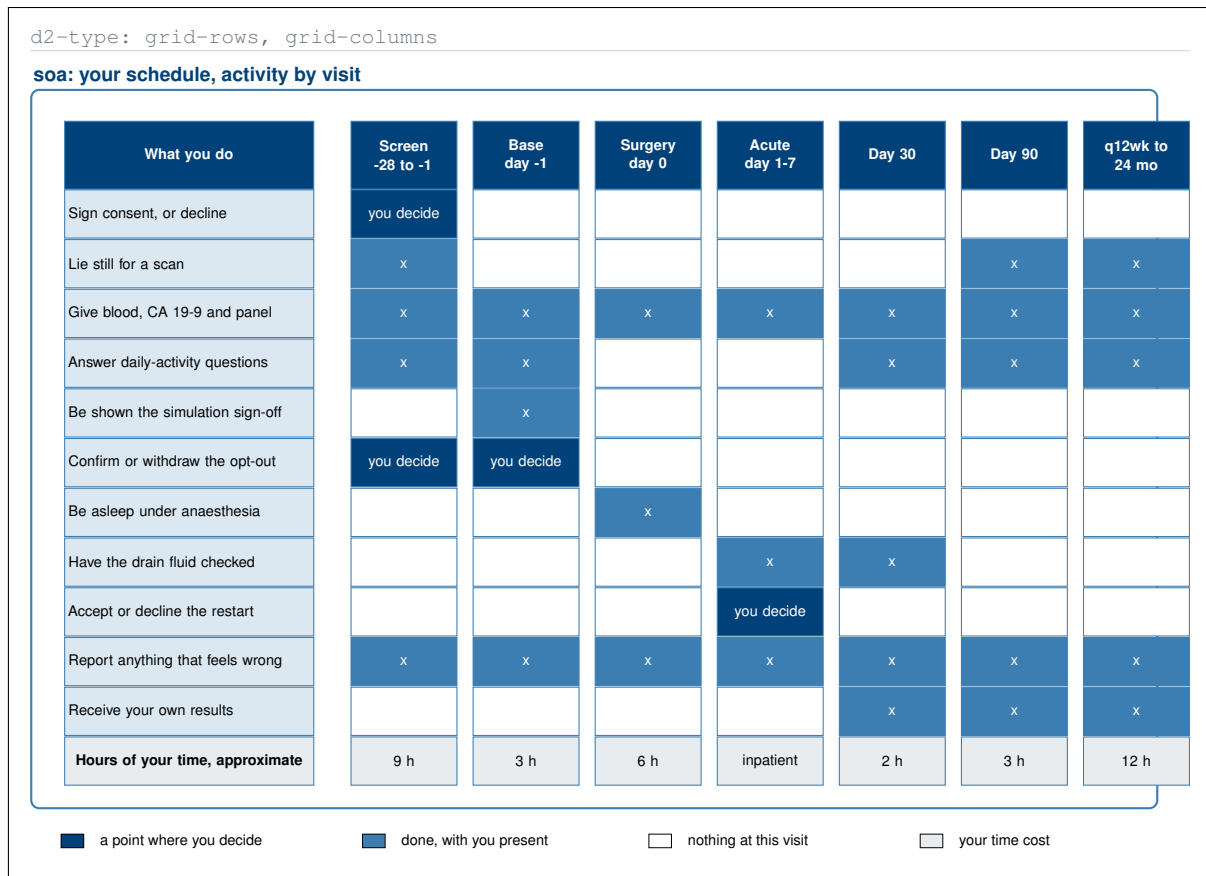


Figure 4. The Schedule of Activities redrawn as a true grid from the participant's chair, with each cell stating what they do rather than what is collected, and a totals row giving the approximate hours of their time each visit will consume.

Activity	Screen -28 to -1	Base Day -1	Surg Day 0	Acute Day 1-7	Day 30	Day 90	LT q12wk
Informed consent, including Physical AI opt-out	X						
Staging and KRAS G12 confirmation	X						
CA 19-9	X	X			X	X	X
ECOG performance status	X	X			X	X	X
Safety laboratory panel	X	X	X	X	X	X	X
Phase 0 simulation sign-off		X					
USL at least 7.0 and safety matrix		X					

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Activity	Screen -28 to -1	Base Day -1	Surg Day 0	Acute Day 1-7	Day 30	Day 90	LT q12wk
Robotic Whipple surgery			X				
Intra-operative telemetry capture			X				
ISGPS fistula grading				X	X		
Daraxonrasib restart advisory				X			
Adverse event review	X	X	X	X	X	X	X
RECIST imaging	X					X	X
PFS and OS assessment						X	X

Abbreviations: ISGPS, International Study Group on Pancreatic Surgery; LT, long-term; OS, overall survival; PFS, progression-free survival; RECIST, Response Evaluation Criteria in Solid Tumors; USL, Unified Safety Level.

2.7 How many people, and when your turn comes

Up to 18 participants are enrolled across three dose levels, three per level and expandable to six if a dose-limiting toxicity appears. That number is not a budget decision. It is what the 3+3 rule needs in order to answer the dose question, and enrolling more people than the question needs is itself a harm.

Enrollment is deliberately slow, and the reason matters to anyone waiting. A dose level does not open until the level below it has been cleared, and within a level the second participant is not dosed until the first participant's sentinel window has closed. A participant who is screened while a sentinel window is open waits for it to close. That wait is not an administrative delay and it is not negotiable; it is the mechanism that makes commitment three true.

The practical consequence is that the calendar in §9.1 describes the participant's own 24 months, not the study's. The study runs longer than any one participant's participation, and results from participants ahead of you are reviewed by the Data and Safety Monitoring Board before your level opens. A participant at DL3 is therefore protected by every safety observation made at DL1 and DL2, and §5.6 states what happens if one of those observations goes the wrong way while that participant is still enrolled.

There is no waiting list in the ordinary sense, because eligibility is re-assessed at the moment a slot opens rather than at the moment of the first call. Disease progresses; a participant who was resectable in March may not be in June. §6.2 states the criteria that are re-checked, and §6.4 states what the site owes a participant who becomes ineligible while waiting, which is a prompt referral back to their own oncologist with the study's imaging in hand.

2.8 What the study cannot tell you

A Phase 1 study of at most eighteen participants is powered for safety and feasibility and for nothing else. It cannot establish that this combination improves survival, and no result it produces should be read as though it had. The National Cancer Institute's guidance on how trials work makes the same point about early-phase studies generally [20], and the FDA's public information on computer-assisted surgical systems states that cancer-related long-term outcomes have not been established for every use of robotically assisted surgical devices [21].

This is stated here, at the front, rather than in a limitations paragraph at the back, because the surveyed literature is consistent that patients discount enthusiasm and trust candour [3, 4]. A paper arguing for participation has to be able to say what participation will not deliver. What this study can deliver is a clean answer to three questions: whether the dose is tolerated, whether the device causes harm within 30 days, and whether the operation achieves a clear margin at a rate consistent with the best contemporary comparator.

3 The Documented Patient Concerns

3.1 Where the twenty-one come from

Two dated research passes on July 28, 2026 supply the evidence base for this section. The first, by Gemini 3.1 Pro, returned six concern families. The second, by ChatGPT 5.6 Thinking Extended, returned sixteen numbered concerns, each with the question a patient is likely to ask and what the trial team should explain, together with thirteen supporting references. The two overlap on five items. De-duplicated, they give twenty-one distinct documented concerns, and those twenty-one are the subject of the rest of this paper.

Neither pass invented its material. Behind them sits human evidence. A 2026 qualitative study of 50 previously operated patients found that 56 percent would consider semi-autonomous robot-assisted surgery, and that willingness depended strongly on which parts of the procedure the robot controlled; the principal themes were informed consent, additional risks, surgeon-patient trust, and anticipated benefit [13]. A cross-sectional survey of 330 oncology patients found approximately half expressing concern about artificial intelligence in cancer care, led by loss of human interaction and by medical error [22]. A systematic review of patient and public perspectives on robotic surgery found that expectation and experience diverge systematically, and that expectation is usually set by promotion rather than by evidence [3]. A survey of public perceptions found that the most common single misconception is that the robot operates by itself [4].

Two conclusions follow, and this paper is built on both. First, patients are not opposed to the technology; they are opposed to being asked to accept it without specifics. Second, the specifics they want are not general. They are per-step, per-limit, per-name.

Concern family	Concerns	Where this paper answers it
Control and the human element	5	§7.3, §7.4; Figures 8, 18, 19
Safety and technical failure	5	§7.4, §7.5, §7.6; Figures 20, 21
Evidence and effectiveness	4	§4.2, §5.1, §10; Figures 13, 27, 28
Data, privacy, and security	3	§9.3, §9.4; Figures 17, 25
Fairness and applicability	2	§6.3, §8.3; Figures 14, 23
Accountability and burden	2	§11, §12; Figures 29, 30

3.2 Control and the human element

Five concerns sit here, and they are the most frequently raised of the twenty-one. Who is actually driving the robot. Whether the surgeon can stop it, and how fast. Whether the surgeon will defer to the model even when their own judgement differs. Whether the participant will still have a doctor who knows them. Whether relying on the system will reduce time with the care team.

The first is answered by classification and by architecture. The system is a Class II collaborative device operating under continuous human oversight, and the on-premises model has no path to an actuator: it emits a candidate plan, a deterministic gate tests that plan, and the operating surgeon signs or declines it. Section 7.3 sets this out and Figure 19 draws every message. The public misconception that the robot operates by itself [4] is not corrected by asserting the opposite; it is corrected by showing a diagram in which the arrow that would be needed is absent.

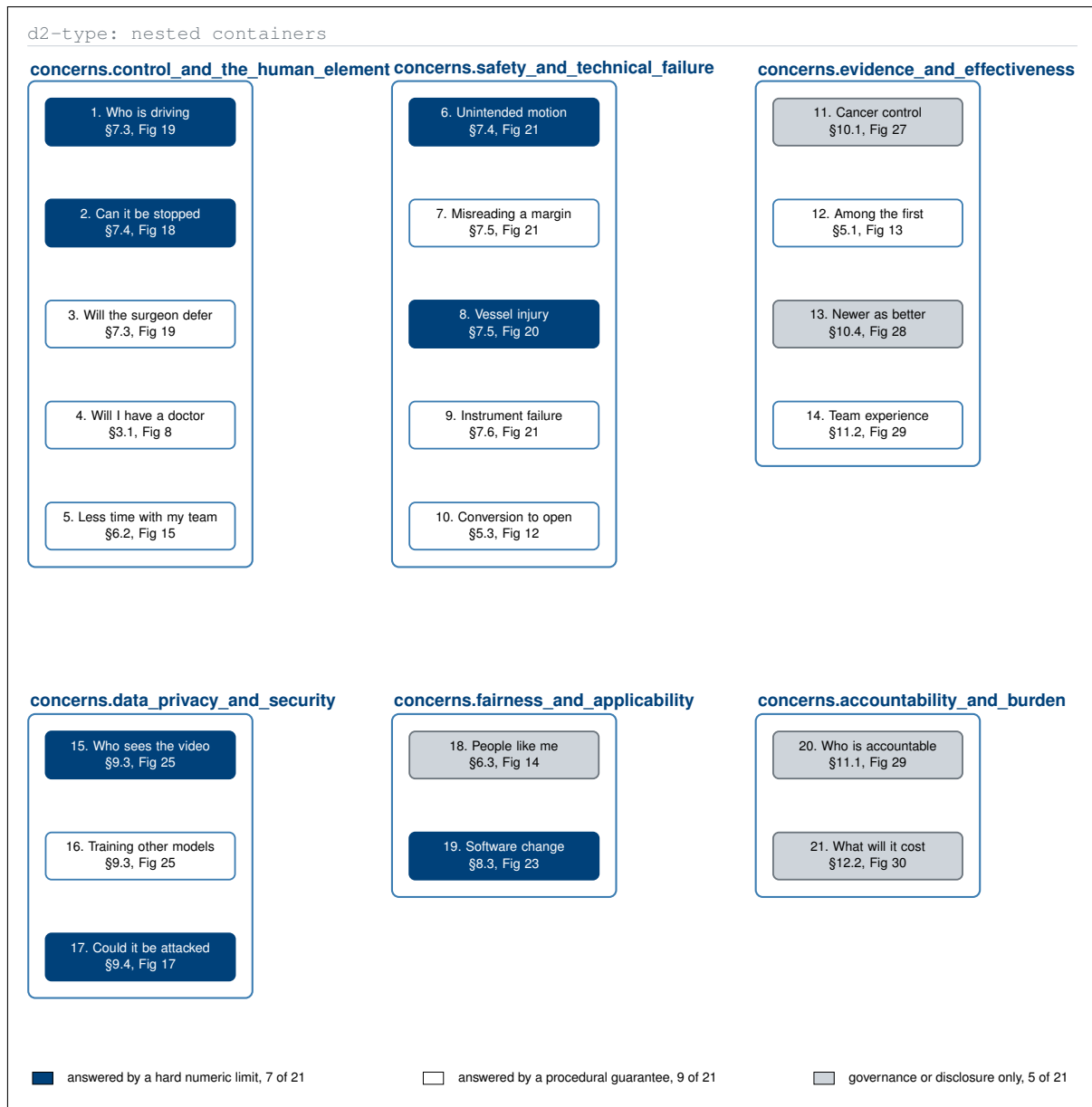


Figure 5. The twenty-one documented concerns grouped into six disjoint families, with the answering section and figure printed inside each item, and a fill key stating how many are answered by a limit, a guarantee, or by governance alone.

The third concern, deference, is the subtlest and the least often addressed. It is not enough that the surgeon may decline; the question is whether they will. This protocol records every override with its reason, publishes the override rate as an exploratory endpoint, and requires no justification for declining, so that declining carries no procedural cost. Automation bias is a measured quantity here rather than an assumed absence [23].

The fourth and fifth concerns are about relationship rather than technology. Every route into the study notifies the participant's own treating oncologist, on every branch, including the branch where the participant turns out to be ineligible. The direct sponsor channel of §6.2 is additional, never a replacement, and Figure 15 draws the notification on each path.

3.3 Safety and technical failure

Five concerns: unintended motion, misreading tissue or a margin, injury to a major vessel, instrument failure during the procedure, and conversion to open surgery.

The first three are answered by limits that are enforced below the level of the software making the decision. Per-arm tip force is capped at 3 N and the sum across arms at 18 N. Positional tolerance is 2 mm and force reproducibility is 0.5 N. A vascular no-fly corridor extends 4 mm either side of the superior mesenteric and portal veins, and a plan that would enter it is rejected by a deterministic gate before the surgeon ever sees it as executable. These are the kinds of limits the international standard for robotically assisted surgical equipment contemplates [10], and the fail-safe design standard for autonomous and semi-autonomous systems frames the stopping behaviour [9].

Instrument failure and conversion are answered differently, because they are not prevented, they are planned for. An open tray and backup instruments are in the room for every procedure. Conversion to open surgery is a recorded outcome, not a failure of the study or of the participant, it does not end participation, and the decision belongs to the operating surgeon alone. Section 7.7 gives the comparator context.

3.4 Evidence and effectiveness

Four concerns: whether the operation removes the cancer as completely as standard surgery, whether the participant is among the first humans to receive it, whether newer is being presented as better, and how experienced the team is.

This is the family this paper answers least completely, and §10 says so with a figure rather than a hedge. Figure 7 places cancer-control effectiveness in the quadrant of concerns that are raised often and answered only by governance or disclosure. The reason is structural: eighteen participants cannot establish an oncologic outcome. What the protocol can do, and does, is report R0 rate, ISGPS grade B/C fistula rate, and 90-day mortality against a contemporaneous nationwide comparator of 1000 robotic pancreaticoduodenectomies [17], and restrict enrollment to a site already past the third phase of that learning curve, so the curve is not paid for by the participant.

On the third concern, the paper takes the position the surveyed literature recommends: use neutral language and present absolute evidence rather than adjectives [3]. Nowhere in this document is the system described as revolutionary, state of the art, or cutting edge. The FDA's own information page states that cancer-related long-term outcomes have not been established for every use of robotically assisted surgical devices [21], and that statement is reproduced rather than argued with.

3.5 Data, privacy, and security

Three concerns: who sees the recording of the operation, whether the data will train other systems, and whether the system can be attacked.

The answers are architectural and are drawn rather than asserted. Section 9.3 publishes a read list per store and a retention period per store, in years. Section 9.4 states the network position: during a procedure there is no route out of the building, so the attack surface is the building. Three things cross the boundary before a procedure, a signed model build, a protocol version, and the pinned registry entry; two cross afterwards, reports to the FDA and the IRB; and nothing crosses during. Figure 17 draws that boundary and Figure 25 draws the pipeline. The relevant FDA guidance on cybersecurity in medical devices [24] and the joint transparency principles for machine learning-enabled devices [23] are the frame.

Data is not used to train a commercial model without a separate consent. That is a statement about something that does not happen, and Figure 25 renders it as an absent edge, which is the only honest way to draw a negative.

3.6 Fairness, applicability, accountability, and burden

Four concerns close the set: whether the system was tested on people like the participant, whether the software can change after consent, who is accountable if something goes wrong, and what participation will cost.

On applicability the honest answer is uncomfortable and is given in §6.3: at $n \leq 18$ subgroup performance is not estimable, and what the protocol commits to is reporting the composition of the enrolled population by age, sex, race, ethnicity, body mass index, and prior therapy, consistent with FDA guidance on reporting such data in device studies [25], together with the real-world performance monitoring the agency has sought comment on [26].

On software change the answer is a hard one: the version is frozen at consent and recorded in a registry, and any change forces a re-consent. Until the participant acts, no study procedure proceeds under the changed version. Figure 23 draws consent as the state machine this implies.

Accountability and cost are the two concerns that land in the honest-gap quadrant alongside effectiveness. Section 11 answers accountability with a rule, one accountable party per failure class, and Figure 29 publishes the matrix. Section 12 answers cost line by line and marks two items as unsettled: lost income, and continued drug access after study closure [27].

3.7 How completely each concern is answered

Figure 7 places sixteen of the documented concerns on two independent axes. The horizontal axis is prevalence, scored from how often the concern appears across the four surveyed human sources. The vertical axis is answer completeness, scored 1.0 where the protocol supplies a numeric limit, 0.7 where it supplies a procedural guarantee, 0.5 where it supplies a governance commitment, and 0.3 where it supplies disclosure only.

Four concerns land in the lower-right quadrant: raised often, answered softly. They are cancer-control effectiveness, accountability for harm, loss of the human relationship, and cost. This paper does not argue them out of that quadrant. Sections 10, 11, and 12 state, for each, exactly what would have to be true for the answer to become a hard one, and what this study can contribute towards making it so.

3.8 The five answered by governance alone, named

Five of the twenty-one are answered by governance or disclosure and by nothing harder. Naming them is more useful than distributing them quietly through the text, because a reader who knows which five they are can weigh them deliberately.

#	Concern	Why the answer cannot be harder than governance
12	Will this work better than the standard operation	A single-arm study of at most 18 participants cannot answer a comparative question. The honest answer is a comparator table and a stated inability, §10.5
13	Long-term cancer outcomes	Progression-free and overall survival are followed to 24 months and reported descriptively. No inference about durability is available at this size, §10.4
18	People like me	The eligibility criteria exclude the frailer patients in whom the question is sharpest. Subgroup performance is not estimable at n at most 18, and §7.3 says so

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#	Concern	Why the answer cannot be harder than governance
20	Who is accountable	Accountability is assigned by a published responsibility matrix and by named parties, not by a numeric limit. It is enforceable, but it is governance, §11
21	What it will cost	Study-only items are paid by the sponsor. Two items, lost income and post-closure data access, remain unsettled and are printed unsettled, §12.3

The distinction being drawn is not between strong answers and weak ones. A governance answer with a named accountable party, a published matrix, and an escalation route is a real answer. The distinction is that a numeric limit fails visibly and a governance commitment fails quietly, so the second kind requires the participant to know where to look. That is what §11.6 is for.

Two of the five could become harder answers, and it is worth being specific about how. Concern 12 becomes answerable only in a randomised study, which this is not and which cannot be run until feasibility is established; that is the argument for this study existing. Concern 21 becomes answerable the moment the two unsettled cost items are settled, which is a decision the sponsor can take and has not yet taken. The other three are structural at this phase.

3.9 The same safeguards, read as capabilities you hold

Every safeguard in the parent protocol is written with the sponsor as the actor: the sponsor verifies, the sponsor reports, the sponsor halts. Figure 8 redraws the same twelve safeguards as a use case diagram with the participant as the primary actor. Nothing in the protocol changed. Only the actor did, and the result reads differently: ten of the twelve are capabilities the participant exercises.

The two that are not are pressing the emergency stop and issuing a software version notice. Both require someone physically present in the room or inside the build system, and neither can be exercised against the participant: the stop can only halt, and the notice can only trigger a re-consent that the participant is free to decline.

3.10 Where each concern physically lives

A concern with no physical address is unanswerable. A concern that resolves to a named cabinet, a named console, or a named cable can be inspected, and inspection is what the surveyed patients asked for. Figure 9 assigns thirteen of the twenty-one to the component that answers them, grouped into the operating room, the on-premises cabinet, the governance bodies, and everything outside the building.

The figure's central claim is again an absence. There is a dashed line between the on-premises model and the network boundary, and it is struck through. During a procedure no route exists out of the building. That single struck line answers concern 17 more completely than any paragraph in this section.

3.11 The twenty-one, one row each

The table below is the paper's index. It states each concern in the form participants state it, the question they are most likely to ask, the answer this protocol gives, and the class of that answer. A reader who wants only one thing from §3 should take this table.

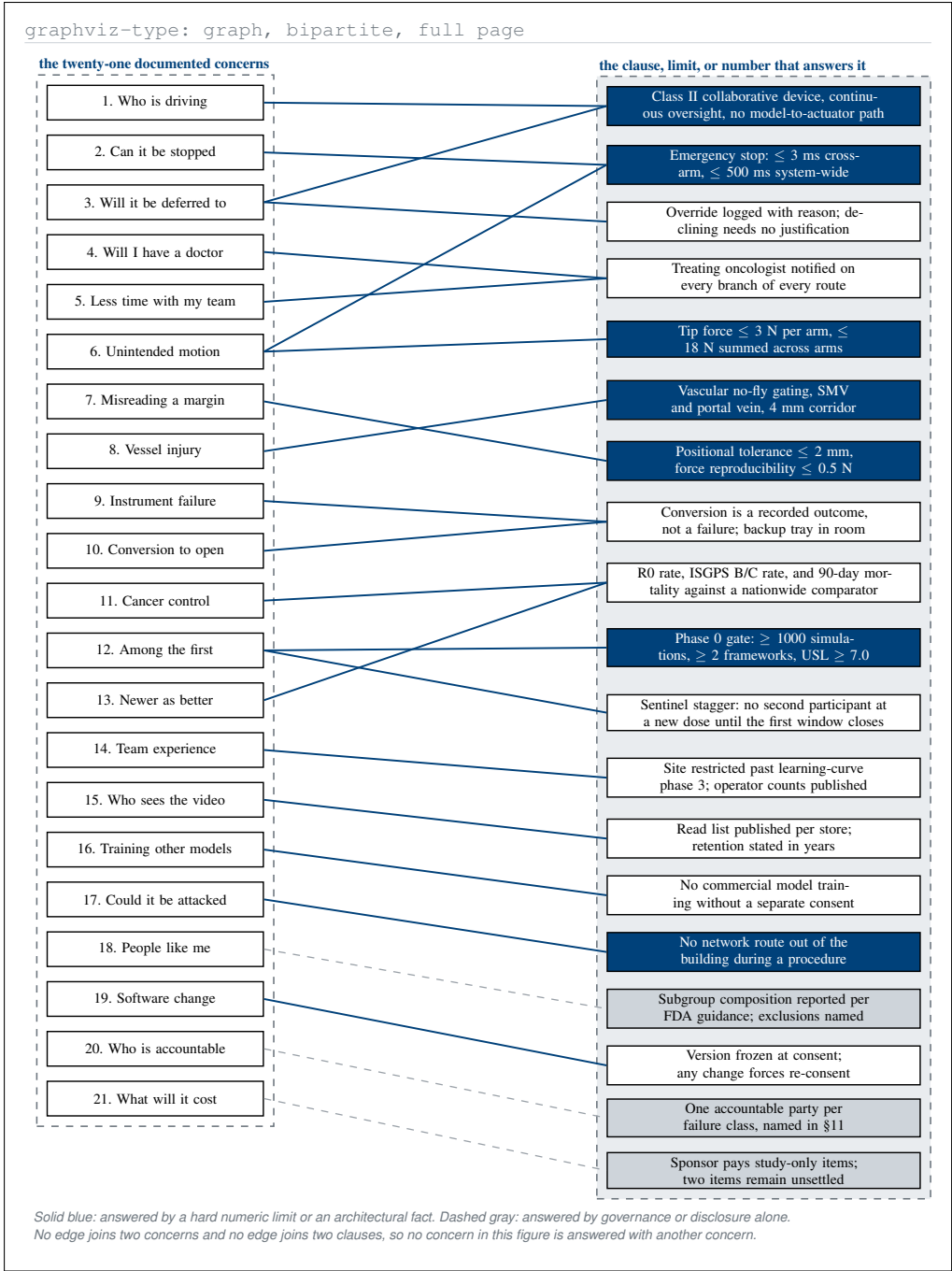


Figure 6. Every documented concern wired to the clause, limit, or number that answers it, drawn as a strictly bipartite graph so that no concern is answered with another concern, and dashed where the answer is governance or disclosure alone.

#	Concern	The question asked	The answer, and its class
#	Concern	The question asked	The answer, and its class
1	Who is driving the robot	Is my surgeon controlling every movement, or approving a plan, or supervising an automated task?	Approving. Every motion is surgeon-signed; the model has no actuator path. Hard architectural limit.
2	Can the surgeon stop it, and how fast	Can the surgeon stop the system immediately, and how quickly is human control restored?	At most 3 ms across arms, at most 500 ms system-wide, measured on every procedure. Hard numeric limit.
3	Will the surgeon defer to the model	Will my surgeon follow the model even when their own judgement differs?	Overrides are logged with a reason and the override rate is a published endpoint. Procedural guarantee.
4	Will I still have a doctor who knows me	Will I still have a surgeon who knows me and is personally responsible?	The operating surgeon is named in your consent and is accountable for every intra-operative motion. Procedural guarantee.
5	Less time with my care team	Will relying on the system reduce my face-to-face time?	The direct sponsor channel is additional; every branch notifies your own oncologist. Procedural guarantee.
6	Malfunction or unintended motion	Could the system move unexpectedly and injure something?	Tip force capped at 3 N per arm and 18 N summed, enforced in hardware. Hard numeric limit.
7	Misreading tissue or a margin	Could it misidentify a structure or a tumour boundary?	Positional tolerance 2 mm, force reproducibility 0.5 N, both gate-enforced. Hard numeric limit.
8	Injury to a major vessel	Could an autonomous movement cause severe bleeding?	A 4 mm no-fly corridor around the SMV and portal vein; a plan entering it is rejected before it is executable. Hard numeric limit.
9	Instrument failure mid-procedure	What if an instrument breaks while I am open?	Backup instruments and an open tray in the room for every procedure; conversion rehearsed in the checklist. Procedural guarantee.
10	Conversion to open surgery	What happens if the robotic approach has to be abandoned?	A recorded outcome, not a failure; participation continues; you are told before discharge. Procedural guarantee.
11	Cancer control	Will this remove my cancer as completely as standard surgery?	R0 rate, fistula rate, and 90-day mortality reported against a 1000-case nationwide comparator. Comparator, not a result.

#	Concern	The question asked	The answer, and its class
12	Being among the first	Am I one of the first people to receive this?	Phase 0 gate of 1000 simulations across two frameworks precedes any patient; sentinel stagger precedes any second dose. Hard gate.
13	Newer described as better	Is this actually better, or just newer?	No superlative appears in this paper; the FDA's own caution is reproduced verbatim. Disclosure.
14	Team experience	How many of these has my surgeon done, on this exact platform?	Case count, platform count, autonomy level, and trainee participation published before consent. Procedural guarantee.
15	Who sees the recording	Who will see the video of my operation?	A published read list per store; operating surgeon, Site PI, and you. Procedural guarantee.
16	Training other models	Will my data be used to train other systems?	Not without a separate consent; drawn in Figure 25 as an absent edge. Procedural guarantee.
17	Could it be attacked	Could someone interfere with the system during my operation?	No network route out of the building during a procedure. Hard architectural limit.
18	People like me	Was it tested on people with my anatomy, age, and history?	Composition reported per FDA guidance; subgroup performance not estimable at $n \leq 18$. Disclosure.
19	Software change after consent	Could the software change after I agree?	Version frozen at consent; any change forces a re-consent before any procedure. Procedural guarantee.
20	Who is accountable	Who answers if I am harmed?	Exactly one accountable party per failure class, named in §11 and published before consent. Governance.
21	What will it cost	What will I pay, during and after?	Line by line in §12; two items, lost income and post-closure drug access, remain unsettled. Governance.

Seven of the twenty-one are answered by a hard numeric or architectural limit, nine by a procedural guarantee, and five by governance or disclosure alone. Those proportions are the honest shape of the answer, and they are the same proportions Figure 5 colours and Figure 7 plots.

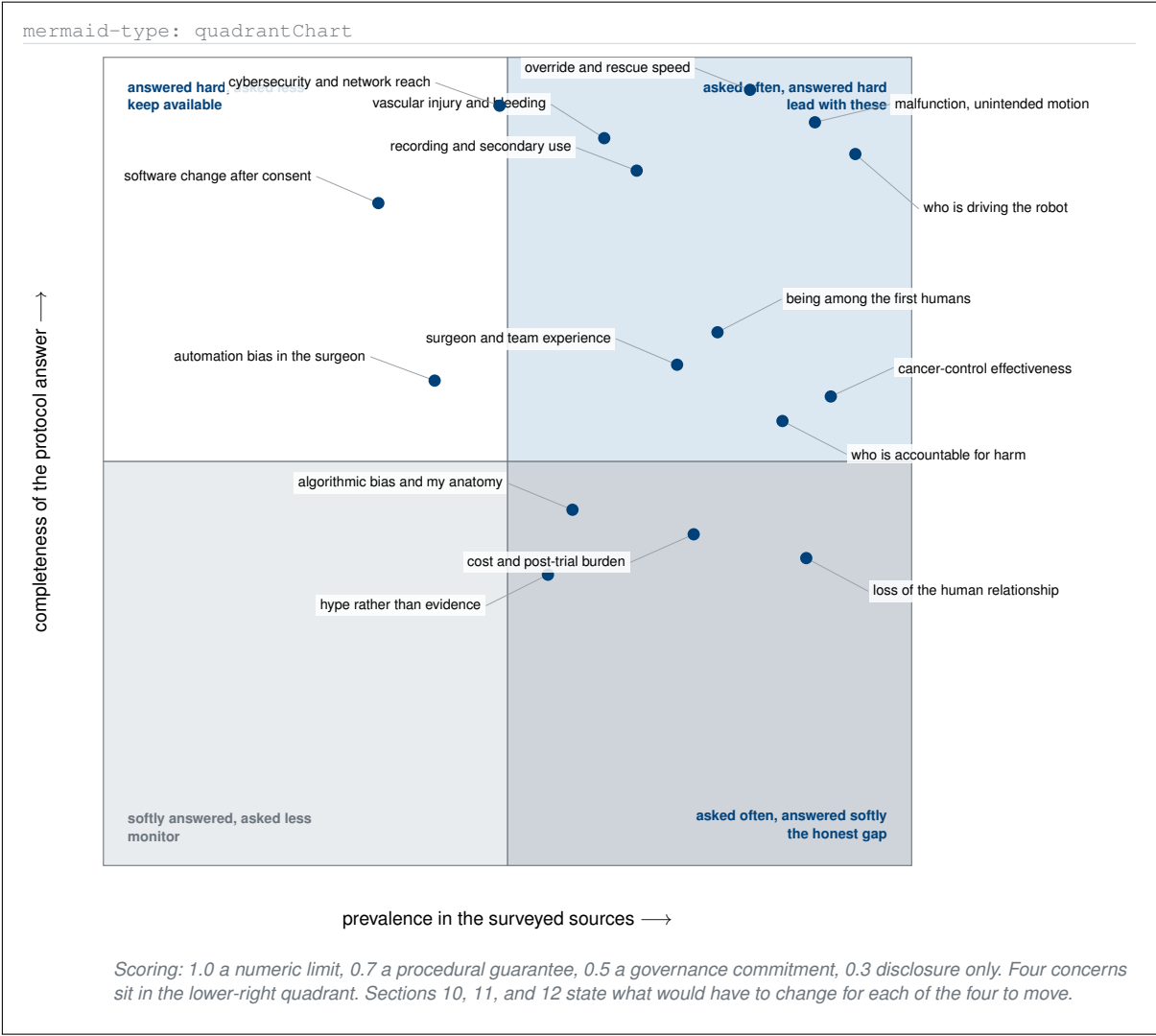


Figure 7. Sixteen documented concerns positioned by how often they are raised in the surveyed literature against how completely this protocol answers them, with the lower-right quadrant collecting the concerns answered by governance or disclosure alone.

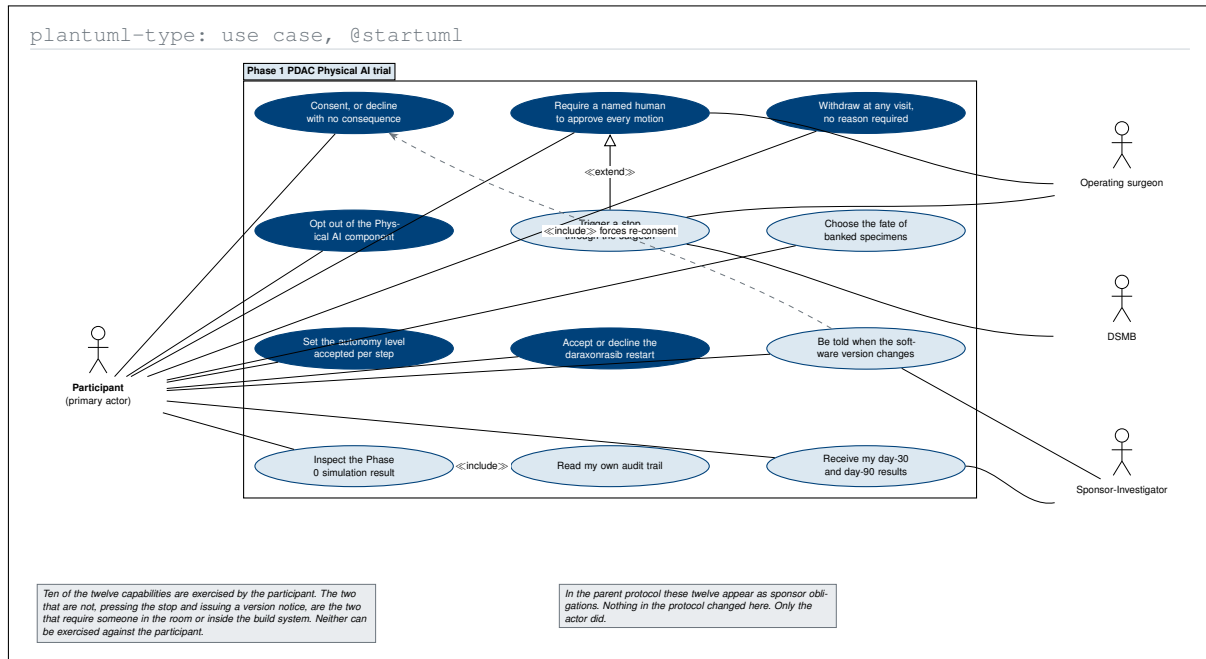


Figure 8. The twelve safeguards of the protocol redrawn as a use case diagram with the participant as the primary actor, so that each one reads as a capability they hold rather than as an obligation the sponsor has undertaken on their behalf.

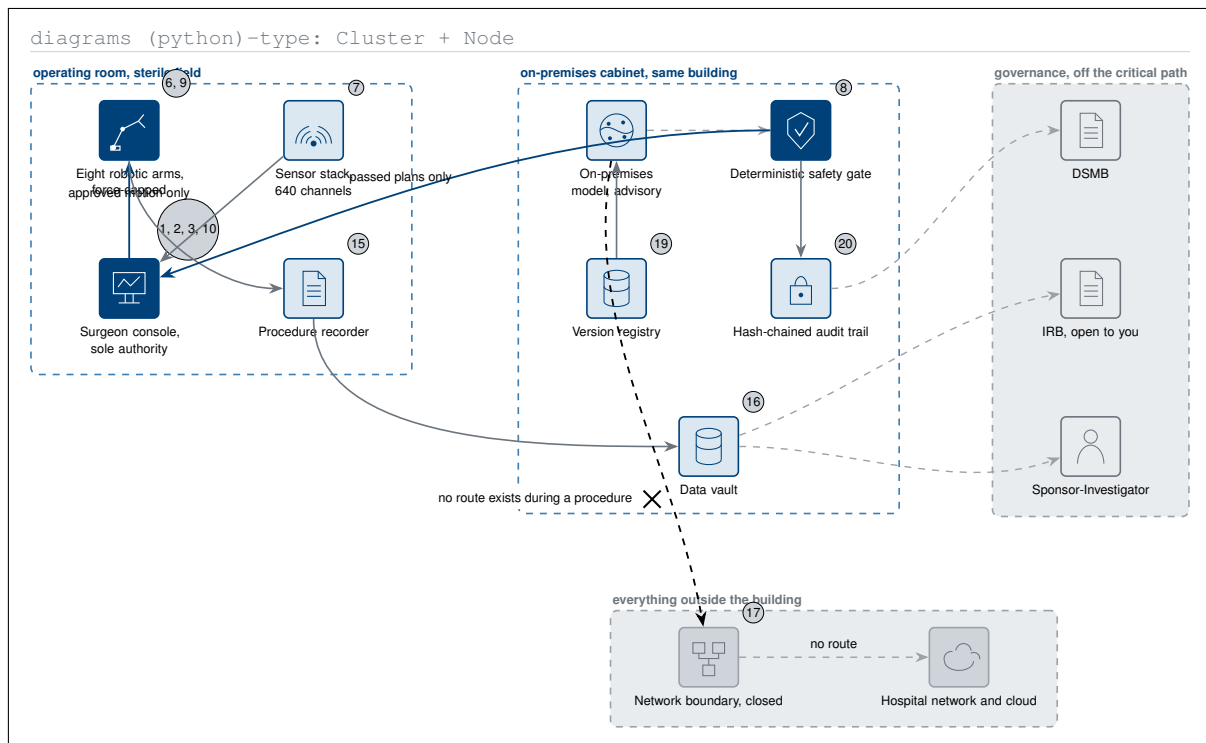


Figure 9. Each documented concern resolved to the physical component that answers it, grouped into the operating room, the on-premises cabinet, the governance bodies, and everything outside, with the intra-procedural route drawn and struck through.

4 Objectives and Patient-Facing Endpoints

4.1 What the study is trying to find out

An objective in a protocol is written as a noun phrase, and a noun phrase is hard to argue with because it does not commit to anything. Restated as questions, the objectives of this study are these.

Primary. Is the combination of perioperative daraxonrasib and an on-premises LLM-directed eight-arm robotic Whipple safe enough, and practical enough, to be carried forward? Safety is measured three ways: the rate of dose-limiting toxicity across the three dose levels, the rate of device-related and procedure-related serious adverse events through day 30, and the rate of Clavien-Dindo grade III or higher complications through day 30.

Secondary. How did the operation and the drug perform on the outcomes that matter to the disease? Four measures: the R0 resection rate at day-90 pathology, the ISGPS grade B/C postoperative pancreatic fistula rate [28], 90-day mortality, and progression-free and overall survival to 24 months.

Exploratory. What did the advisory layer actually contribute? Three measures: how often the surgeon accepted the model's proposed step, how often they declined it, and how often the deterministic gate rejected a proposal before the surgeon saw it. These are reported whether or not they are favourable, because a system whose override rate is never published cannot be audited for automation bias [23].

4.2 Every endpoint, and the sentence it will let someone say

The parent protocol stops at the endpoint definition [1]. A participant who reads only that definition learns that the study measures an ISGPS grade B/C postoperative pancreatic fistula rate, and is no better informed than before. Figure 10 carries each endpoint one rank further, to the plain sentence that endpoint will authorise a clinician to say to the participant or their family once it reads out.

Three of those sentences are marked apart in the figure, because they are the three that patients in the surveyed literature ranked highest: the margins came back clear, you were alive at three months, and the cancer had not returned by this visit [13, 22]. The others matter, but those three are what people actually asked to be told.

4.3 The endpoint table, with a column the protocol does not carry

Class	Endpoint	Timepoint	What it means for you
Primary	Dose-limiting toxicity rate, DL1 to DL3	Day 30	The dose you were given was tolerated at your level
Primary	Device- and procedure-related serious adverse event rate	Day 30	Nothing the device did caused you serious harm
Primary	Clavien-Dindo grade III or higher rate	Day 30	Your recovery needed no unplanned major intervention
Secondary	R0 resection rate	Day 90	The margins came back clear
Secondary	ISGPS grade B/C fistula rate	Day 30	Your pancreas join healed without a draining leak
Secondary	90-day mortality	Day 90	You were alive at three months

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Class	Endpoint	Timepoint	What it means for you
Secondary	Progression-free and overall survival	24 months	The cancer had not returned by this visit
Exploratory	Advisory concordance and override rate	Per procedure	Your surgeon disagreed with the model here, and was free to

4.4 The endpoint registry as a typed record

An endpoint is not a sentence, it is a record. It has a definition, a timepoint, an analysis population it is computed over, a class of evidence, and, in this paper, a plain meaning. Drawing the registry as a schema rather than a list forces every one of those fields to be filled for every endpoint, and exposes any endpoint where one is empty. Figure 11 does that, with foreign keys to the analysis populations and to the four provenance classes that §10 uses.

The ‘plain meaning’ row is the one field no clinical protocol carries. Treating it as a required field of the record, rather than as a nice addition, is the editorial position of this paper expressed as a schema constraint.

4.5 What a positive result would and would not establish

Suppose the study completes all three dose levels with no more than one dose-limiting toxicity in six at any level, no device-related serious adverse event through day 30, an R0 rate consistent with the nationwide robotic comparator, and an ISGPS grade B/C fistula rate at or below the phase-4 figure from that comparator [17]. That is the best realistic outcome.

It would establish that the combination can proceed to a study that could test survival. It would not establish that the combination improves survival, and it would not establish that the robotic approach is superior to an open operation for cancer control. Eighteen participants cannot support either claim, the FDA’s own information page states that long-term cancer outcomes have not been established for every use of robotically assisted surgical devices [21], and the National Cancer Institute’s guidance says the same thing about early-phase studies in general [20].

The reason to say this plainly, in a paper whose purpose is to argue for the trial, is that the surveyed literature shows patients discount claims that outrun their evidence and trust the sources that mark their own limits [3]. An advocacy document that overstated the primary endpoint would undermine the twenty-one specific answers the rest of the paper gives.

4.6 Risk and benefit, stated as a balance rather than a list

A risk-benefit section in a protocol lists risks and then lists benefits. The list format hides the only question that matters to the person deciding, which is how the two compare for them specifically.

The risks that are new. Three risks in this study do not exist in a standard Whipple. An unintended robotic motion, bounded by the force caps and the no-fly corridor. A model-proposed step that is wrong and is nonetheless approved, bounded by the surgeon’s obligation to review confidence and rationale and by the published override rate. And a systemic agent given across the perioperative window, bounded by the dose-escalation rule and by the pause across the healing period.

The risks that are not new. Everything else. Pancreatic fistula, delayed gastric emptying, bleeding, infection, and the mortality of a major upper abdominal operation are risks of the operation, not of the

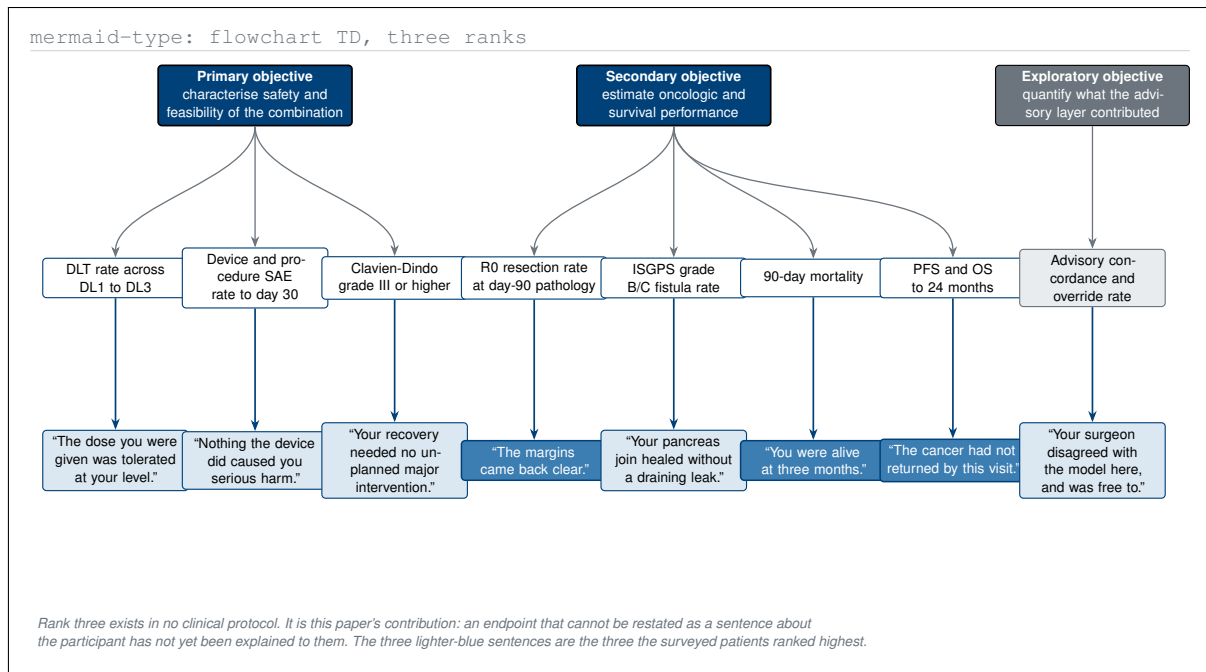


Figure 10. Each objective carried through its endpoints to the plain sentence that the endpoint will authorise, with the three sentences patients in the surveyed literature ranked highest, clear margins, being alive, and no return, marked apart from the rest.

study, and they are the reason the comparator in §10.2 is a series of the same operation performed conventionally with a robot rather than a series of open surgery.

The benefits that are plausible. Precision at the tip within a 2 mm tolerance, force bounded below the level at which pancreatic tissue tears, a recorded rationale for every operative decision, and a perioperative drug schedule optimised to a measured fistula grade rather than a fixed calendar.

The benefit that is not claimed. Longer survival. This study is not powered for it, will not test it, and this paper does not assert it.

Risk	New to this study	What bounds it
Unintended robotic motion	Yes	3 N per arm, 18 N summed, 4 mm no-fly corridor, 3 ms cross-arm halt
An incorrect model proposal approved	Yes	Confidence and rationale shown; override rate published; declining needs no justification
Perioperative systemic agent	Yes	3+3 escalation, sentinel stagger, pause across the healing window
Pancreatic fistula	No	ISGPS grading, drain surveillance, restart advisory keyed to the grade
Delayed gastric emptying	No	Standard management; the commonest cause of a longer inpatient stay
Bleeding and infection	No	Standard of care at a phase-3-or-better site

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Risk	New to this study	What bounds it
Operative mortality	No	Site restricted past learning-curve phase 3; 90-day mortality is a reported endpoint

4.7 What each endpoint costs you to measure

Every endpoint has a price, paid by the participant in time, in blood, or in discomfort, and a protocol that lists endpoints without their price is describing only half of what it is asking for. The table below prices them.

Endpoint	What it costs you	Whether the cost is avoidable
Dose-limiting toxicity rate	Safety laboratory draws at seven timepoints	No. It is the primary endpoint and the reason the study exists
Device-related serious adverse events to day 30	The day-30 visit, and a symptom diary	No, but the diary is short and may be completed by telephone
Emergency-stop latency	Nothing. It is measured from the machine	Not applicable. No participant burden at all
Force-envelope compliance	Nothing. It is measured from the machine	Not applicable. No participant burden at all
Surgeon override rate	Nothing. It is read from the audit trail	Not applicable. No participant burden at all
R0 margin rate	The day-90 pathology review, already part of care	No, and the result is reported back to you directly
ISGPS fistula grade	Drain surveillance during the inpatient stay	No. Drain management is standard care after a Whipple
Progression-free and overall survival	Imaging every 12 weeks to 24 months	Partly. Imaging may be done at your own centre and read centrally
Patient-reported experience	Four questionnaires, about 20 minutes each	Yes. Declining a questionnaire has no consequence for your participation

Three of the nine endpoints cost the participant nothing at all, because they are read from the machine and from the audit trail rather than from the person. That is a design property worth stating: the endpoints that carry the most weight in answering the robot-specific concerns are precisely the ones that require nothing from the participant. The endpoints that do carry a cost are, with one exception, endpoints that

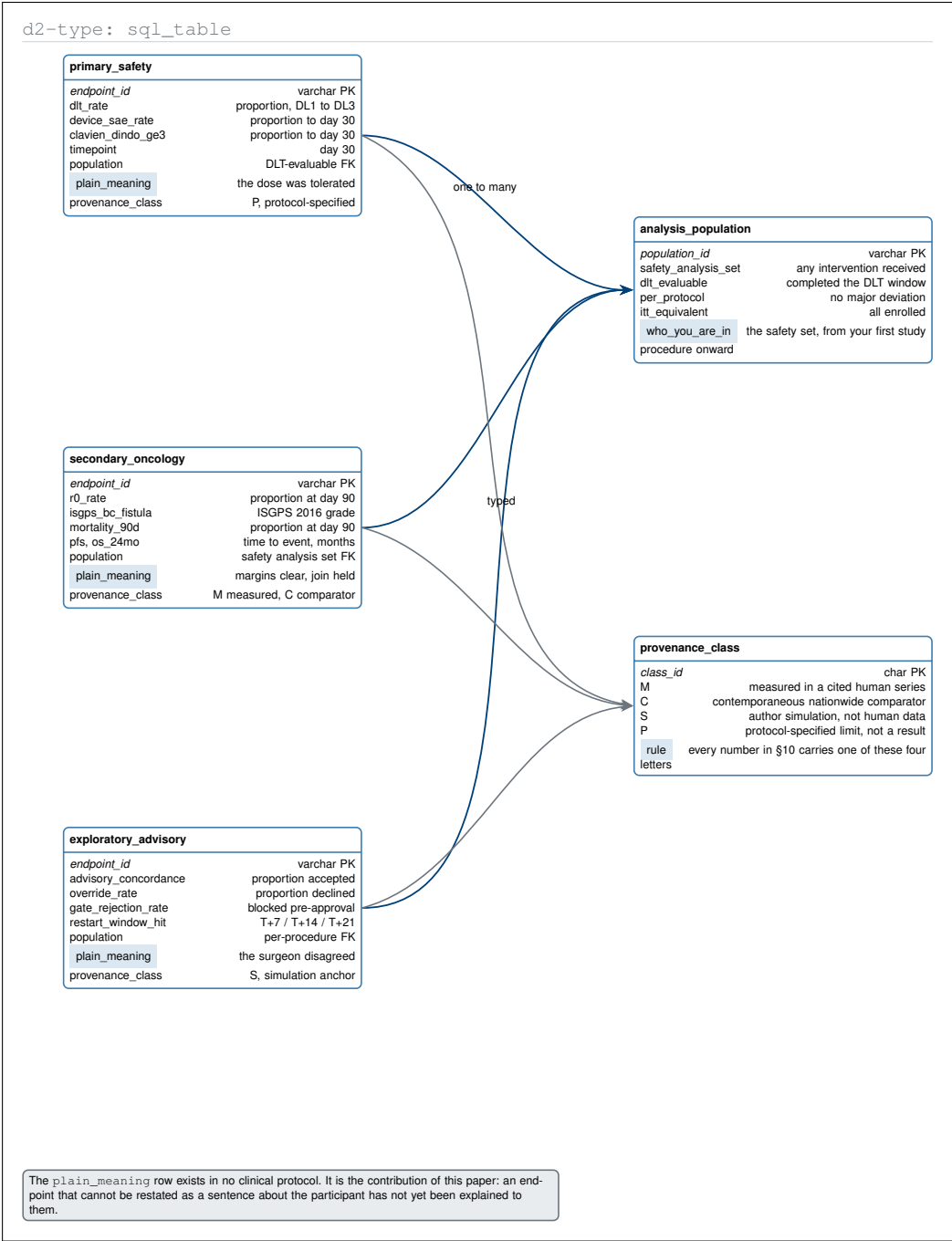


Figure 11. The endpoint registry drawn as typed records with foreign keys to the analysis populations and to the provenance classes, and with a plain-meaning row that no clinical protocol carries but that this paper treats as a required field of the record.

a standard Whipple would generate anyway. The exception is the patient-reported experience set, and it is optional.

4.8 Which endpoints are reported back to you, and when

An endpoint that only ever appears in a sponsor's summary table is invisible to the person who supplied it. Four of the nine are reported back to the individual participant, on a stated schedule, and the rest are reported to the cohort.

The day-90 pathology result, including the R0 determination, is reported to the participant directly and to their own oncologist, because it is the result that changes what happens next in their care. The ISGPS fistula grade is reported during the inpatient stay, as it is assigned. Any device-related adverse event affecting the participant is reported to them as an affirmative obligation, which is one of the three edges pointing back to the root in Figure 2. Any study pause or halt is reported to every enrolled participant, whether or not it arose from their own case.

The remaining five are cohort-level and are reported in aggregate at study closure, with the individual participant's own rows available on request through the audit extract in §11.5. The distinction is not about secrecy. A single participant's emergency-stop latency distribution means very little on its own, and reporting it individually would invite exactly the over-reading this paper argues against elsewhere.

5 Study Design Explained

5.1 Open-label, single-arm, and what that means for you

Three technical terms describe the design, and each has a consequence a participant can act on.

Open-label means nothing about the treatment is concealed. The participant knows the dose level they are on, the software version that will operate, and the identity of the operating surgeon. There is no masking to maintain and therefore nothing that has to be withheld to protect the study.

Single-arm means there is one treatment group and no allocation. The concern reported most often about trial mechanics is being assigned to a treatment the participant would not have chosen [20, 13]. In this study that cannot happen: every participant who enrolls receives the same intervention, and the only thing that varies between them is the daraxonrasib dose level, which is set by what happened to the people ahead of them rather than by chance.

Dose-finding means the study is looking for the highest dose that is tolerated, not for the largest effect. A participant at DL1 is not receiving a lesser treatment; they are receiving the dose that the study is confident enough to start with, and their tolerance of it is what permits DL2 to open.

5.2 Your status in the study, and how it can change

Figure 12 draws participant status as a state machine, and the reason to use a state machine rather than a flowchart is the guard. Every transition in the figure carries a named condition that must evaluate true before the transition can be taken, and the protections live in the guards rather than in the states. A participant who wants to know what stands between them and the operating room should read the guard on the transition into the operative state: [Phase 0 gate signed], [USL $\geq$ 7.0 verified], and [pre-procedure matrix clear], all three, conjunctively.

One state in that figure exists only for the participant's benefit. The cohort hold is entered when a prior participant is still inside their sentinel window, and it costs the study calendar time. It would be tidier to fold it into a transition guard, and it is drawn as an explicit state because it is the clearest single piece of evidence that this design accepts delay in exchange for safety.

5.3 The 3+3 rule, and why you are never the untested dose

The rule is usually stated statistically and is easier to check operationally. Three participants are treated at a dose level. If none has a dose-limiting toxicity, the next level opens. If one does, the level is expanded to six rather than escalated; if that sixth participant brings the total to one of six, the next level opens, and if two or more of six have a toxicity, escalation stops and the level below becomes the recommended dose. If two or three of the first three have a toxicity, escalation stops immediately.

Layered on top is the sentinel stagger. The second participant at a new dose level is not dosed until the first participant's safety window has closed. That delay is the difference between a rule that protects the cohort and a rule that protects the individual, and it is why the answer to "am I the untested dose" is no: the dose a participant receives was cleared by the people ahead of them, and the study expands rather than escalates the moment anyone is harmed.

Figure 13 draws the rule as a decision tree, which is the right structure because there is exactly one path from the root to each leaf. Two of the leaves are stops, and one of those convenes the Data and Safety Monitoring Board and is reported to every enrolled participant. A dose-escalation figure with no stopping leaf would misrepresent the rule.

5.4 The dose levels

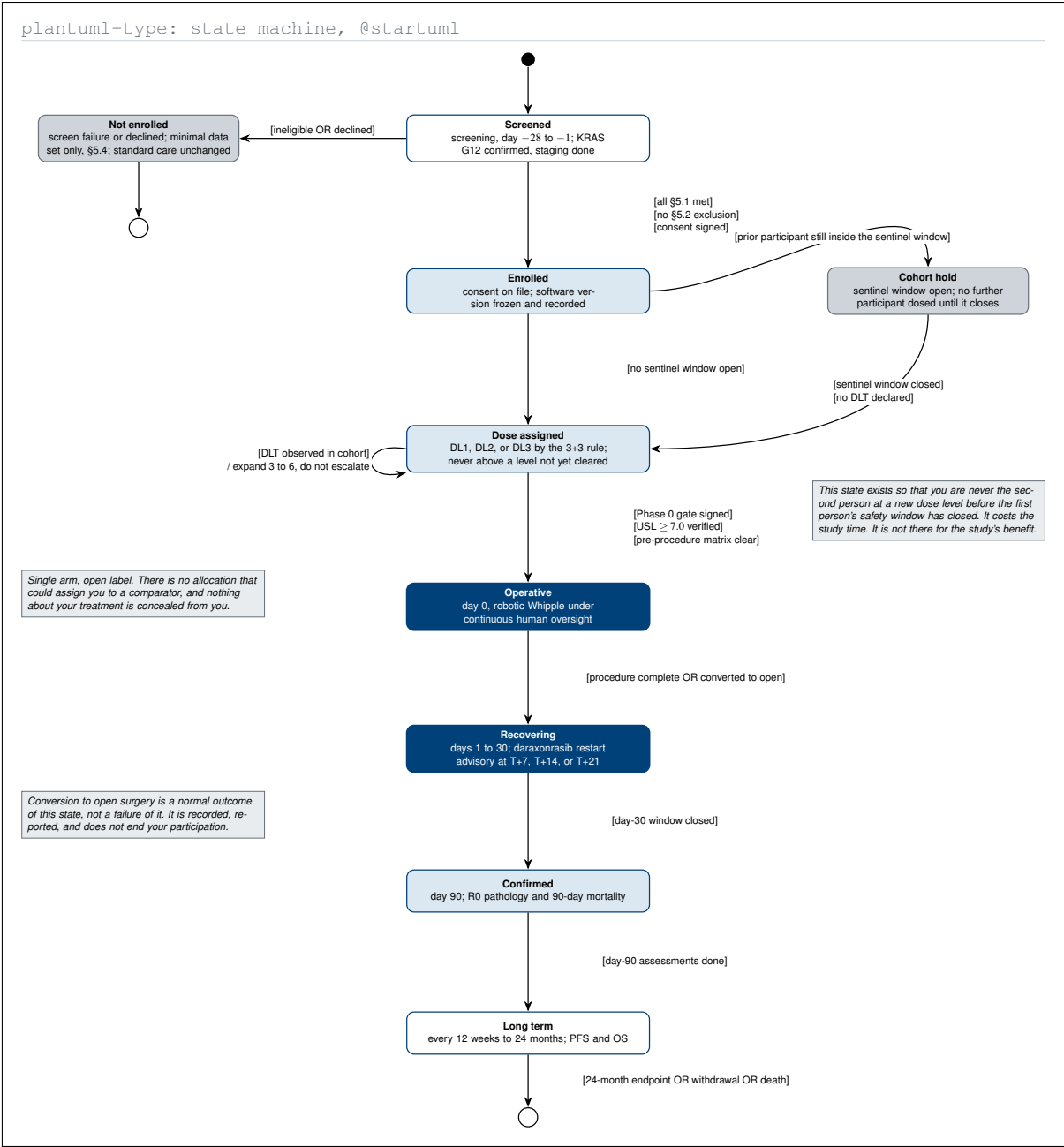


Figure 12. Participant status as a guarded state machine, screening through the 24-month endpoint, with the sentinel cohort hold drawn as an explicit state, because it is the clearest evidence that the design accepts calendar delay in exchange for safety.

Level	Daraxonrasib dose	Cohort size	What has to be true before this level opens
DL1	Starting dose level	3, expandable to 6	Phase 0 gate signed, USL at least 7.0, IND in effect, IDE approved by both the FDA and the IRB

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Level	Daraxonrasib dose	Cohort size	What has to be true before this level opens
DL2	One step above DL1	3, expandable to 6	DL1 cleared: 0 of 3, or 1 of 6, dose-limiting toxicities, and every sentinel window at DL1 closed
DL3	Two steps above DL1	3, expandable to 6	DL2 cleared on the same rule, plus a DSMB review at the cohort boundary

Planned enrollment is up to 18 participants across the three levels. Enrollment is staggered: no participant enters a level while a sentinel window at that level remains open. The dose anchor for the starting level is taken from the pan-KRAS programme in previously treated metastatic disease [15, 16], adjusted for the perioperative setting.

5.5 The Phase 0 gate that runs before any patient

No patient-facing robotic activity occurs until a simulation-validation gate has been signed. The gate has three components, and all three must pass.

Volume and independence. At least 1000 simulated procedures across at least two independent simulation frameworks. Independence matters more than volume: two frameworks that share a physics engine share its errors, and the protocol requires that they do not.

Fidelity. Motion reproduced within a 2 mm positional tolerance and a 0.5 N force tolerance across frameworks. These are the credibility questions that the ASME verification and validation standard for computational modelling in medical devices [29] and the NASA models-and-simulations standard [30] were written to frame.

Readiness. A documented Unified Safety Level of at least 7.0 for surgical deployment, under the Physical AI overlay adapted for end-to-end oncology investigations [7]. The simulation source for the operative values is the author's 60-second robotic Whipple and daraxonrasib simulation [8].

The gate sign-off is a dated document and the participant is shown it at the baseline visit. Section 10.3 reports what happened across those 1000 runs, including the 8.2 percent that were halted by a safety mechanism, and states in the same sentence that a simulated worst case is not a human worst case.

5.6 Stopping rules, stated before they are needed

Prespecified triggers pause or stop the study. Four of them are device-specific, and those are the ones a participant worried about the robot will look for first. Any breach of a per-arm or cumulative force cap. Any entry into a vascular no-fly corridor. Any emergency-stop latency measured above budget. Any loss of the heartbeat bus that is not resolved by the automatic halt.

Trigger	Effect	Who decides, and who is told
Two or more DLTs at a dose level	Escalation stops	Sponsor-Investigator declares; DSMB reviews; every enrolled participant is told
Any device-related death	Immediate study halt	DSMB; FDA and IRB notified within the reporting window

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Trigger	Effect	Who decides, and who is told
Force-cap breach at the tip	Procedure halt, then study pause	The gate halts automatically; the DSMB decides on resumption
Entry into a vascular no-fly corridor	Procedure halt, then study pause	The gate halts automatically; Physical AI Safety Review Committee re-gates
Emergency-stop latency above budget	Study pause pending re-gate	Physical AI Safety Review Committee; USL reassessed
Heartbeat loss not resolved by halt	Study pause pending re-gate	Physical AI Safety Review Committee; USL reassessed
Grade B/C fistula rate above the phase-1 comparator level	DSMB review at the cohort boundary	DSMB; the rate is published with the safety data

In every row, the participant is told. That is the point of publishing the table before it is needed: a stopping rule that a participant only learns about after it fires is a governance mechanism, and a stopping rule they were shown in advance is a commitment.

5.7 How an adverse event is reported, and how fast

Reporting timelines are usually written for the regulator. Restated for the participant, they answer a different question: how long after something happens to me does somebody outside this hospital know about it.

Event class	Reported within	Reported to	What you are told, and when
Death or life-threatening event, related and unexpected	7 calendar days	FDA, IRB, DSMB	Immediately, and in writing before discharge
Other serious, related, and unexpected event	15 calendar days	FDA, IRB, DSMB	Within 24 hours, by the Site PI
Unanticipated adverse device effect	10 working days	FDA, IRB, DSMB	Within 24 hours, by the Site PI
Any safety-limit event, force cap or no-fly	Same procedure	Physical AI Safety Review Committee	Before you leave hospital, with the measured value
Any emergency-stop activation	Same procedure	Sponsor-Investigator, DSMB	Before you leave hospital, with the measured latency
Protocol deviation affecting you	5 working days	IRB	Within 5 working days, in writing

The last three rows are the device-specific ones and they have no counterpart in a conventional drug protocol. They exist because a safety-limit event is not an adverse event in the regulatory sense: nothing may have happened to the participant at all. The protocol reports them anyway, and tells the participant, because a limit that is never reported when it fires is indistinguishable from a limit that does not exist.

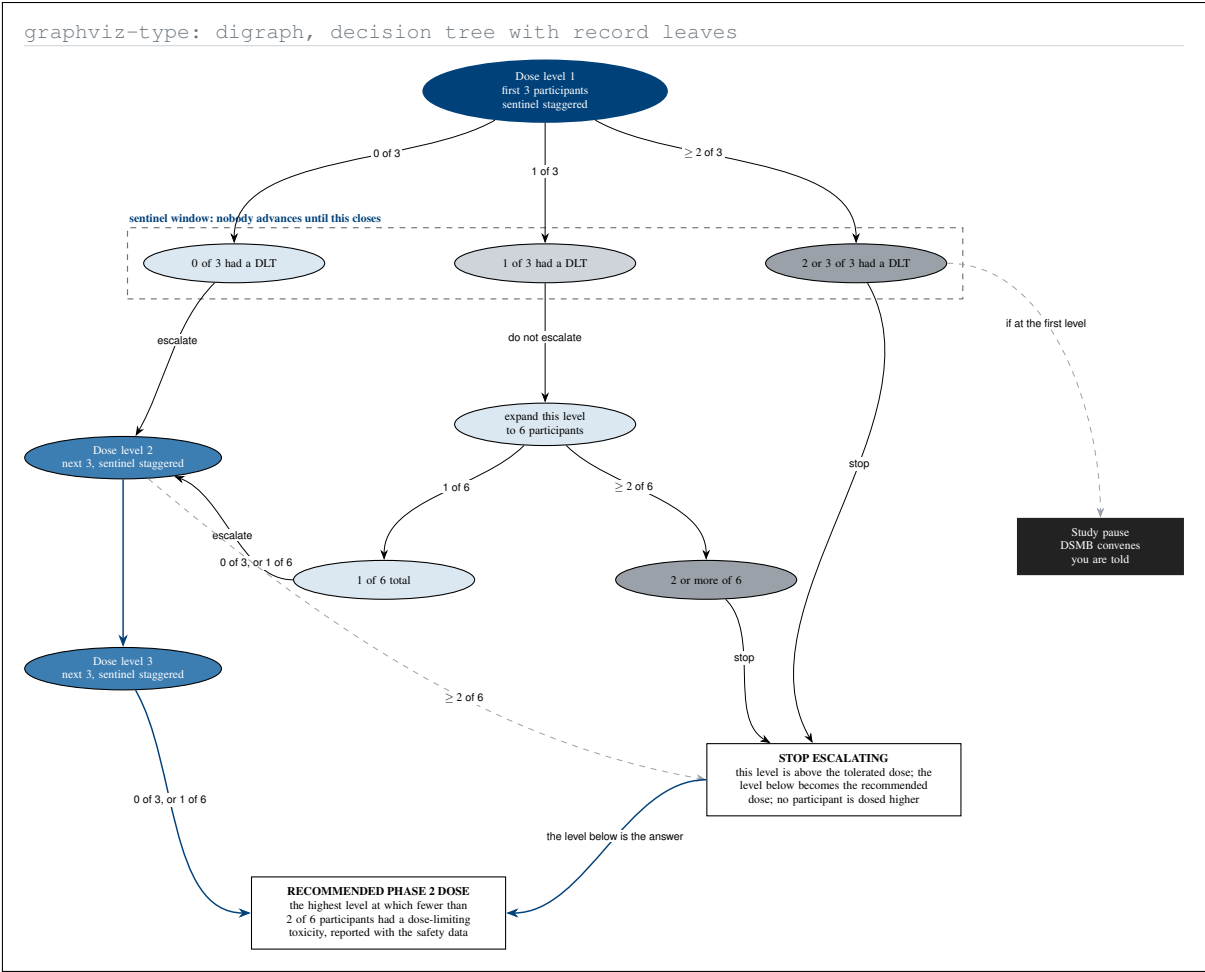


Figure 13. The 3+3 dose-escalation rule drawn as a decision tree with exactly one path from the root to each leaf, including the two stopping leaves and the study pause that convenes the Data and Safety Monitoring Board and is reported to every participant.

5.8 What a dose-limiting toxicity is, in this study

The term decides whether a dose level opens, so it needs a definition a participant can check rather than a category name. A dose-limiting toxicity in this study is any of the following, occurring within the day-30 window and attributed to daraxonrasib rather than to the operation.

Event	Why it counts
Grade 4 haematologic toxicity lasting more than 7 days	Marrow suppression at this duration delays adjuvant therapy
Grade 3 or higher non-haematologic toxicity, excluding controllable nausea and fatigue	The exclusions are the toxicities that respond to supportive care

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Event	Why it counts
Any toxicity delaying the planned operation by more than 14 days	A delay of this length changes the oncologic calculus
Any toxicity preventing the restart of the drug by T+21	The restart window is the endpoint the advisory layer targets
Grade 3 or higher hepatic transaminase elevation	The operation loads the liver; the drug must not compound it

Attribution between drug and operation is adjudicated by the Site Principal Investigator with DSMB review at the cohort boundary, and the attribution is recorded with its reasoning. That reasoning is in the participant's own audit extract, described in §11.5.

5.9 Why there is no control group, and what replaces one

The most frequent design objection a participant raises is the absence of a control group, and it deserves a direct answer rather than a methodological one. Without a control group there is nothing to compare the study's results against, which appears to make the results uninterpretable.

The answer has three parts. First, a Phase 1 study is not asking a comparative question. It is asking whether the combination can be delivered at all and whether it hurts people, and both of those are answerable from the single arm. Second, randomising a participant to a conventional Whipple when the platform's feasibility is unknown would expose the randomised group to a study's burden without any prospect of the study's benefit, which is the reason ethics review does not permit it at this stage. Third, and most practically, an external comparator replaces the internal one.

That comparator is a published, contemporary, nationwide robotic pancreaticoduodenectomy series, and it is used in one specific way: not to declare superiority, which §10.5 shows is impossible at this size, but to set the boundary beyond which the study stops. If the R0 rate falls materially below the comparator, or the ISGPS grade B/C fistula rate rises materially above it, the stopping rules in §5.6 apply. The comparator is therefore a floor rather than a target, and every number taken from it is labelled C for comparator in the provenance table in §10.6, never presented as a result of this study.

The corollary is worth stating in the participant's terms. A participant in this study is not being compared with anyone. They are being watched against a published floor, and if the study drops through it, the study stops before the next person is enrolled.

5.10 What happens if the study is stopped while you are in it

Stopping rules are written from the sponsor's side, and read from the sponsor's side they describe an orderly wind-down. From the participant's side the question is narrower: what happens to me, specifically, on the day it stops.

If the study halts before the operation, the participant returns to standard care with the study's staging imaging and tissue review in hand, and the site makes the referral rather than leaving it to the participant. Nothing about the halt changes the participant's eligibility for a conventional resection, and the work-up already done does not have to be repeated.

If the study halts after the operation, the operation is complete and is not undone by the halt. Follow-up continues, because the safety endpoints that matter to the participant, the day-30 and day-90 assessments,

are the ones that protect them. Daraxonrasib is stopped or continued according to the reason for the halt: a halt for a device reason does not by itself stop a drug the participant is tolerating, and a halt for a drug reason does.

In both cases every enrolled participant is told, whether or not the halt arose from their own case, and the notification states the reason rather than the fact alone. That is one of the three affirmative obligations in Figure 2, and it is not discretionary.

Data already collected remains in the study record unless the participant withdraws it under the routes in §8, and §8.1 states which of those routes can and cannot remove data that has already been analysed.

6 Who Can Join, and Who Decides

6.1 Eligibility as a two-way gate

An eligibility flowchart normally runs in one direction. The protocol tests the patient against a list and returns one of two results: eligible, or screen failure. The patient is the object of the process.

Under the author's proposed legislation the process runs in both directions. The participant is also doing the selecting, and their decisions leave a record. H. R. 9510 v5 [2], which supersedes the earlier H. R. 9501 to H. R. 9507 numbering used in the antecedent analysis [5], would establish participant self-selection, participant choice of autonomy level, and participant data self-custody as recorded rights rather than as informal courtesies. This protocol adopts all three voluntarily, in advance of any enactment.

Figure 14 draws both directions as facing columns of decisions converging on a single enrollment node that requires both to be satisfied. The asymmetry that remains is real and is stated in the table below it: a protocol criterion can only be changed by amendment, while a participant's answer can be revisited at any visit without a reason.

	Protocol column	Participant column
Who authors the criteria	Sponsor-Investigator, reviewed by the IRB	The participant, informed by §3 of this paper
What a "no" produces	Screen failure; minimal data set retained under §5.4	Opt-out recorded; standard-of-care pathway unchanged
Can it be revisited	Only by protocol amendment	At any visit, without giving a reason
Is it recorded	Yes, in the screening log	Yes, in the consent record, per H. R. 9510 v5

6.2 Inclusion and exclusion, in plain terms

You may join if	Why this criterion exists
You are 18 or older	The study has no paediatric safety data and does not enrol children
Your tumour is confirmed PDAC on tissue	The drug and the operation are both specific to this disease
Your tumour carries a KRAS G12 mutation	Daraxonrasib acts on RAS(ON); without the mutation there is no rationale
Your tumour is resectable or borderline resectable	The operation must be able to achieve a clear margin
Your performance status is ECOG 0 or 1	A Whipple is a major operation and the study will not expose a frailer participant to it
Your organ function is adequate	Both the drug and the operation load the liver, kidneys, and marrow

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You may join if	Why this criterion exists
You can attend the day-30 and day-90 visits	The primary and confirmatory endpoints are measured at those two visits
You are able to give informed consent yourself	Consent under 21 CFR §50.20 is personal to you
You may not join if	Why this criterion exists
Your disease is metastatic	A curative-intent resection is not the right operation for you
You have had a prior pancreatic resection	The anatomy the platform is validated against is not present
You have an uncontrolled intercurrent illness	Perioperative risk could not be attributed cleanly to the intervention
You are pregnant or breastfeeding	The drug has no safety data in pregnancy
You have had prior upper abdominal radiation to the field	Tissue planes are altered in a way the no-fly geometry does not model
You cannot attend the follow-up schedule	The safety endpoints depend on the day-30 and day-90 visits
You are receiving another investigational agent	Attribution of an adverse event would not be possible

6.3 Was this tested on people like me?

This is the concern where an advocacy paper is most tempted to reassure and least able to. A first-in-human study of at most eighteen participants cannot establish subgroup performance for any subgroup, and any claim that it could would be false. The honest answer has three parts.

First, what is reported. The protocol reports the composition of the enrolled population by age, sex, race, ethnicity, body mass index, prior therapy, tumour location, vessel contact, and comorbidity burden, consistent with FDA guidance on evaluating and reporting age, race, and ethnicity data in device clinical studies [25]. Composition is reported even though performance by subgroup is not estimable, because a reader can then judge for themselves how close the enrolled population is to them.

Second, what is not claimed. No subgroup analysis is prespecified, no subgroup finding will be reported as a result, and any post hoc subgroup observation will be labelled as hypothesis-generating.

Third, what happens afterwards. Real-world performance monitoring for AI-enabled devices is an active area of FDA attention [26], and the protocol commits the sponsor to contributing this study's device performance data to any such monitoring programme in a form that preserves the de-identification described in §9.3.

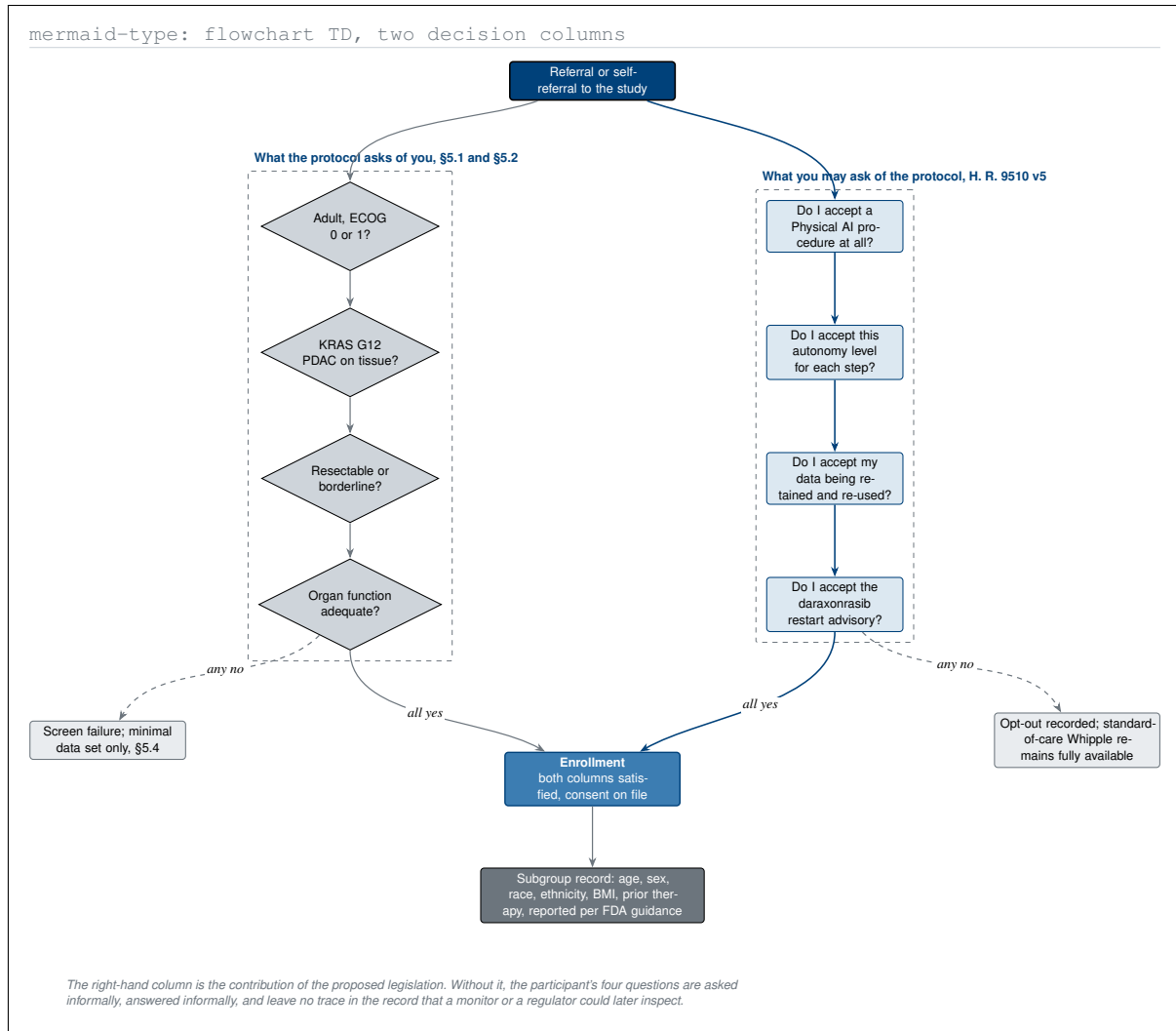


Figure 14. Eligibility drawn as two decision columns running against each other; what the protocol asks of the participant and what the participant may ask of the protocol, converging on a single enrollment node that requires both columns to be satisfied.

Characteristic	Reported	What the study can and cannot say
Age, sex, race, ethnicity	Yes, per participant	Composition is reported; subgroup performance is not estimable at $n \leq 18$
Body mass index	Yes, per participant	Relevant to port placement and to force at the tip
Prior therapy	Yes, per participant	Prior radiation changes tissue planes and is an exclusion in this study
Tumour location and vessel contact	Yes, per participant	Determines how much of the no-fly corridor is in play
Comorbidity burden	Yes, per participant	Feeds the Clavien-Dindo interpretation at day 30

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Characteristic	Reported	What the study can and cannot say
Operator case count	Yes, per procedure	Published to participants before consent, per §11.3

6.4 Booking a slot yourself, and getting an answer from a person

Under current practice a patient learns that a trial exists from a clinician who happens to know about it, and every subsequent question routes back through that clinician. That works when the clinician has time and fails when they do not. The direct channel proposed in H. R. 9510 v5 [2] and modelled in the author's automated sponsor platform [31] adds a second route with a stated service level: a screening slot may be requested at any hour without a referral, a provisional hold is held for 72 hours so that nobody is rushed into confirming, and a named human at the sponsor answers a direct question within one business day with the answer logged verbatim in the participant's record. The target from first request to a confirmed screening visit is five business days.

The protection that makes this acceptable rather than threatening is drawn on every branch of Figure 15: the participant's own treating oncologist is notified whichever way the eligibility pre-check resolves, including the branch where the participant turns out to be ineligible. The channel is additional. It is never a replacement, and the surveyed literature is clear that displacement of the existing clinician relationship is the fear attached to exactly this kind of mechanism [13, 22].

6.5 The five choices you keep

A consent form presents every choice at once, at the moment of greatest stress, and the predictable result is that most participants recall almost none of them [12, 32]. The same five choices, revealed as layers that accumulate rather than replace, are each live at a different point in the study.

Figure 16 draws them in order, with the interval each is live in and what happens if the answer is no. The conclusion the layout makes visible is stated at its foot: not one of the five is exercised on the day of surgery. Every one is live before the operating room, and three of the five are still live afterwards. No decision that matters is taken at the moment of least capacity to take it.

6.6 What the consent conversation should cover

The surveyed literature converges on ten items that a consent process for a Physical AI surgical trial should address, and this paper is designed to be read before that conversation so it can be spent on the items a document cannot settle. The ten are set out below with the section of this paper that prepares each one.

#	Item the consent process should cover	Where this paper prepares you for it
1	The exact role of the robot and the model, step by step	§7.3 and Figure 19: the model proposes, the gate tests, the surgeon signs
2	The division of control between the system and the team	§7.3 and §7.4: who initiates, who approves, who may interrupt

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#	Item the consent process should cover	Where this paper prepares you for it
3	The evidence supporting the procedure	§10.2 and §10.5: the comparator, and what this study cannot establish
4	Known, anticipated, and unknown risks	§4.5 and §7.6: the risk balance and the fault tree with its one gap
5	The failure and conversion plan	§7.4 and §7.7: the stop budget, the backup tray, and conversion as a recorded outcome
6	The experience of the surgeon and the team	§11.4: case count, platform count, autonomy level, trainee participation
7	Data governance and cybersecurity	§9.3 and §9.4: read lists, retention in years, and the closed boundary
8	Population representativeness	§6.3: composition reported, subgroup performance not estimable
9	Software lifecycle management	§8.2 and §8.4: version frozen at consent, change forces re-consent
10	Trial mechanics and alternatives	§5.1 and §8.1: single arm, no allocation, four routes out

The items are drawn from the minimum-information list in the surveyed concern set [13, 12, 23] and re-ordered to match the order this paper takes them in. A participant who arrives having read §3 and §7 will spend the conversation on item 6, the experience of the particular team in front of them, which is the one item no document can answer in advance.

6.7 Second opinions, and how to get one that is worth having

Nothing in this protocol discourages a second opinion, and §8 makes declining after consent free of consequence. The practical question is not whether to seek one but how to make it useful, because a second opinion given without the protocol in front of it is an opinion about robotic surgery in general rather than about this study.

Three documents make the difference, and the site will provide all three on request: the protocol synopsis, the Phase 0 gate sign-off with its Unified Safety Level, and the eligibility determination that names which criteria the participant meets and which were borderline. A surgeon reading those three can answer the question the participant is actually asking, which is whether this operation at this site is a reasonable choice for this disease.

Two questions are worth putting to the second opinion directly. Whether the tumour is resectable on the imaging as read by an independent radiologist, since resectability is the eligibility criterion most open to interpretation and §7.2 excludes borderline disease. And whether the participant's own comorbidity profile makes a longer operative time a material risk, since a platform-assisted Whipple is not shorter than a conventional one during a learning phase.

A second opinion that disagrees is not a problem for the study. It is information the participant is entitled to have before consent rather than after, and a site that treats disagreement as an obstacle has misunderstood which party the protocol serves.

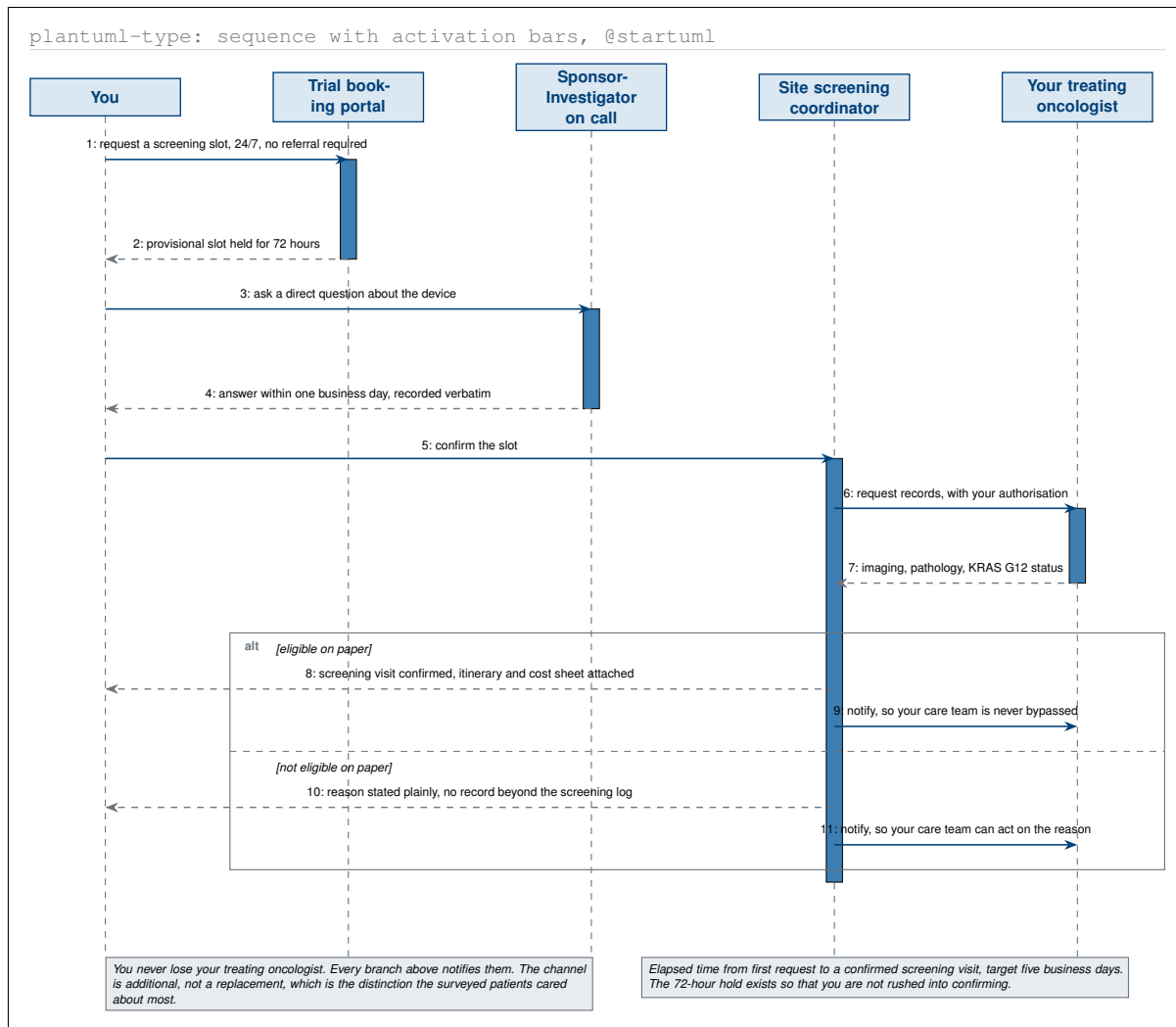


Figure 15. Booking a screening slot directly and receiving an answer from a named human at the sponsor, with activation bars showing who is blocked at each step and every branch of the eligibility fragment notifying the participant's own oncologist.

6.8 Bringing someone with you, and who may decide

Consent is a conversation, and a participant may bring anyone they choose to it. The protocol places no limit on that, and the site is expected to schedule the conversation so that a family member or advocate can attend rather than at whatever time is convenient for the clinic.

The legal position is narrower and should be stated exactly. Consent for this study must be given by the participant personally. There is no provision for a legally authorised representative, for the reason that the eligibility criteria already require an ECOG performance status of 0 or 1 and full decisional capacity, so the situation in which a representative would be needed is one in which the participant is not eligible. A participant who loses capacity after enrollment does not lose the protections of the study, and §8.4 states how their data and their withdrawal rights are handled in that case.

A companion may ask questions, may take notes, and may hold the copy of this paper. What a companion may not do is consent on the participant's behalf, and a site that accepts a companion's signature has committed a deviation that is reportable under §11.6.

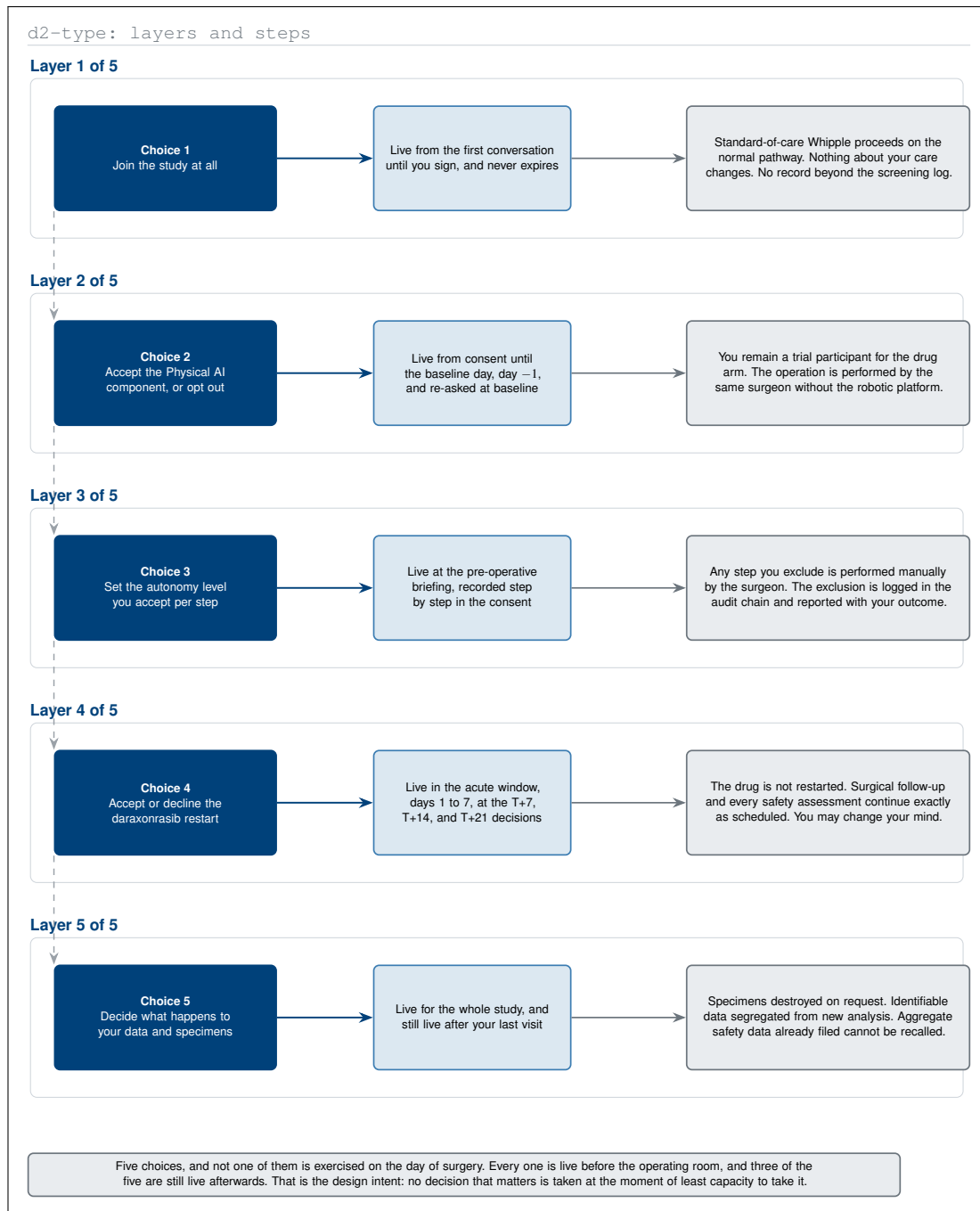


Figure 16. The five choices the participant keeps, revealed as accumulating layers with the choice, the interval in which it is live, and what happens if the answer is no, so that no decision that matters is taken at the moment of least capacity.

Interpretation is provided where it is needed, by a qualified medical interpreter rather than by a family member, because a family member translating a consent conversation is being asked to carry both roles at once and neither is well served.

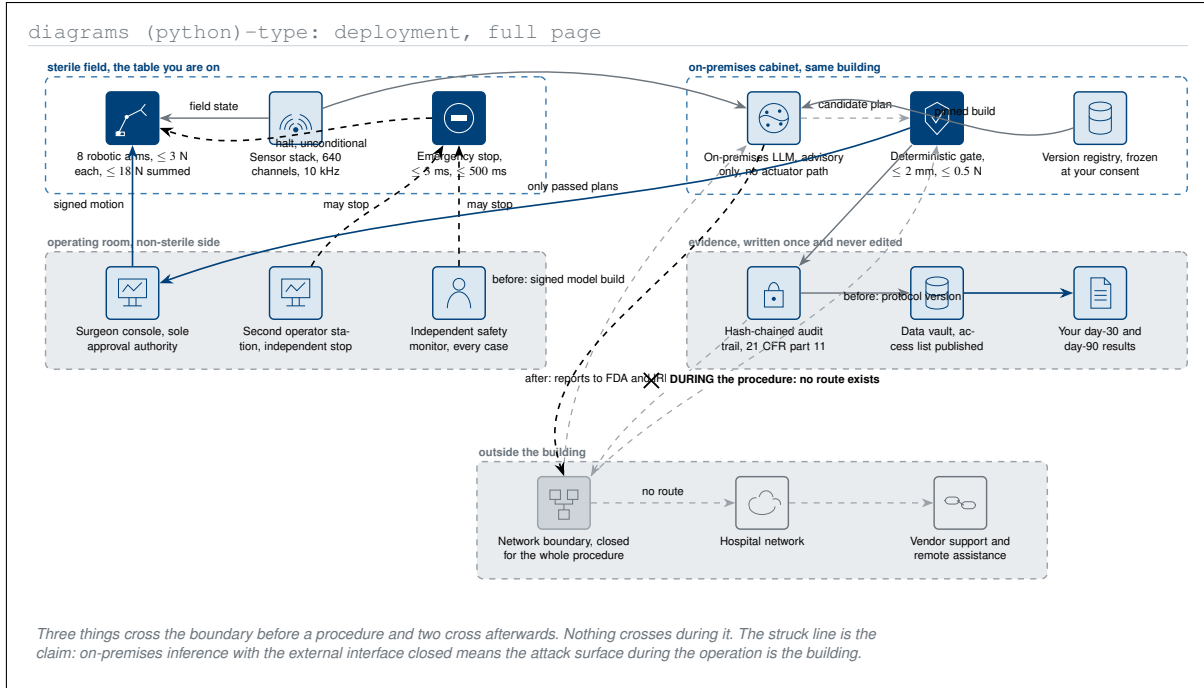


Figure 17. The operating room and the on-premises stack drawn as a deployment architecture, with the sterile field, the non-sterile side, the cabinet, the evidence store, and the network boundary that carries no intra-procedural route out of the building at all.

7 What Happens in the Operating Room

7.1 The room, the cabinet, and the cable that is not connected

Three things cross the network boundary of the operating suite before a procedure: a cryptographically signed model build, the protocol version, and the pinned registry entry that binds the two. Two things cross afterwards: the required reports to the FDA and to the reviewing Institutional Review Board. During the procedure itself nothing crosses, because during the procedure no route exists.

That is the whole answer to the cybersecurity concern, and it is architectural rather than reassuring. If there is no path out of the building while the operation is running, the attack surface during the operation is the building. The relevant FDA guidance frames the expectation for premarket cybersecurity documentation [24]; the design that satisfies it here is on-premises inference with the external interface closed for the duration [1].

Figure 17 draws the stack. The topology is adapted from the author’s autonomous single-patient journey simulation [19], which was non-small cell lung cancer, and it is re-scoped here to PDAC. Four differences matter: eight arms rather than four, three anastomoses rather than one bronchial closure, a vascular exclusion envelope around the superior mesenteric and portal veins that the thoracic configuration has no equivalent of, and a drain that stays in for days rather than hours.

7.2 The eight arms, and what limits them

The limits below are enforced beneath the level of the software that makes the decision. A model that proposed an out-of-envelope motion would have that proposal rejected by a deterministic gate; a controller that commanded an out-of-envelope force would have the arm halted by the force sensing itself. The distinction matters, because a limit implemented in the same layer as the decision is not a limit, it is a preference.

Limit	Protocol cap	Worst observed in simulation	What it protects
Tip force, any single arm	3.0 N	2.1 N	Tissue crush injury at a single instrument
Tip force, all arms summed	18.0 N	12.4 N	Cumulative traction on the mesentery and the vessels
Positional tolerance	2.0 mm	1.3 mm	Margin accuracy and corridor integrity
Force reproducibility	0.5 N	0.31 N	Consistency of the anastomotic suture line
Heartbeat period	0.1 ms	0.1 ms, held	Detection of any loss of control before motion continues
Cross-arm emergency stop	3.0 ms	1.7 ms	Time from the surgeon pressing stop to all arms held
System-wide emergency stop	500 ms	310 ms	Time to a fully quiescent, safe-pose system
No-fly corridor	4 mm either side	no entries	Superior mesenteric vein and portal vein

The observed column is class S: it comes from the author’s simulation set [8] and not from human procedures. A simulated worst case is not a human worst case, and §10 repeats that on the same line as the numbers. The standard the architecture is built against is the international standard for robotically assisted surgical equipment [10].

7.3 Who is controlling the operation

This is the most frequently raised of the twenty-one concerns and the one a paragraph answers worst. The chain has four links and one absence.

The sensor stack streams the current field state: force vectors, vessel proximity, instrument pose, across 640 channels at 10 kHz. The on-premises model reads that stream and emits a candidate motion plan with a confidence value and a stated rationale. The deterministic gate tests the candidate against every limit in the table above; if any limit would be violated, the plan is rejected, the rejection is logged with the violated limit named, and the plan is never presented to the surgeon as executable. If every limit is satisfied, the plan is presented for approval, with its confidence and rationale attached. The operating surgeon accepts, modifies, or declines it. Declining requires no justification. Only an approved, surgeon-signed command reaches the arms.

The absence is that there is no message from the model to the arms. Not a throttled one, not a supervised one, not a fallback one. Figure 19 draws five lifelines and the arrow that would be needed is not there. That is what a systematic review of FDA-cleared surgical robots describes as task-level autonomy under supervision, the highest level any cleared system reaches [18], and it is what the FDA’s own description of computer-assisted surgical systems assumes [21].

Deference is handled separately, because the architecture does not address it. If a surgeon always accepts, the human in the loop is a formality. The protocol therefore records every override with its reason and

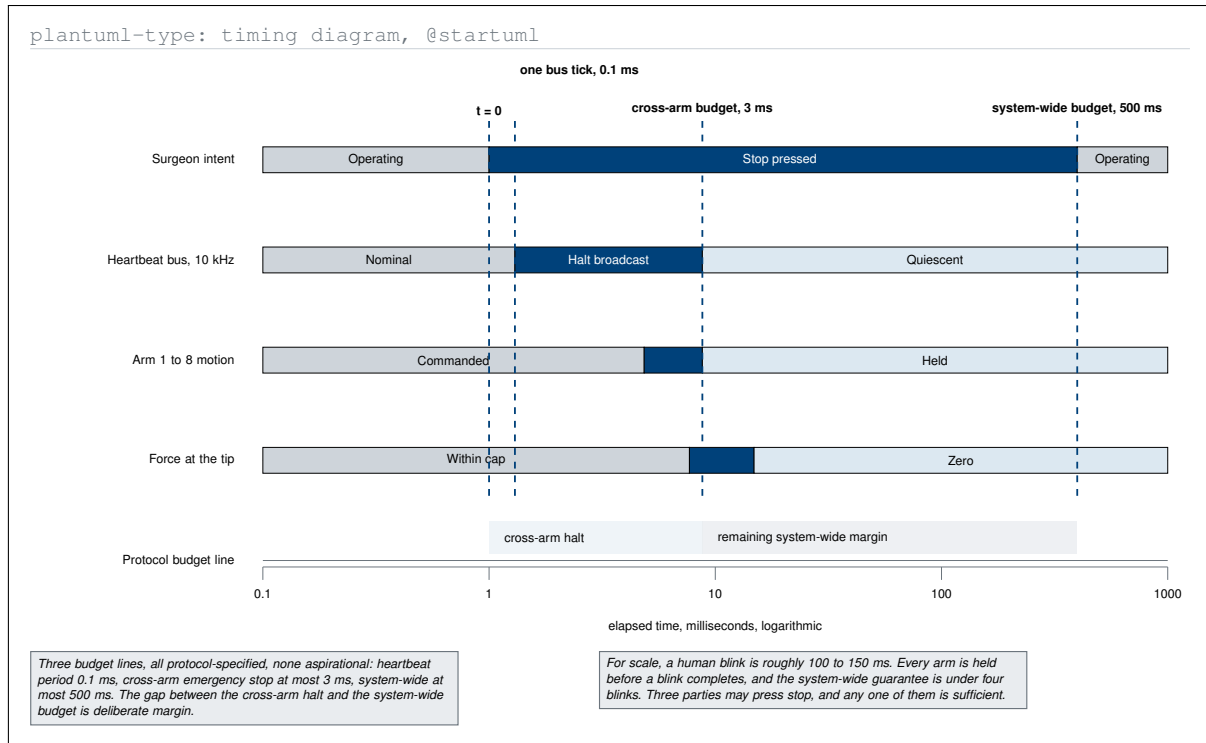


Figure 18. The emergency-stop chain on a logarithmic millisecond axis, from the instant the surgeon presses stop through the bus broadcast and arm deceleration to zero tip force, with all three protocol budget lines marked across every one of the timelines.

publishes the override rate as an exploratory endpoint, so automation bias is a measured quantity rather than an assumed absence [23].

7.4 How fast stopping actually is

Three budgets are specified, and Figure 18 puts all three on one logarithmic millisecond axis because they span four orders of magnitude and a linear axis would render the first two invisible.

The heartbeat bus broadcasts at 10 kHz, a 0.1 ms period. Loss of the heartbeat halts every arm without any human involvement. When a human does press stop, the cross-arm budget is 3 ms from assertion to all arms held, and the system-wide budget is 500 ms to a fully quiescent state. The gap between the two is deliberate margin, not unaccounted latency.

For scale: a human blink takes roughly 100 to 150 ms. Every arm is held before a blink completes, and the system-wide guarantee is under four blinks. Three parties may press stop: the operating surgeon at the console, the second operator at an independent station, and the independent safety monitor who is present for every procedure. The fail-safe design standard for autonomous and semi-autonomous systems frames what such a guarantee has to demonstrate [9].

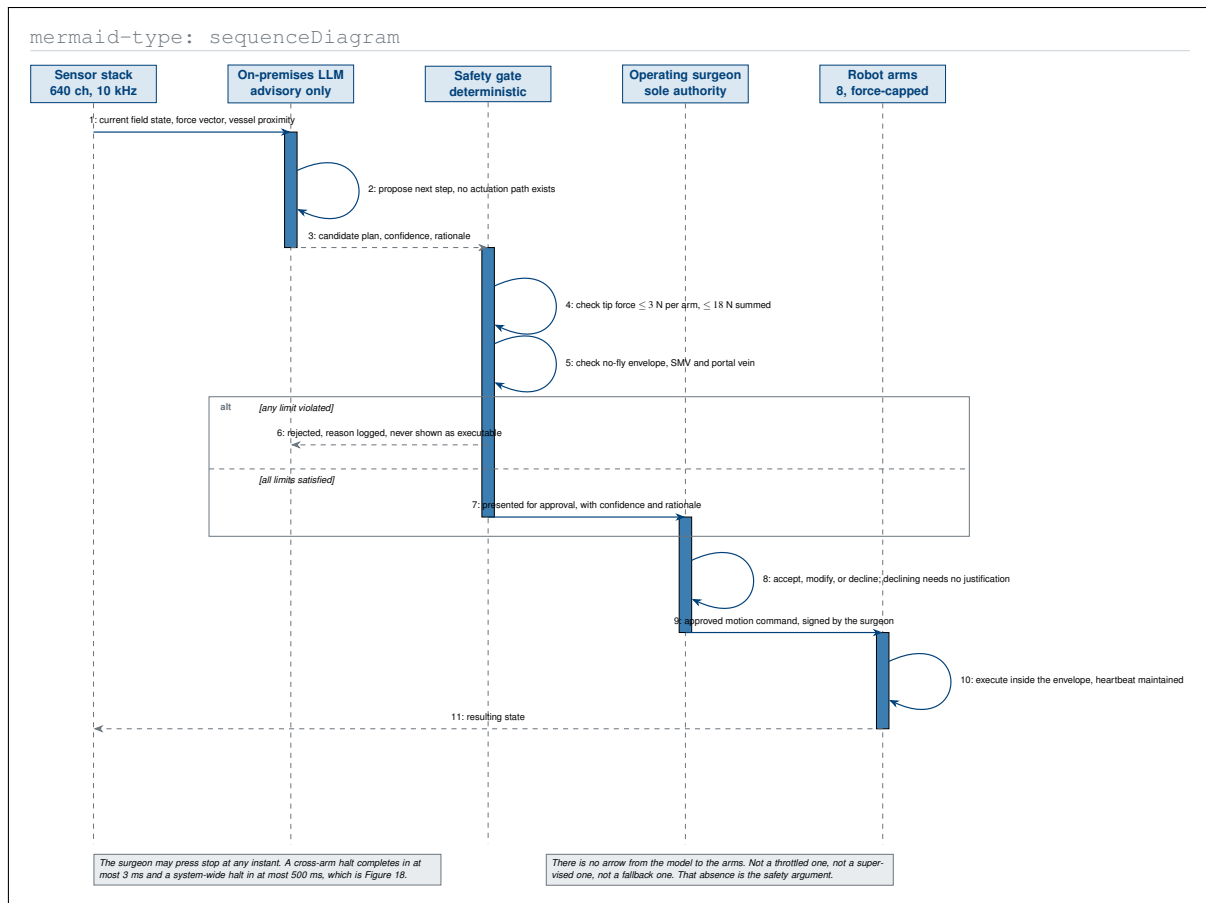


Figure 19. One operative step message by message across five lifelines, with the deterministic gate sitting between advice and approval, and no arrow at all from the model to the arms, which is the safety argument this figure makes by an absence rather than words.

7.5 The vascular no-fly corridor and the force envelope

A force cap does not prevent an arm entering a vascular corridor, and a corridor does not cap the force applied inside permitted tissue. Neither limit is sufficient alone, and the pairing is the safety property. Figure 20 draws them as sibling containers inside one envelope for that reason, with the deterministic gate as the single element that reads both.

The corridor is 4 mm either side of the superior mesenteric and portal veins, and the approach angle to the superior mesenteric artery is constrained separately. Proximity is tested at the heartbeat rate, so a drifting instrument is caught within one tick rather than at the end of a motion.

7.6 What could fail, and what catches it

Figure 21 is a fault tree, and the gate at its apex is an AND gate. Harm reaches the participant only if all four intermediate events occur together: a mechanical or control fault, a failure of the deterministic gate to reject it, no human intervening in time, and the fault not being visible in the telemetry. Each of those four is itself decomposed, and each has a named barrier.

Four barriers are intact. The force caps are hardware-limited. The no-fly corridor is enforced by the gate. Surgeon approval is required for every motion, with a second station and an independent safety monitor able to stop. The 10 kHz heartbeat halts all arms within 3 ms without a human if it is lost.

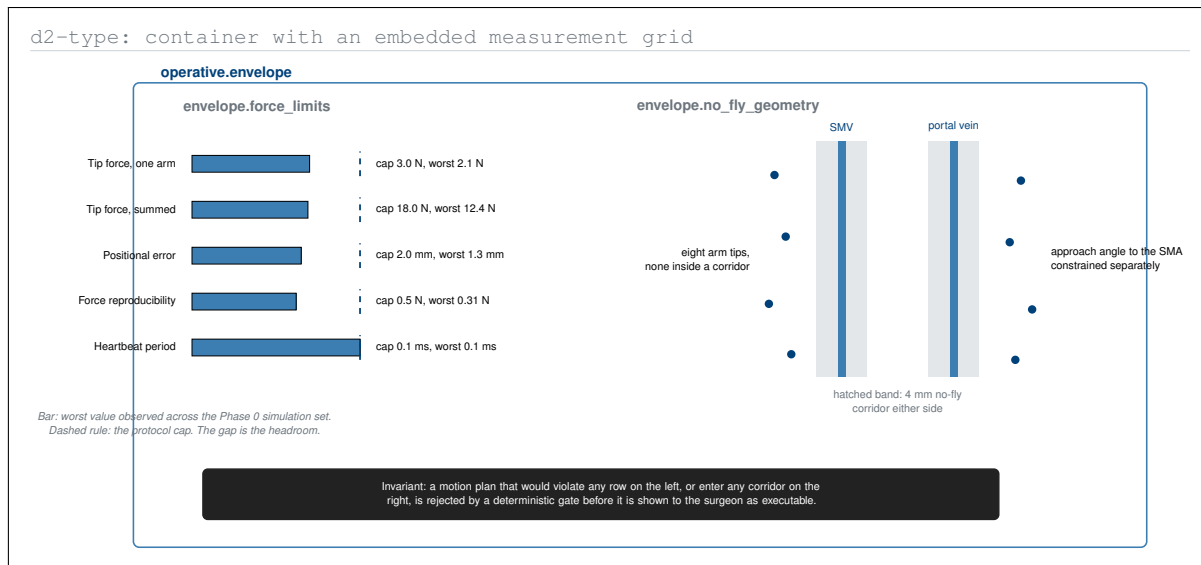


Figure 20. The operative envelope drawn as two sibling containers, the force limits with their observed headroom on the left and the vascular no-fly geometry on the right, joined by the deterministic gate that rejects any plan violating either of them.

One is not, and the figure says so in medium-dark gray: registration drift beyond 2 mm is caught by the deterministic gate and by nothing else. It is a single point of failure in the barrier set. Naming it is what a fault tree is for, and suppressing it would have made the other four less credible.

7.7 Conversion to open surgery

Conversion from the robotic approach to an open operation is a recorded outcome of the procedure, not a failure of the study, of the device, or of the participant. It does not end participation, it does not remove the participant from any analysis population, and the decision belongs to the operating surgeon alone and is not subject to review during the procedure by anyone else.

An open tray and backup instruments are in the room for every procedure, and the conversion sequence is rehearsed as part of the pre-procedure checklist. In the contemporaneous nationwide series of 1000 robotic pancreatoduodenectomies, conversion was most frequent in the earliest learning-curve phase and fell with institutional experience [17]; this study enrolls only at sites already past the third phase of that curve, which is stated in §10.2 with the numbers attached.

If a conversion occurs, the participant is told before they leave hospital, the reason is recorded in the audit chain, and the event is reported to the DSMB at the next cohort boundary. None of those three is discretionary.

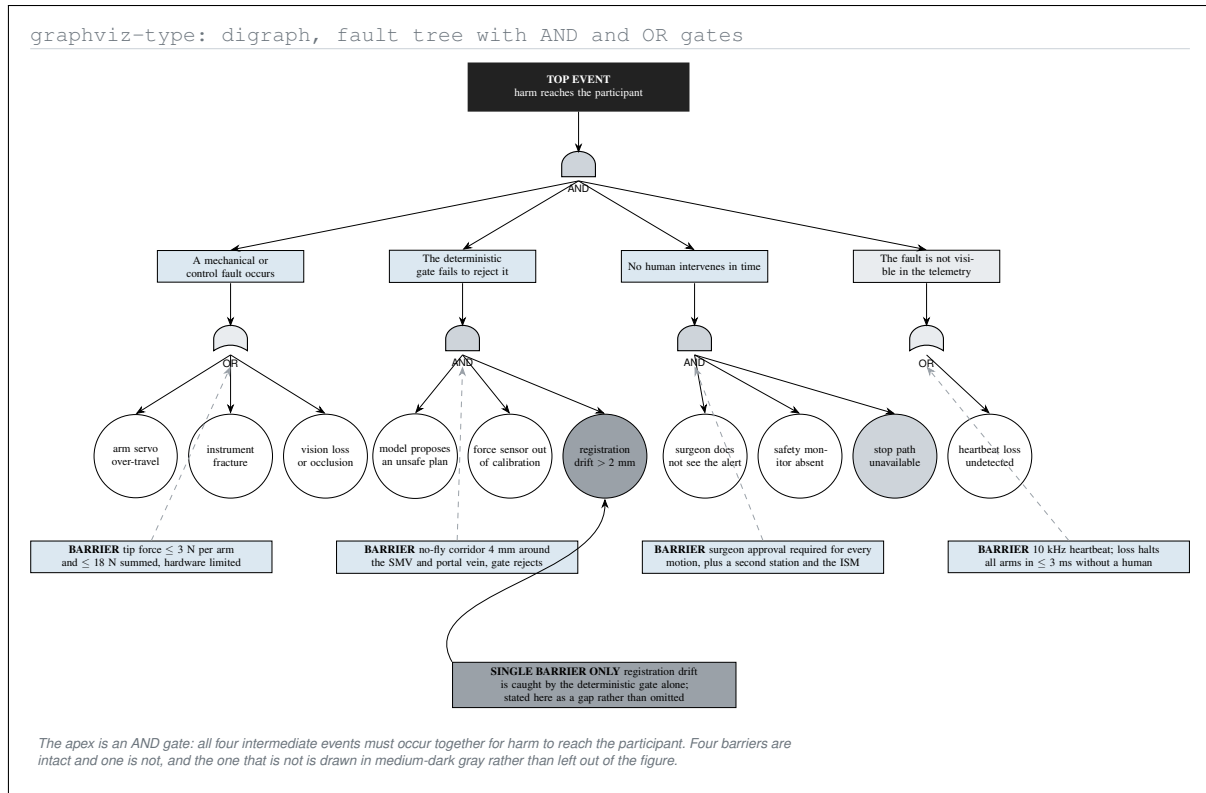


Figure 21. The hazard fault tree with the top event behind an AND gate, so that harm requires all four intermediate events together, and with the single-barrier gap at registration drift marked in medium-dark gray rather than being quietly left out of it.

7.8 The drug, paused and restarted around the operation

Daraxonrasib is given before the operation, paused across it, and restarted afterwards on an advised schedule. Each of those three phases exists for a reason a participant can check.

Before. The preoperative course runs through the screening and baseline window at the assigned dose level. Its purpose in this study is not to shrink the tumour to resectability; that is not what a Phase 1 safety study can claim. Its purpose is to establish tolerance at the assigned level in the individual participant before the operation adds its own physiological load.

The pause. The drug is held before surgery for a protocol-specified interval. The reason is anastomotic healing: three new connections are made during a Whipple, a pancreaticojejunostomy, a hepaticojejunostomy, and a gastrojejunostomy or duodenojejunostomy, and the first of those three is the one whose failure defines the dominant early complication of the operation [28]. Holding a systemic agent across the healing window is the conservative choice, and this protocol takes it.

The restart. The drug is restarted at T+7, T+14, or T+21 days. Which of the three is advised depends on two measured quantities: the ISGPS fistula grade at the time of assessment, and the drug trough. The advisory is produced by the same on-premises model that advises intra-operatively, it is reviewed by the Site Principal Investigator, and it is presented to the participant as an advisory rather than an instruction. The participant may decline it, and declining does not end participation or change any other part of the schedule.

Restart window	Advised when	Who decides	What you are told
T+7 days	No fistula, or ISGPS biochemical leak only, and trough in range	Advisory from the model, review by the Site PI, decision by you	The grade, the trough, and why the earliest window was advised
T+14 days	ISGPS grade B fistula resolving, or trough below range at T+7	Same	What changed since T+7, and what would move it to T+21
T+21 days	ISGPS grade B fistula not yet resolving, or any grade C	Same	Why the drug is being held longer and what the plan is if it does not resolve
Not restarted	Grade C fistula unresolved at T+21, or a dose-limiting toxicity	Site PI, with DSMB notification	That surgical follow-up and every safety assessment continue unchanged

7.9 The three connections, and why the first one matters most

A pancreaticoduodenectomy removes the head of the pancreas, the duodenum, the gallbladder, the distal bile duct, and, in the classical form, the gastric antrum. Reconstruction then rebuilds three connections in sequence.

The pancreaticojejunostomy joins the remnant pancreas to the small bowel. It is the connection whose leak produces a postoperative pancreatic fistula, and the ISGPS grading of that leak, biochemical, grade B, or grade C, is the secondary endpoint that carries the most weight in the early window [28]. It is also the connection the force caps most directly protect: soft, friable pancreatic tissue is the tissue least tolerant of traction, and the 3 N per-arm cap exists for it.

The hepaticojejunostomy joins the bile duct to the bowel. Its leak rate is far lower and its consequences are usually manageable with drainage.

The gastrojejunostomy or duodenojejunostomy restores gastric outflow. Delayed gastric emptying after it is common, uncomfortable, and rarely dangerous, and participants should expect to be told about it because it is the complication most likely to lengthen their inpatient stay without being serious.

The order matters to the participant for one practical reason: the drain that stays in for days after the operation is there to detect a leak from the first of the three connections, and the drain fluid check described in §9.1 is what converts that drain into the ISGPS grade.

7.10 The day of the operation, hour by hour

The protocol describes the operation as a procedure. The participant experiences it as a day, and the two descriptions do not overlap much. What follows is the day, with the study-specific steps marked, so that a participant can tell which parts are the operation they would have had anyway and which parts exist because this is a study.

Roughly when	Study-specific	What is happening, and where you are
Day -1, afternoon	Partly	Baseline visit. Bloods, performance status, and the Phase 0 gate sign-off is shown to you. Daraxonrasib is held from this point
Day 0, before dawn	No	Admission, marking, and the anaesthetic conversation. Nothing about the platform changes this
About 60 min before	Yes	Room readiness check: heartbeat bus verified at 10 kHz, emergency-stop latency measured and logged for your case, no-fly geometry loaded from your own imaging
Induction	No	General anaesthesia, lines, and positioning. You are asleep before the arms are brought to the table
First 30 to 60 min	No	Access, staging laparoscopy, and the decision that the tumour is still resectable. If it is not, the operation stops here and no platform step occurs
The resection	Yes	Docking of the eight arms, then the resection itself, step by step. Every motion is proposed, gated, and signed before it executes
The reconstruction	Partly	The three reconnections. The pancreatic anastomosis is the one §7.9 discusses; the surgeon may complete any of the three by hand
Closure	No	Drains placed, closure, and the count. Standard care
Recovery, hours 0 to 4	No	Post-anaesthesia care unit. Your family is told when you arrive there
Ward, day 0 evening	Yes	The audit trail for your case is sealed and hashed. Your emergency-stop and force-envelope numbers exist from this point and may be requested

Two things in that table are worth pulling out. The room readiness check is the only study-specific step before induction, and it happens whether or not it finds a problem; if it finds one, the operation proceeds without the platform. And the audit trail is sealed the same evening, which is why the extract in §11.5 can be requested from the following day rather than at study closure.

7.11 Who is in the room, and what each of them does

A participant asked to consent to an eight-arm platform reasonably wants to know whether the room is more crowded than it would otherwise be, and whether the additional people are clinical or technical.

The clinical team is unchanged from a conventional robotic Whipple: the operating surgeon at the console, an assistant surgeon at the table, the anaesthetist, the scrub practitioner, and the circulating practitioner. Those five make every clinical decision, and the operating surgeon is the one named in the participant's consent form.

Two roles are added by the study. An independent safety monitor watches the gate decisions and the force telemetry and holds an unconditional authority to halt, which they may exercise without agreeing it with the surgeon first. A platform engineer maintains the stack and has no clinical role, no console access, and no authority over any motion. The distinction is deliberate and is visible in the audit trail: no motion in the record can carry an engineer's identity, and a record that did would be a protocol violation rather than an unusual event.

The safety monitor is the role most often misread, so it is worth being exact. They are not a second surgeon and they do not offer an opinion on the operative plan. Their entire function is to stop the platform

when a limit is approached or a gate behaves unexpectedly, and their stop does not require the surgeon's agreement. A participant who wants to know whether anyone in the room can override enthusiasm has a specific answer: yes, and that person's only job is to be willing to.

7.12 What is different from a conventional robotic Whipple

Most of what happens is not new. The comparison below separates what the platform changes from what it does not, because a participant weighing the study is weighing the difference and not the whole operation.

Aspect	Conventional robotic Whipple, and what this study changes
The operation performed	Identical. Same resection, same specimen, same three reconnections
Who moves the instruments	Identical. The surgeon moves them from a console; no motion is autonomous in either case
Number of arms	Eight rather than four. The additional arms hold retraction and camera positions a human assistant would otherwise hold
Force at the tip	New. A hardware cap at 3 N per arm and 18 N summed, which a conventional platform does not enforce
Vascular protection	New. A 4 mm no-fly corridor computed from your own imaging, enforced by the gate rather than by vigilance alone
Step planning	New. The model proposes the next step with a rationale; the surgeon accepts, edits, or declines it
Stopping	Stronger. A measured 3 ms cross-arm and 500 ms system-wide budget, verified for your case before induction
Record	New. Every advisory, gate decision, and approval is hash-chained and available to you
Operative time	Longer during a learning phase, and §10.2 gives the learning-curve figures that make the difference material
Anaesthesia and recovery	Identical. Nothing in the platform changes the anaesthetic or the ward pathway

Read down the table, the platform adds four constraints, one record, and one advisory layer, and changes nothing about what is cut or how it is sewn. That is a narrower claim than the phrase Physical AI usually suggests, and it is the accurate one.

8 Stopping, Withdrawing, and Changing Your Mind

8.1 Four routes out, and what each does to your record

Participants are routinely told they may withdraw at any time, and are almost never told what withdrawal does to the data already collected. The honest answer depends on which route out is taken, and there are four. Route A stops the drug only. Route B stops study procedures but keeps follow-up. Route C is full withdrawal from participation. Route D withdraws consent for future data use as well.

None of the four requires a reason, and none of the four is recorded as a reason unless the participant chooses to give one. The right is unconditional under 21 CFR part 50 [6], and the National Cancer Institute's guidance on consent describes it in the same terms [12]. What differs between the routes is the consequence for the record, and Figure 22 draws all four side by side with a lane for the participant, a lane for the site team, and a lane for the data and specimens.

Two of those consequences are unwelcome and are stated rather than smoothed. Under route C, data already collected cannot be removed from the safety analysis, because removing one participant's adverse events from a safety data set biases that data set for everyone else who is still enrolled. Under route D, aggregate safety data already reported to the FDA cannot be recalled. Both are true. Saying so is what makes routes A and B believable.

Route	What stops	What happens to data and specimens
A	Daraxonrasib only	Everything already collected stays in the safety analysis; new data continues to be collected
B	Study procedures, follow-up continues	Existing data retained; new data limited to survival status; specimens retained unless you ask for destruction
C	All study participation	Data already collected cannot be removed from the safety analysis; nothing new is collected; specimens destroyed on request
D	Participation and future data use	Identifiable data segregated from any new analysis; aggregate safety data already filed with the FDA cannot be recalled

8.2 Consent as a state, not a signature

A consent form implies a single act at a single moment. In a Physical AI trial that model is wrong, because the thing consented to can change while the participant is still enrolled. An AI-enabled device can be updated, and the joint transparency principles for machine learning-enabled medical devices identify version change as a matter requiring explicit communication [23]. Real-world performance can also drift without any version change at all [26].

This protocol handles both by freezing the software version at consent and recording it in a registry that the participant is shown. Any change to that version, any newly identified risk, and any protocol amendment forces a re-consent. Until the participant acts on that re-consent, no study procedure proceeds under the changed version. Declining the change ends participation and returns the participant to standard care with no penalty.

Figure 23 draws consent as the state machine this implies, with a nested enrolled composite, withdrawal reachable from every state, and a re-consent transition the participant did not cause.

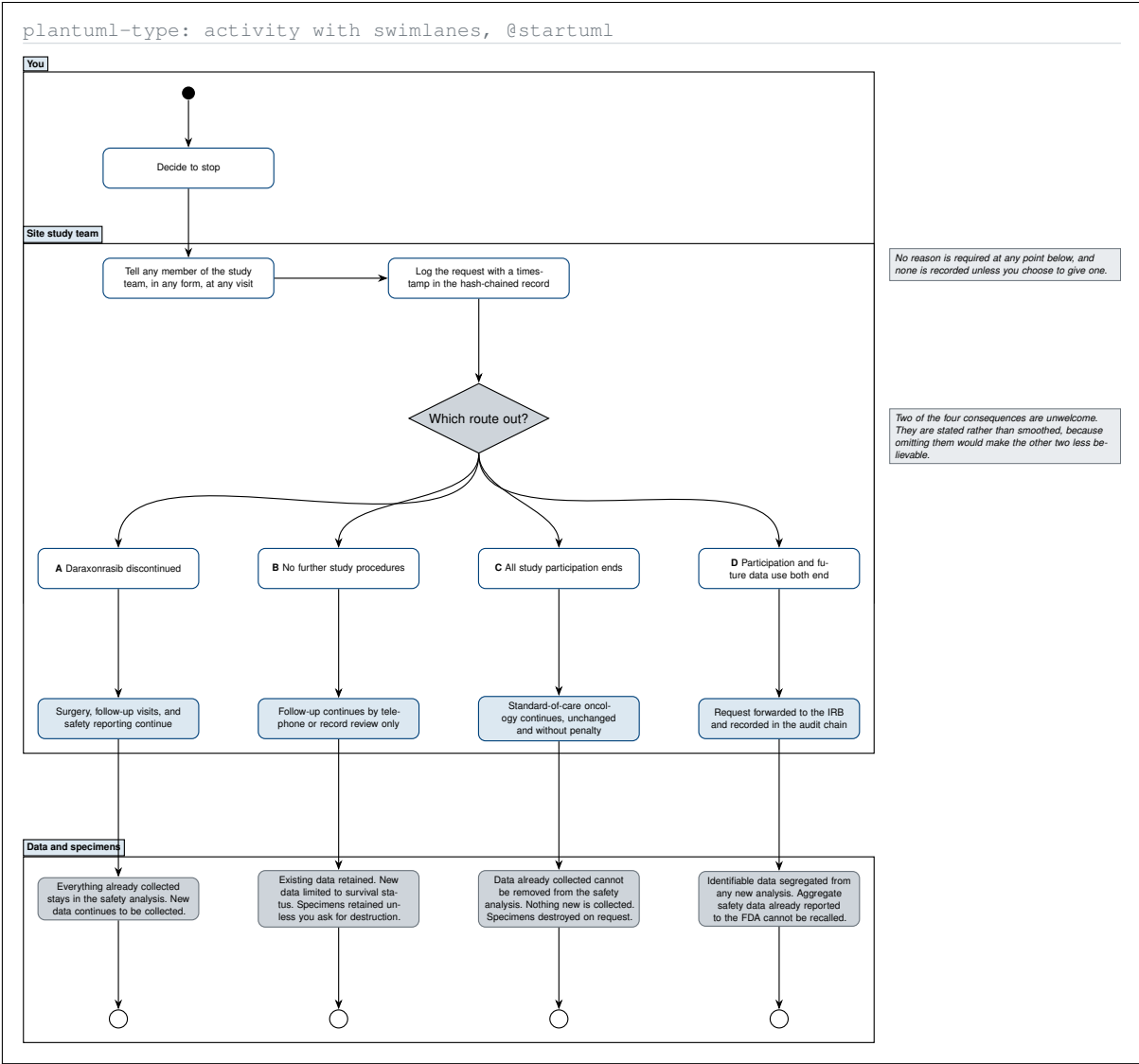


Figure 22. The four routes out of the study drawn as a swimlaned activity, with one lane for the participant, one for the site team, and one for the data and specimens, so each route’s consequence for the record is visible directly beside the route itself.

8.3 The two transitions the parent protocol does not draw

Transition	Why it matters to the participant
Enrolled to re-consent required	An AI-enabled device can be updated. If the version that operates on the participant is not the version they consented to, the consent is stale. This protocol freezes the version at consent and forces a re-consent on any change, so nobody is operated on by software they have not been told about.

Table continued on the next page.

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Transition	Why it matters to the participant
Re-consent required to withdrawn	Declining a change must be a real option rather than a formality. Declining ends participation and returns the participant to standard care with no penalty and no loss of any care they were already receiving.

8.4 Discontinuation the study may initiate

The participant stopping and the study stopping the participant are different things, and this subsection is about the second. In every case below the participant is told the reason in writing and returns to standard care.

Trigger	What follows for you
Dose-limiting toxicity at your level	The drug is held or discontinued; surgical follow-up and every safety assessment continue as scheduled
Disease progression before surgery	A curative-intent resection is no longer the right operation; you are referred back for systemic therapy without delay
Intercurrent illness precluding surgery	Participation ends; the standard-of-care pathway resumes without penalty
Study-wide pause or halt	You are told, in writing, within the timeframe the protocol specifies; follow-up continues throughout the pause
Withdrawal of the IND or the IDE	Participation ends; the sponsor's obligation for research-related injury care continues unchanged
Loss of the site's eligibility to enrol	Your care transfers with a written handover; you are not asked to travel to another site

8.5 Lost to follow-up

A participant is recorded as lost to follow-up only after three documented contact attempts across at least 30 days, by at least two different means, plus one written attempt to the last known address. That threshold is deliberately high, because a participant recorded as lost is a participant whose survival status is unknown, and an unknown survival status weakens the endpoint for everyone else in the study.

A participant who reappears at any point is resumed rather than excluded. They re-enter at the visit that is next due, their earlier data is unaffected, and no explanation is required. There is no penalty of any kind for a missed visit.

8.6 Changing your mind is not a failure of the study

The word withdrawal carries a suggestion of something going wrong, and the suggestion is worth removing because it changes behaviour. Participants who believe that leaving damages the study, or embarrasses the surgeon who recruited them, stay in studies they would rather leave, and the surveyed literature records that pressure in exactly those terms.

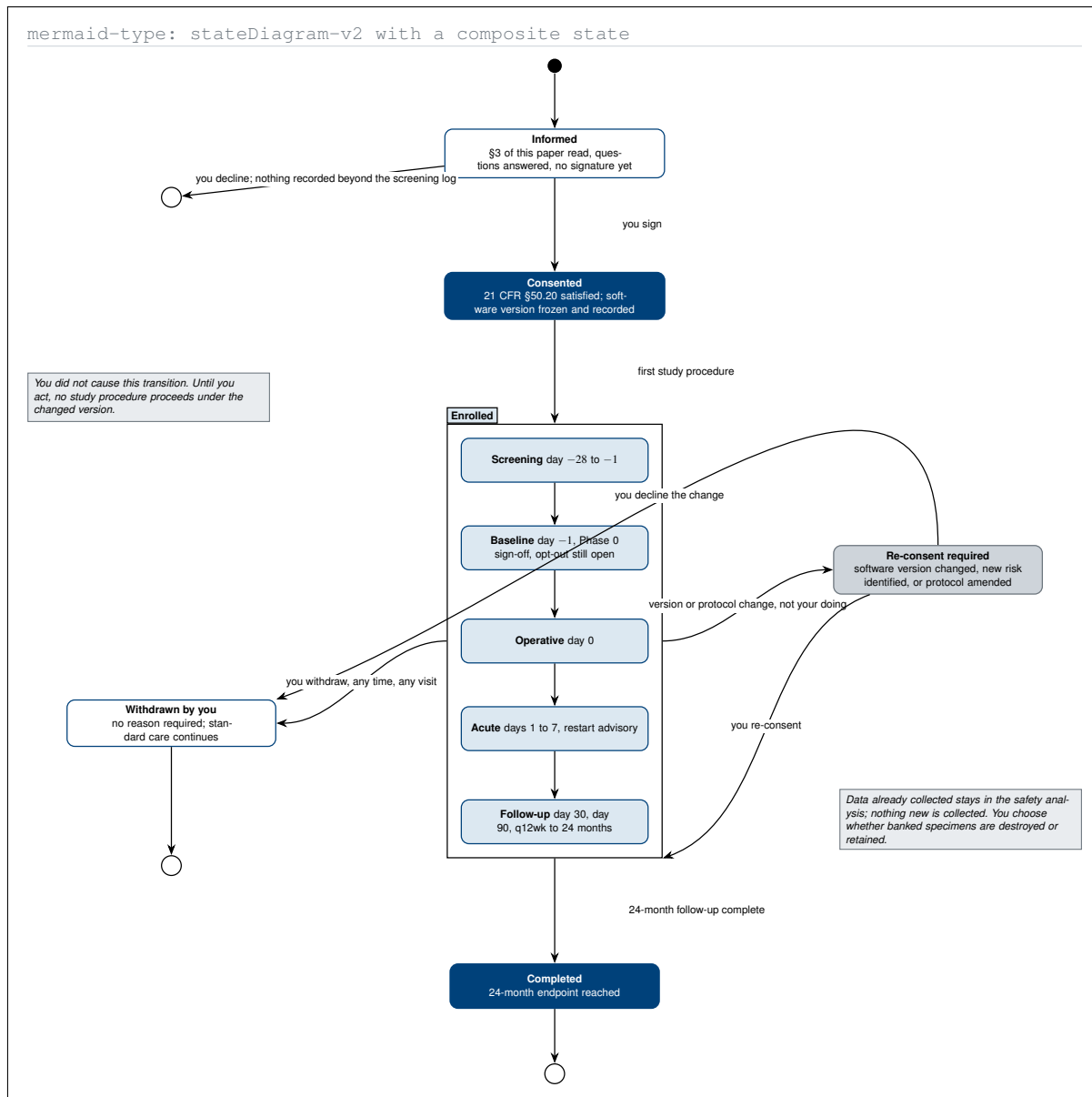


Figure 23. Consent drawn as a state machine with a nested enrolled composite, withdrawal reachable from every state, and a re-consent transition the participant did not cause, triggered by a software version change, a new risk, or a protocol amendment.

The design position is the opposite. A study that cannot tolerate withdrawal is a badly designed study, and this one is sized with that in mind: the 3+3 rule counts participants who complete the dose-limiting toxicity window, and a withdrawal simply means the slot is refilled. No individual's departure invalidates a cohort, and no individual's departure is reported in a way that identifies them to the other participants. Nor does withdrawal end the relationship with the site. A participant who leaves the study remains a patient of the hospital, keeps their surgeon, and keeps every appointment that is part of standard care. §8.1 sets out which of the four routes stop study assessments only and which stop the drug as well, and in none of the four does standard oncology care stop.

The one thing withdrawal cannot do is reverse an operation that has happened or un-analyse data that has already been included in a locked analysis. §8.1 states that limit exactly rather than softening it, because

a participant told that they may withdraw completely, and who then discovers that a locked analysis still contains their data, has been misled about the only thing they were promised.

A participant who is unsure is entitled to pause instead of leaving. A pause is not one of the four formal routes, but the sentinel and window mechanics make it practical: assessments can be deferred, the drug can be held, and the decision can be revisited at the next visit. The site is expected to offer that option rather than presenting the choice as binary.

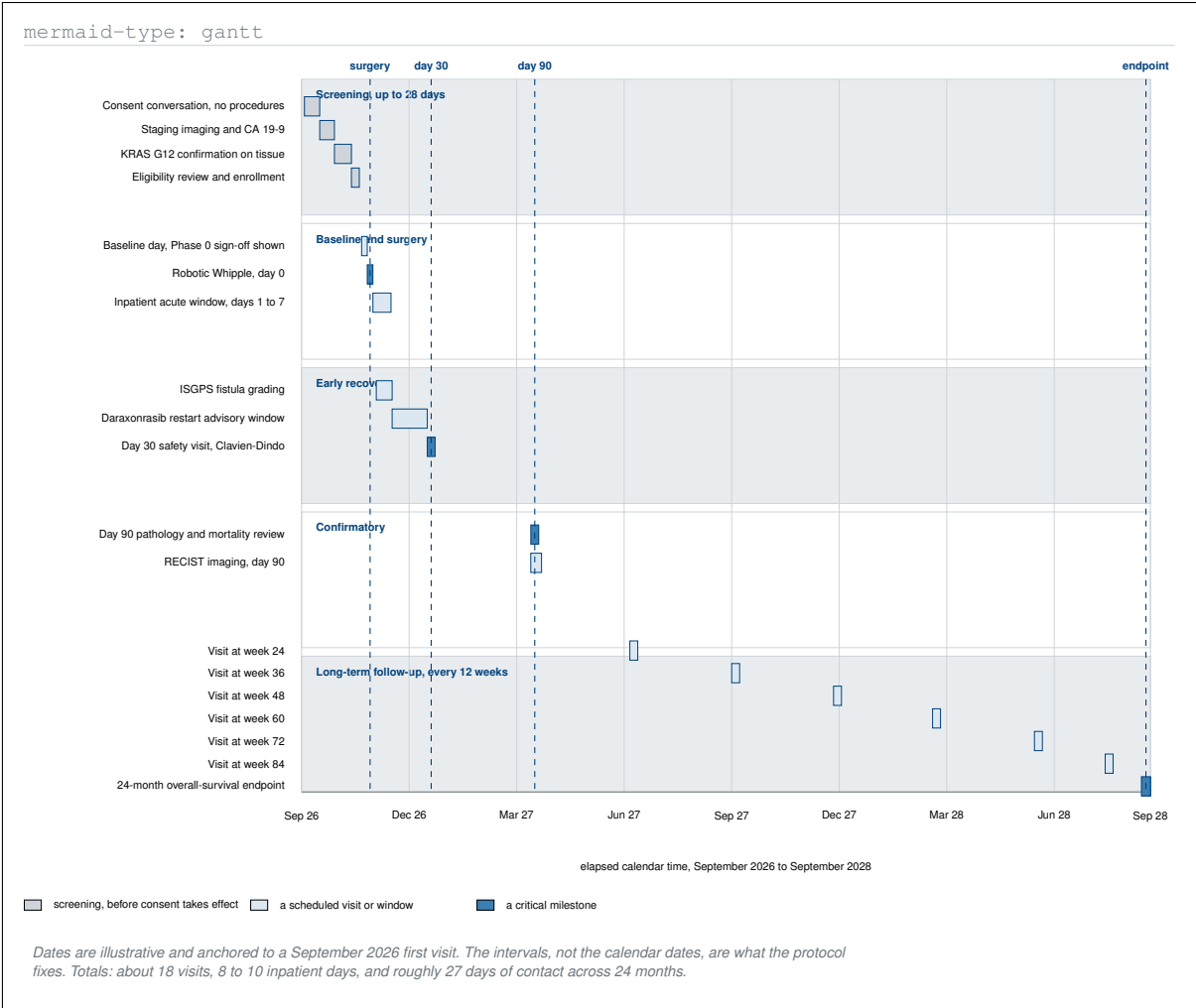


Figure 24. The participant calendar from screening through the 24-month endpoint, drawn with real durations rather than crosses in a matrix, and with the four critical milestones, surgery, day 30, day 90, and the endpoint, marked down to the axis.

9 Your Visits, Your Data, Your Robot Instructions

9.1 The calendar, and what it costs you

Figure 24 converts the Schedule of Activities into a calendar with durations, and then totals it. Roughly 18 visits, 8 to 10 inpatient days, and about 27 days of contact spread across 24 months. The dates on the figure are illustrative and anchored to a September 2026 first visit; the intervals, not the calendar dates, are what the protocol fixes.

Totalling the burden is not a courtesy. The National Cancer Institute’s guidance on trial costs identifies travel, time away from work, and caregiving as the practical burdens participants most often underestimate before enrolling [27], and none of the three can be estimated from a matrix of crosses.

Phase	Visits	Inpatient days	Clinic days	Notes for planning
Screening	4	0	about 4	Imaging and tissue work; no study procedure occurs before consent
Baseline and surgery	2	8 to 10	1	The single inpatient stay of the whole study
Early recovery	3	0	3	The restart advisory is a conversation, not a procedure
Confirmatory	2	0	2	Day 90 carries the two results most people want
Long-term	7	0	7	One half-day every 12 weeks for 24 months
Total	18	8 to 10	about 17	Roughly 27 days of contact over 24 months

9.2 Everything recorded about you

Four kinds of data are captured. Arm motion, tip force, and vessel proximity across 640 sensor channels at 10 kHz. Operative video and the endoscope feed. Clinical measurements: vital signs, laboratory values, CA 19-9, and drain output. And the case report form, which is the structured clinical record of the study.

All four are written into a hash-chained record that cannot be edited after the fact, under the electronic records and electronic signatures rule [11]. From there the pipeline splits into an identified store and a de-identified store, and the split is one-way: nothing flows back from de-identified to identified.

9.3 Who can read what, and for how long

Store	Retention	Who may read it
Raw telemetry	10 years	Site PI, Sponsor-Investigator, and you on request
Operative video	10 years	Operating surgeon, Site PI, and you on request
Case report form	15 years	Site PI, CRO monitor, and you on request
Hash chain	15 years	Sponsor-Investigator, FDA on inspection, and you on request
Identified store	15 years	Site PI, and you on request
De-identified store	10 years	Sponsor-Investigator, DSMB, and the study statistician
Analysis outputs	15 years	Everyone above, and the public at publication

Your data is not used to train a commercial model without a separate consent. That is a statement about something that does not happen, and Figure 25 draws it as a struck edge rather than asserting it in a caption, because a negative claim rendered as an absence is checkable and a negative claim rendered as a sentence is not.

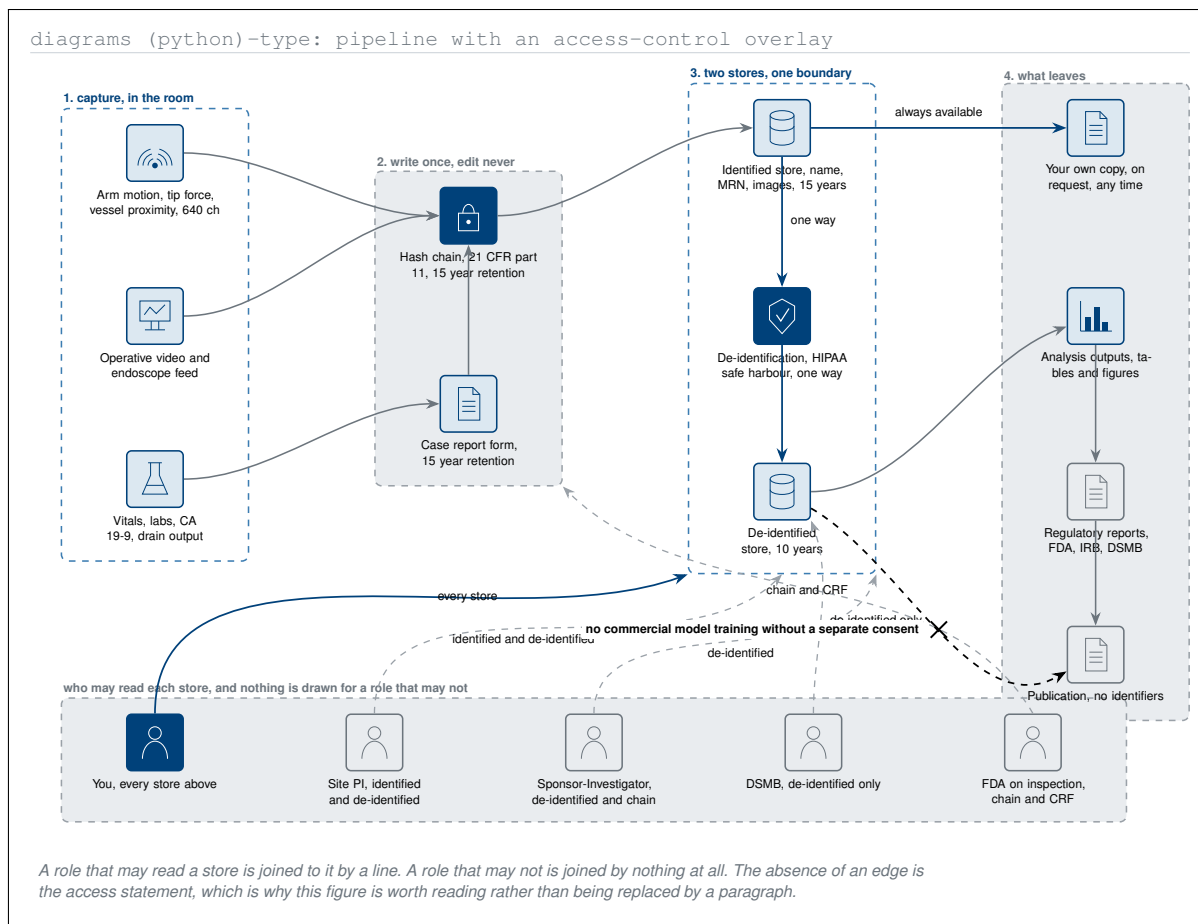


Figure 25. Your data through four stages, capture, a write-once hash chain, the identified and de-identified stores, and what leaves, with the read list attached to every store and no edge at all drawn for a principal who may not read that particular store.

9.4 Cybersecurity, in the terms the question is asked

The question participants ask is "could someone hack the system", and the answer is architectural. During a procedure there is no network route out of the building. The external interface is closed for the duration, so the attack surface during the operation is the building itself, and the building is a controlled environment with the usual physical access controls.

Three things cross the boundary before a procedure: a cryptographically signed model build, the protocol version, and the registry entry that pins the two together. Two cross afterwards: the required reports to the FDA and to the IRB. If hospital connectivity fails entirely, mid-procedure or otherwise, the procedure is unaffected, because the procedure does not depend on it. The FDA's premarket cybersecurity guidance frames what such a design has to document [24]; §7.1 and Figure 17 show the design.

9.5 The robots you will meet, and the ones you will not

The author's instruction set covers ten robot types [33]. In this protocol, two are certain, four are possible depending on recovery, and four are not used at all. Knowing which is which is itself an answer to the question of what the participant is agreeing to, and it is why the four unused types are listed rather than silently dropped: a participant who has read about steerable needle robots elsewhere needs to know that this study does not use one.

d2-type: grid of cards, full page

1. Eight-arm surgical system, you will meet this one 1. Lie face up, arms secured, and stay still for about two minutes while anaesthesia begins. 2. You are asleep for the whole 240 to 420 minute Whipple; eight arms work through 8 to 12 mm ports while your surgeon approves every motion from the console. 3. You wake in recovery 30 to 60 minutes after the arms retract; keep your abdomen relaxed and follow the nurse's breathing checks. <i>Re-scoped from the mediastinal sheet: PDAC pancreaticoduodenectomy, far longer, three anastomoses, a drain placed.</i>	2. Imaging robot, screening, day 90, and every 12 weeks 1. Lie still on the table; the gantry moves around you and never touches you. 2. Hold your breath when asked, for 10 to 20 seconds at a time. 3. Tell the technologist at once if you feel unwell; the sequence pauses without losing the study. <i>Re-scoped: pancreatic-protocol CT and RECIST assessment, with CA 19-9 drawn on the same visit.</i>
3. Bedside cobot, likely, days 1 to 7 1. The cobot works beside you at reduced speed and stops on contact; you may touch it without harm. 2. Keep your drain line on the side the nurse indicates so the arm's path stays clear. 3. Say stop at any time; the arm halts and a nurse takes over the task. <i>Re-scoped: the acute-window tasks here are drain handling and early mobilisation after a Whipple.</i>	4. Needle-placement robot, only if a collection needs drainage 1. You are awake with local anaesthetic; tell the team if you feel sharp pain rather than pressure. 2. Stay still and breathe shallowly while the needle advances, about 60 seconds. 3. The robot holds the trajectory; the radiologist advances it and can withdraw it instantly. <i>Re-scoped: used for a post-operative collection or a suspected grade B fistula, not a lung biopsy.</i>
5. Companion robot, optional, inpatient stay 1. It reminds, records, and calls a human; it performs no clinical task and touches nothing. 2. Ask it to fetch a nurse; it will, and it logs that it did so. 3. You may switch it off, and switching it off changes nothing at all about your care. <i>Re-scoped: reminders are Whipple-specific, drain output, early mobilisation, diet advancement.</i>	6. Humanoid transport assistant, possible, ward transfers 1. It moves at walking pace with a human escort at all times; it never transports you alone. 2. Keep your hands inside the chair rails while it is moving. 3. Any staff member can halt it with the panel stop within one second. <i>Re-scoped: post-Whipple transfers with a drain and an epidural line, which constrain route and speed.</i>
7. Rehabilitation exoskeleton, optional, week 2 onward 1. A physiotherapist fits and starts it; it will not move until they release the brake. 2. Walk at your own pace; the device assists and does not lead. 3. Stop when your abdomen aches, not when the session timer ends. <i>Re-scoped: abdominal-incision recovery, so the gait programme is low load and short duration.</i>	8. Radiotherapy positioning robot, not used in this protocol 1. The couch moves in six axes and pauses if you move. 2. Breathe normally unless a breath hold is requested. 3. The intercom is always open, and speaking pauses the beam. <i>Not part of this protocol. Included because you may receive radiotherapy outside the study.</i>
9. Radiotherapy motion-tracking robot, not used here 1. Markers on your skin or an implanted fiducial let the system follow your breathing. 2. Do not adjust the markers between fractions. 3. Coughing interrupts the beam automatically and does not spoil the treatment. <i>Not part of this protocol. Retained for completeness of the ten-sheet source set.</i>	10. Steerable needle robot, not used in this protocol 1. Not used in this study at any timepoint. 2. If offered outside this study, ask which steps are autonomous and who approves each one. 3. The same questions in §3 of this paper apply to it unchanged. <i>Explicitly excluded here. Listed so you can tell what this protocol does not involve.</i>

Of the ten robot types in the source instruction set, **two** are certain in this protocol, the eight-arm surgical system and the imaging robot; **four** are possible depending on your recovery; and **four** are not used at all. Knowing which is which is itself an answer to the question of what you are agreeing to. The two certain cards carry a heavier border.

Figure 26. Ten robot-type instruction cards re-scoped from general oncology to this PDAC Whipple protocol, each with three numbered instructions and a re-scoping note, and with the four robot types this protocol does not use marked plainly as unused.

The source sheets are mixed-oncology and are illustrated with photographs. Both change here. The clinical content is re-scoped to this PDAC Whipple protocol, and no raster image is carried across: Figure 26 is a card grid drawn as vector line art like every other figure in this paper.

9.6 Software changes during your participation

The software version is frozen at the moment of consent and recorded in a version registry that the participant is shown. Any change to it forces a re-consent, and until the participant acts on that re-consent no study procedure proceeds under the changed version.

Declining the change ends participation and returns the participant to standard care with no penalty. That is a real option rather than a formality, and §8.2 and Figure 23 draw it as a transition of the consent state machine rather than as a paragraph. The regulatory frame for treating version change as a communication obligation is the joint transparency guidance [23], and the frame for monitoring performance change without a version change is the FDA's real-world performance work [26].

9.7 The ten instruction sheets, in full

Figure 26 is a grid because ten parallel items are best compared side by side. The full text of each sheet is reproduced here so that a participant can carry the relevant one, and so that the four unused types remain visibly unused.

1. Eight-arm surgical system, certain. Lie face up with your arms secured, and stay still for about two minutes while anaesthesia begins. You are asleep for the whole of the 240 to 420 minute operation. Eight arms work through 8 to 12 mm ports while your surgeon approves every motion from the console. You wake in recovery 30 to 60 minutes after the arms retract. Keep your abdomen relaxed, follow the nurse's breathing checks, and expect a drain.

2. Imaging robot, certain. Lie still on the table; the gantry moves around you and never touches you. Hold your breath when asked, for 10 to 20 seconds at a time. Tell the technologist at once if you feel unwell, because the sequence pauses without losing the study. You will meet this at screening, at day 90, and at each 12-week visit.

3. Bedside cobot, likely. The cobot works beside you at reduced speed and stops on contact, so you may touch it without harm. Keep your drain line on the side the nurse indicates so the arm's path stays clear. Say stop at any time; the arm halts and a nurse takes over the task.

4. Needle-placement robot, possible. You are awake with local anaesthetic; tell the team if you feel sharp pain rather than pressure. Stay still and breathe shallowly while the needle advances, about 60 seconds. The robot holds the trajectory, the radiologist advances it, and it can be withdrawn instantly. This is used only if a post-operative collection needs drainage.

5. Companion robot, optional. It reminds, records, and calls a human; it performs no clinical task and touches nothing. Ask it to fetch a nurse and it will, and it logs that it did. You may switch it off, and switching it off changes nothing about your care.

6. Humanoid transport assistant, possible. It moves at walking pace with a human escort at all times and never transports you alone. Keep your hands inside the chair rails while it is moving. Any staff member can halt it with the panel stop within one second.

7. Rehabilitation exoskeleton, optional. A physiotherapist fits and starts it, and it will not move until they release the brake. Walk at your own pace; the device assists and does not lead. Stop when your abdomen aches, not when the session timer ends.

8. Radiotherapy positioning robot, not used here. Retained because you may receive radiotherapy outside this study. The couch moves in six axes and pauses if you move; breathe normally unless a breath hold is requested; the intercom is always open and speaking pauses the beam.

9. Radiotherapy motion-tracking robot, not used here. Retained for the same reason. Markers on your skin or an implanted fiducial let the system follow your breathing; do not adjust the markers between fractions; coughing interrupts the beam automatically and does not spoil the treatment.

10. Steerable needle robot, not used here. Not used in this study at any timepoint. If one is offered to you outside this study, ask which steps are autonomous and who approves each one. The questions in §3 of this paper apply to it unchanged.

Robot type	In this study	When	Who is in control
Eight-arm surgical system	Certain	Day 0	Operating surgeon, every motion signed
Imaging robot	Certain	Screening, day 90, q12wk	Radiographer; the gantry never contacts you
Bedside cobot	Likely	Days 1 to 7	Ward nurse; contact stops it
Needle-placement robot	Possible	Only if a collection needs drainage	Interventional radiologist
Companion robot	Optional	Inpatient stay	You; it may be switched off
Humanoid transport assistant	Possible	Ward transfers	Human escort, always present
Rehabilitation exoskeleton	Optional	Week 2 onward	Physiotherapist releases the brake
Radiotherapy positioning robot	Not used	n/a	n/a in this protocol
Radiotherapy motion-tracking robot	Not used	n/a	n/a in this protocol
Steerable needle robot	Not used	n/a	n/a in this protocol

9.8 Missing a visit, and what actually happens

The calendar in §9.1 has eighteen visits over 24 months, and a participant reading it for the first time usually wants to know the consequence of missing one. The consequence is not uniform, and the honest answer distinguishes three kinds of visit.

Four visits are load-bearing. The baseline visit at day -1, the day-30 safety visit, the day-90 pathology review, and the first restart assessment after the operation each carry a primary or secondary endpoint that cannot be reconstructed later. Missing one is not a protocol violation on the participant's part, but it does remove a data point, and if the day-30 visit is missed entirely that participant does not count towards the device-safety endpoint. The site will work hard to reschedule these four within their windows, and the windows are wider than the nominal dates suggest.

Nine visits are routine surveillance: the every-twelve-week imaging and bloods through 24 months. Missing one shifts the schedule rather than breaking it, and the imaging may be done at the participant's own centre with the images read centrally, which is the arrangement most often used by participants who live at a distance.

Five visits are administrative or optional, including the patient-reported experience questionnaires. These may be declined outright with no consequence whatever.

A participant who cannot attend should say so rather than not appear, and the reason is practical rather than moral: an unexplained absence starts a lost-to-follow-up process described in §8.5, which involves contact attempts the participant may find intrusive and which a single message would have avoided.

9.9 The direct channel, and what it is not

Participants in this study have a direct channel to the sponsor for questions about the platform. It exists because the accountability literature is consistent that a participant who can only ask the person who recruited them does not, in practice, ask.

The channel is additional and never a substitute. Every clinical question is routed to the participant's own care team, and every branch of the channel notifies the participant's oncologist, which is the point Figure 3 makes with an edge rather than a sentence. A participant who uses it does not lose the relationship with the site, and the site is told that a question was asked, not what the answer implied about them.

Three things the channel cannot do are worth stating so that the promise stays honest. It cannot give medical advice, because the sponsor is not the treating clinician. It cannot change a protocol requirement for one participant, because that is an amendment and requires the Institutional Review Board. And it cannot adjudicate a complaint about care, which goes to the escalation route in §11.6 and, if unresolved there, to the Institutional Review Board directly.

What it can do is answer a platform question with a number, retrieve the participant's own audit extract, and escalate a safety concern to the independent monitor within one working day.

10 The Numbers Behind the Reassurance

10.1 The provenance rule this section obeys

Every quantitative claim in this section carries one of four class letters, and the letter appears on the same line as the number.

Class	Meaning	What a reader should do with it
M	Measured in a cited human series	Treat as evidence about people; check the citation
C	Contemporaneous nationwide comparator	Treat as the benchmark this study will be judged against
S	Author simulation, not human data	Treat as a design target that has not yet met a patient
P	Protocol-specified limit, not a result	Treat as a commitment, checkable in the audit record

Two of the nine headline values in this section are class S: the proportion of Phase 0 simulated procedures completing inside every envelope, and the worst observed tip force. Both are simulated. Neither has met a patient. Figure 28 traces all nine to their sources.

The discipline matters because the surveyed literature is consistent on the point: expectation and experience diverge systematically in robotic surgery, and expectation is usually set by promotion rather than by evidence [3, 4]. A paper that mixed simulated and measured values without marking which was which would be doing exactly what those studies describe.

10.2 Robotic pancreatoduodenectomy, by learning-curve phase

The single most useful comparator for this study is a contemporaneous nationwide series of 1000 robotic pancreatoduodenectomies analysed across four phases of the institutional learning curve [17]. Both the serious fistula rate and 90-day mortality fall monotonically with volume.

That finding has a direct protocol consequence. If outcomes improve with institutional experience, then enrolling at an inexperienced site means the participant pays for the learning curve. This study therefore restricts enrollment to sites already past the third phase, and publishes the operating surgeon's case count to the participant before consent.

Learning-curve phase	ISGPS B/C fistula (C)	90-day mortality (C)	What this protocol does about it
Phase 1, cases 1 to 80	24.0 percent	8.0 percent	Not eligible to enrol; a site must be past this phase
Phase 2, cases 81 to 180	18.5 percent	6.0 percent	Not eligible to enrol
Phase 3, cases 181 to 400	14.0 percent	4.5 percent	Minimum eligible experience for a participating site

Table continued on the next page.

Table continued from the previous page.

Learning-curve phase	ISGPS B/C fistula (C)	90-day mortality (C)	What this protocol does about it
Phase 4, cases 401 and above	11.0 percent	3.5 percent	Preferred; operator case counts are published to participants

10.3 The reassurance dashboard

Figure 27 puts five panels on one page. Panel A is the learning-curve comparator above. Panel B reports where the 1000 Phase 0 simulated procedures ended, and 8.2 percent of them were stopped by a safety mechanism before completion. Panel C plots each protocol cap against the worst value observed across that simulation set. Panel D is the survival context with its honest denominator. Panel E counts how many of the twenty-one documented concerns get which class of answer.

Panel B is the one a reader should linger on. A dashboard that reported only the 918 completions would be marketing. The argument this paper makes is not that the system never stops; it is that when it stops, it stops safely, and it did so 82 times before any human was involved.

10.4 Safety-limit headroom

Limit	Protocol cap (P)	Worst observed (S)	Headroom
Tip force, any single arm	3.0 N	2.1 N	30 percent
Tip force, all arms summed	18.0 N	12.4 N	31 percent
Cross-arm emergency stop	3.0 ms	1.7 ms	43 percent
System-wide emergency stop	500 ms	310 ms	38 percent
Positional error	2.0 mm	1.3 mm	35 percent
Force reproducibility	0.5 N	0.31 N	38 percent

The observed column is class S throughout. It comes from the author's simulation set [8] and from no human procedure. Headroom of 30 to 43 percent in simulation is a design margin, not a clinical result, and the first human procedure is the first datum in the M column for any of these six rows.

10.5 The survival context, with the honest denominator

Five-year overall survival for pancreatic cancer across all stages in the United States is about 13 percent [14]. Resected disease treated with adjuvant FOLFIRINOX does substantially better [34]. Achieving a microscopically clear margin does substantially better than not achieving one, by a magnitude that varies between series.

This study will not move any of those numbers. Eighteen participants cannot, and no analysis of eighteen participants should be presented as though it could. The FDA's public information on computer-assisted surgical systems makes the general version of the same point about robotically assisted devices [21], and the National Cancer Institute's description of early-phase studies makes it about trial phase [20].

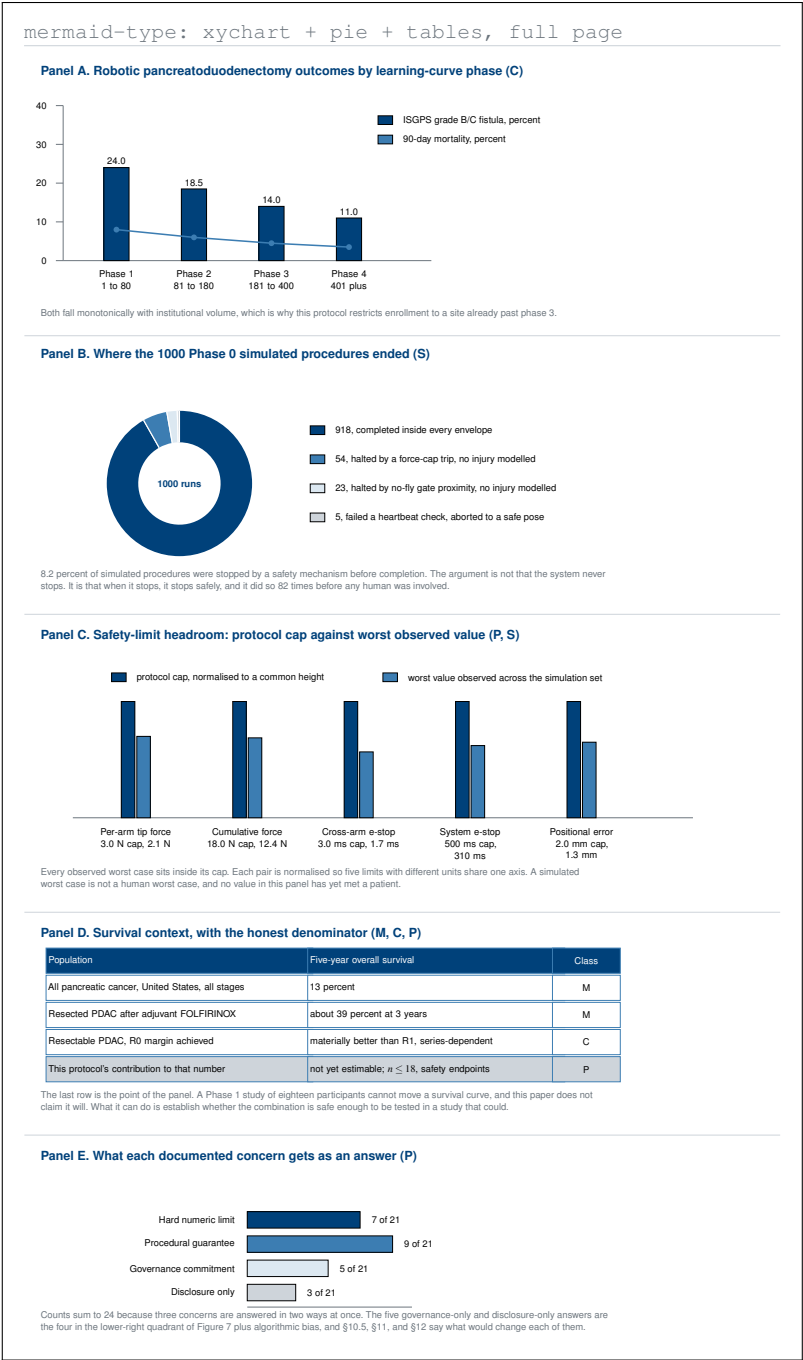


Figure 27. Five panels of quantitative reassurance on one page, learning-curve outcomes, Phase 0 simulation endings, safety-limit headroom, the survival context with its honest denominator, and the class of answer each documented concern actually receives.

What this study can establish is narrower and still worth establishing: whether a combination that pairs a RAS(ON) inhibitor with a hard-limited robotic platform is safe enough, and practical enough, to be tested in a study that is powered for survival. That is the claim this paper defends, and it is the only one it defends.

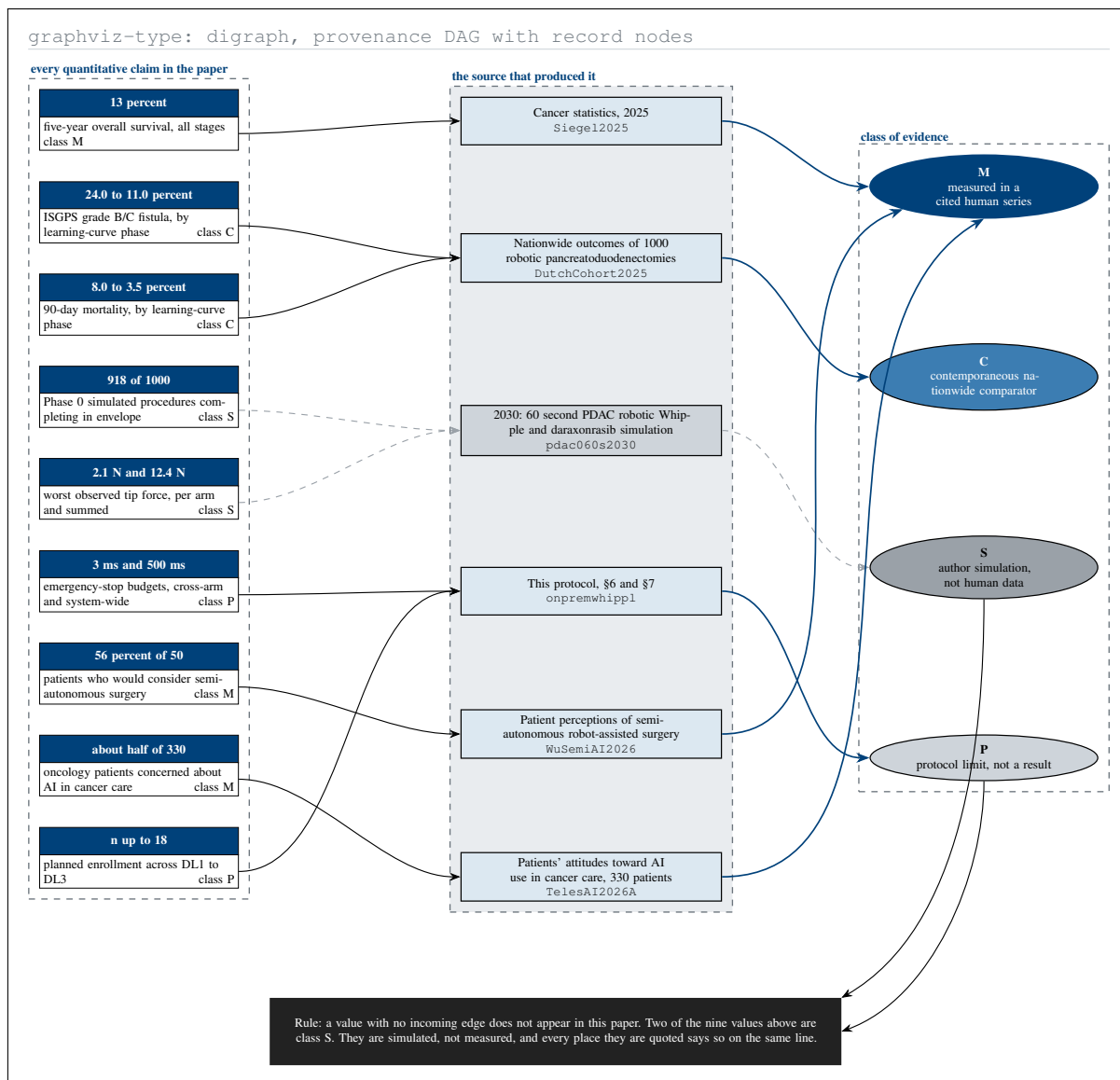


Figure 28. Every quantitative claim in this paper traced through its source to its class of evidence, with the two class-S values, which are simulated rather than measured in humans, marked by dashed edges so a reader can separate them at a glance.

10.6 Where the numbers come from

Figure 28 traces every quantitative claim in this section through its source to its class of evidence. Nine values, six sources, four classes, and one rule: a value with no incoming edge does not appear in this paper. Two of the nine are class S and their edges are dashed, so a reader can separate the simulated from the measured at a glance rather than by reading a footnote.

10.7 What would change the conclusion

An advocacy paper should be able to name the findings that would make it wrong. Four would.

A device-related serious adverse event rate above the prespecified halt threshold would end the argument that the hard limits are sufficient. An R0 rate below the range reported in the nationwide comparator would end the argument that the platform achieves an oncologically adequate resection. An ISGPS grade

B/C fistula rate at or above the phase-1 learning-curve level, at a site restricted to phase 3 or better, would indicate that the platform introduces risk the comparator does not carry. And any emergency-stop latency measured above budget in a human procedure would falsify the single most load-bearing number in this paper.

Each of those four has a prespecified stopping rule attached, set out in §5.6 and published before enrollment rather than after. That is the difference between a stopping rule and a retrospective explanation.

10.8 How to read a number in this paper, in four steps

Every number in this section carries a provenance class letter, and the letter is the most important part of the number. A reader who takes nothing else from §10 should take the following four steps, which apply to any figure quoted anywhere in the paper.

Step one: find the letter. M means measured in humans on this platform. C means taken from a published comparator series. S means produced by simulation. P means a protocol limit, which is a commitment rather than an observation. Where a number appears without a letter it is arithmetic on numbers that carry them, and the derivation is stated.

Step two: ask what an M number is measured on. At the time of writing there are no M numbers from human procedures on this platform, because no human procedure has been performed. Every performance claim in this paper is therefore S or P, and any future version of this paper that converts one to M should be read as materially different from this one.

Step three: check the denominator. A rate quoted from a comparator series is only as good as the series, and §10.2 gives the learning-curve phase of each because a phase-1 site and a phase-4 site produce different rates for the same operation. A comparator rate without a phase is not usable.

Step four: ask what would falsify it. A P number falsifies visibly: the force cap is either enforced or it is not, and the audit trail shows which. An S number falsifies when a human procedure disagrees with the simulation, which is one of the four events in §10.7 that would change the conclusion. A C number does not falsify at all, because it is somebody else's result.

The reason for the discipline is specific to this paper's purpose. An advocacy document that mixed simulated and measured performance would be doing exactly what the surveyed literature says patients detect and discount, and the whole value of §3's twenty-one answers depends on this section being unmixed.

d2-type: grid matrix, responsibility assignment

accountability.matrix

If this goes wrong	Operating surgeon	Sponsor-Investigator	Site PI	DSMB	IRB and FDA	You call
A motion during your operation	A	C	R	I	I	the surgeon, by name, recorded in your consent
A device malfunction or a halt	R	A	R	C	I	the Sponsor-Investigator
A drug dose or a restart decision	C	A	R	C	I	the Site PI first, then the Sponsor
A serious adverse event	R	A	R	C	I	the Site PI, on the 24-hour line
A decision to pause the study	I	R	I	A	C	you do not need to call; you are told
Your data being mishandled	I	R	A	I	C	the Site PI, then the IRB directly
A software version change	I	A	R	C	C	you are re-consented before anything proceeds
A bill you did not expect	I	A	R	I	I	the study coordinator, then the Sponsor
A complaint about how you were treated	I	C	R	I	A	the IRB directly, without going through the site

■ A, accountable, exactly one per row ■ R, responsible □ C, consulted □ I, informed ■ the call you make

Nine failure classes, nine single accountable assignments. Two accountable parties in one cell would be the same as none, which is the failure the concern describes. The far-right column is the only black fill in the figure, because it is the only instruction in it.

Figure 29. The responsibility matrix with exactly one accountable party in every row and a final column, the only black fill in the figure, naming the person the participant should actually call when that particular thing goes wrong for them.

11 Accountability, Oversight, and Who Answers

11.1 The question, stated as patients state it

The accountability concern in the surveyed literature is not that nobody is responsible. It is that everybody is, which amounts to the same thing [3, 13]. When a human surgeon errs, the chain of liability is familiar to the person harmed and to their family. When a semi-autonomous movement or a model-generated suggestion contributes to harm, participants report being told that the surgeon, the hospital, the software developer, the device manufacturer, and the sponsor are all involved, which answers nothing and leaves them with no one to call.

This protocol answers with a rule rather than a list: exactly one accountable party per failure class, named in advance, published before consent. The rule is checkable, because a matrix with two accountable parties in a single row would fail it and the matrix is published. The oversight package this is drawn from is the author's site documentation set [35].

11.2 One accountable party per failure class

Figure 29 renders the matrix. Nine failure classes, nine single accountable assignments, and a final column giving the person the participant should actually call. That last column is the only black fill in the figure, because it is the only part of the matrix that is an instruction rather than a description.

Two accountable parties in one cell would be the same as none. The invariant is stated in the source and verified: nine rows, nine A assignments, no row with two and no row with zero.

If this goes wrong	Accountable party	Who you call
A motion during your operation	Operating surgeon	The surgeon, by name, recorded in your consent
A device malfunction or a halt	Sponsor-Investigator	The Sponsor-Investigator
A drug dose or a restart decision	Sponsor-Investigator	The Site PI first, then the Sponsor
A serious adverse event	Sponsor-Investigator	The Site PI, on the 24-hour line
A decision to pause the study	DSMB	You do not need to call; you are told
Your data being mishandled	Site PI	The Site PI, then the IRB directly
A software version change	Sponsor-Investigator	You are re-consented before anything proceeds
A bill you did not expect	Sponsor-Investigator	The study coordinator, then the Sponsor
A complaint about how you were treated	IRB	The IRB directly, without going through the site

11.3 The oversight bodies, and what each may actually do

Authority and advice are different things, and a governance section that lists bodies without distinguishing them is not informative. The strongest action each body may take is stated below.

Body	Independent of the sponsor	Strongest action it may take
Data and Safety Monitoring Board	Yes	Halt the study at a cohort boundary, or immediately
Physical AI Safety Review Committee	Yes	Require re-gating and a Unified Safety Level reassessment
Institutional Review Board	Yes	Suspend or withdraw approval; receives your complaints directly
Contract research organisation monitor	Contracted by the sponsor	Raise a finding to closure; cannot stop a procedure
Independent safety monitor, in the room	Yes, per procedure	Stop the procedure; cannot change the protocol
Operating surgeon	No	Stop, convert, or decline any model-proposed motion
Sponsor-Investigator	No	Pause enrollment; cannot overrule a DSMB halt

The two rows a participant should note are the last two. The Sponsor-Investigator, who is accountable for six of the nine failure classes, cannot overrule a halt by the Data and Safety Monitoring Board. And the independent safety monitor, who can stop any procedure, cannot change the protocol. Neither of those constraints would be visible in a list of bodies.

11.4 Team experience, disclosed rather than implied

The concern reported in the literature is not general surgical experience; it is experience with this exact platform, this software version, this autonomy level, and this operation [13, 3]. General experience is what gets offered, and it is not the same thing.

Four figures are published to the participant before consent. The operating surgeon’s total robotic pancreatoduodenectomy case count. The site’s position on the learning curve defined by the nationwide comparator [17]. The number of procedures the operating surgeon has performed on this specific platform at this specific autonomy level. And whether a trainee will participate in any part of the procedure, and if so which part.

Case count is the right question because the comparator series shows outcome varying with it, which §10.2 quantifies. A participant who is told only that their surgeon is experienced has been given an adjective; a participant who is told a number can compare it to the phase boundaries in that table.

11.5 The audit trail, and your right to read it

Every model advisory, every gate decision, every surgeon approval, and every safety-limit event is written to a hash-chained record bound to a deterministic seed and a specific repository commit, under 21 CFR part 11 [11, 7]. The chain is append-only: an entry cannot be altered after it is written without breaking the hash of every entry after it, which is the property that makes requesting an extract worth doing.

A participant may request their own extract at any time. The extract contains every advisory issued during their procedure with its confidence value, every gate decision including every rejection and the limit that caused it, every surgeon approval with its timestamp and signature, and every safety-limit event with the measured value. It does not contain any other participant’s data, the model weights, or the source code, and it is provided in a form a non-specialist can read alongside a plain-language key.

The chain either verifies or it does not, and the verification tool is provided with the extract. That is a stronger commitment than a promise of accuracy, because it can be checked by someone who does not trust the person making it.

11.6 What happens if a rule in this section is broken

Three breaches are the most likely, and each has a route that ends at a body independent of the sponsor and is available to the participant without going through the site.

Breach	First route	Independent route, always available to you
An adverse event affecting you is not reported to you	Site PI, on the 24-hour line	The IRB directly; the IRB may suspend approval
Your data is accessed by someone not on the read list	Site PI, who must log the request	The IRB directly, and the access is visible in the hash chain
A procedure proceeds under a software version you did not consent to	Sponsor-Investigator	The IRB and the Physical AI Safety Review Committee; the version registry is the evidence

In each case the evidence needed to establish the breach is already in the participant’s own extract. That is the practical reason the audit trail is worth having: it converts a dispute about what happened into a question about what the record says.

11.7 What accountability does not mean here

The word accountability is used in this section in a narrow sense, and the narrowness is deliberate, because the broad sense promises something no protocol can deliver.

It does not mean that someone will be punished. A named accountable party is a person who must answer for a failure class, explain what happened, and carry the decision about what changes. Whether anyone is disciplined is a matter for the institution's own processes and for the law, and this protocol has no authority over either. A participant told that a protocol makes someone accountable, who then discovers that accountable meant only that a name appears in a matrix, has been oversold.

It does not mean compensation. §12.3 states what the sponsor pays and what remains unsettled, and no clause in the responsibility matrix creates a right to payment. Injury compensation in a trial is governed by the site's own agreement and by the consent form, and a participant should read that paragraph of the consent form specifically rather than infer it from this section.

It does not mean that the accountable party was at fault. The matrix assigns answerability in advance, before anyone knows what will happen, precisely so that the assignment is not made afterwards by whoever is least able to object. The party accountable for a device failure class answers for it whether the failure was foreseeable or not.

What it does mean is that for every failure class there is exactly one party who must answer, that the party is named before the study opens rather than found afterwards, that no party reviews its own work, and that the participant has a route to that party which does not run through the person who recruited them. Those four properties are checkable, and Figure 29 is the check.

12 Patient Rights, Costs, and H. R. 9510 v5

12.1 The premise, stated as legislation would state it

The cancer patient, not the doctor, not the nurse, not the trial sponsor, not the Institutional Review Board, and not the regulator, is the priority participant in a United States oncology clinical trial. That premise is the subject of the author's proposed legislation, cited throughout this paper as H. R. 9510 v5 [2], which supersedes the earlier H. R. 9501 to H. R. 9507 numbering used in the antecedent analysis [5]. The case for enactment is developed separately [36, 37].

The bill matters to this paper for a specific reason. Six of its provisions describe things this protocol already does voluntarily. The difference between doing them voluntarily and doing them under statute is that a voluntary commitment can be withdrawn by the party that made it, and a statutory one cannot. A participant reading the table below should understand the right-hand column as a description of current practice at one site, and the left-hand column as what would make it general.

What the bill would establish	What this protocol already does voluntarily
Participant self-selection of trials, with round-the-clock booking	Direct booking channel with a 72-hour provisional hold and a five-business-day target, §6.4 and Figure 15
Participant choice of robot and of autonomy level	Autonomy level set per operative step and recorded in the consent, §6.5 and Figure 16
Participant procedural modification authority	Any excluded step is performed manually, and the exclusion is logged and reported with the outcome
Verification before generation, with published gates	Phase 0 gate at 1000 simulations across two frameworks, USL at least 7.0, §5.5 and Figure 13
Real-time participant to sponsor communication	A named human at the sponsor answers within one business day, logged verbatim in your record
Participant health data self-custody	Per-store read lists published, and your own copy available at any time, §9.3 and Figure 25

12.2 Your rights during the study, enumerated

Some of the rights below are already required by federal regulation. Others go beyond the regulatory minimum. The distinction is marked, because a document that presented a legal minimum as a favour would deserve exactly the distrust the surveyed literature reports.

Right	Source	Note
To be informed of the study's purpose, procedures, risks, and alternatives before consenting	21 CFR §50.25	The regulatory minimum [6, 12]

Table continued on the next page.

Table continued from the previous page.

Right	Source	Note
To withdraw at any time without penalty	21 CFR §50.25	The regulatory minimum
To be told of significant new findings affecting your willingness to continue	21 CFR §50.25	The regulatory minimum
To contact the IRB directly about the conduct of the study	21 CFR §50.25	The regulatory minimum
To be shown the Phase 0 simulation sign-off at the baseline visit	This protocol	Beyond the minimum
To set the autonomy level accepted for each operative step	This protocol	Beyond the minimum
To be re-consented before any software version change takes effect	This protocol	Beyond the minimum
To request your own audit-trail extract, with a verification tool	This protocol	Beyond the minimum
To receive your own day-30 and day-90 results directly, not on request	This protocol	Beyond the minimum
To be told the operating surgeon's case count on this platform	This protocol	Beyond the minimum
To have your treating oncologist notified on every branch of every route	This protocol	Beyond the minimum

12.3 What this costs you

The sponsor pays for the drug, the platform, study-only imaging and laboratory work, protocol-required visits, research-related injury care, and travel to a stated cap. The participant's usual arrangement covers standard surgical care and the inpatient stay, exactly as it would outside the study.

Two items are not settled by this protocol. Lost income is not compensated. And continued access to daraxonrasib after study closure is not guaranteed. Both are stated in the table, both are marked, and neither is buried. The National Cancer Institute's guidance on who pays for clinical trials identifies exactly these as the questions participants most often fail to ask in advance [27].

Item	Who pays	Note
Daraxonrasib	Sponsor	Supplied for the whole study period
Robotic platform use	Sponsor	No charge reaches you at any point

Table continued on the next page.

Table continued from the previous page.

Item	Who pays	Note
Study-only imaging and laboratory work	Sponsor	Anything the protocol adds beyond routine care
Protocol-required visits	Sponsor	The 18 visits of Figure 24
Research-related injury care	Sponsor	Stated in the consent form, not only here
Travel and lodging	Sponsor, to a cap	The cap is stated in your consent form
Standard surgical care	Your usual arrangement	Exactly as it would be outside the study
Standard inpatient stay	Your usual arrangement	The 8 to 10 day stay is standard care
Long-term survivorship care	Your usual arrangement	On the routine pathway after handback
Lost income	Not settled	This protocol does not compensate it; H. R. 9510 v5 would address it
Continued drug access after closure	Not settled	Not guaranteed by this protocol

12.4 After your last visit

Care is handed back to the participant's own oncology team in writing, and a full copy of the record goes to both the participant and that team. Survivorship follow-up returns to the routine pathway. The device support obligation ends at study closure, and the figure draws that ending rather than omitting it. The data vault remains retrievable for fifteen years. Study results are sent to the participant whether or not they completed, which is a commitment about the sponsor's behaviour after the participant has stopped being useful to the study.

The longer-term proposal, a national platform that would carry post-trial obligations across studies rather than leaving each sponsor to define its own, is developed separately [38]. Figure 30 draws it as a dotted edge rather than a solid one, because it is a proposal and not a provision.

12.5 If you take one page to the consent conversation

Everything above is the argument. What follows is the page to carry, because a participant reading forty pages the night before a consent conversation will remember five things at most and should choose which five.

Ask who is accountable for the failure class you care about most, and ask for the matrix. It is published, it has one accountable party per row, and if the answer names a committee rather than a person, the answer is wrong.

Ask for the two numbers measured in your own operating room. The emergency-stop latency, cross-arm and system-wide, is measured for your case before induction. They are the only performance numbers in this whole paper that will be about you specifically.

Ask what happens if you say no today and yes in six weeks. The answer should be that eligibility is re-assessed at the moment a slot opens, that nothing about declining now prejudices later enrollment, and that your standard care does not change either way.

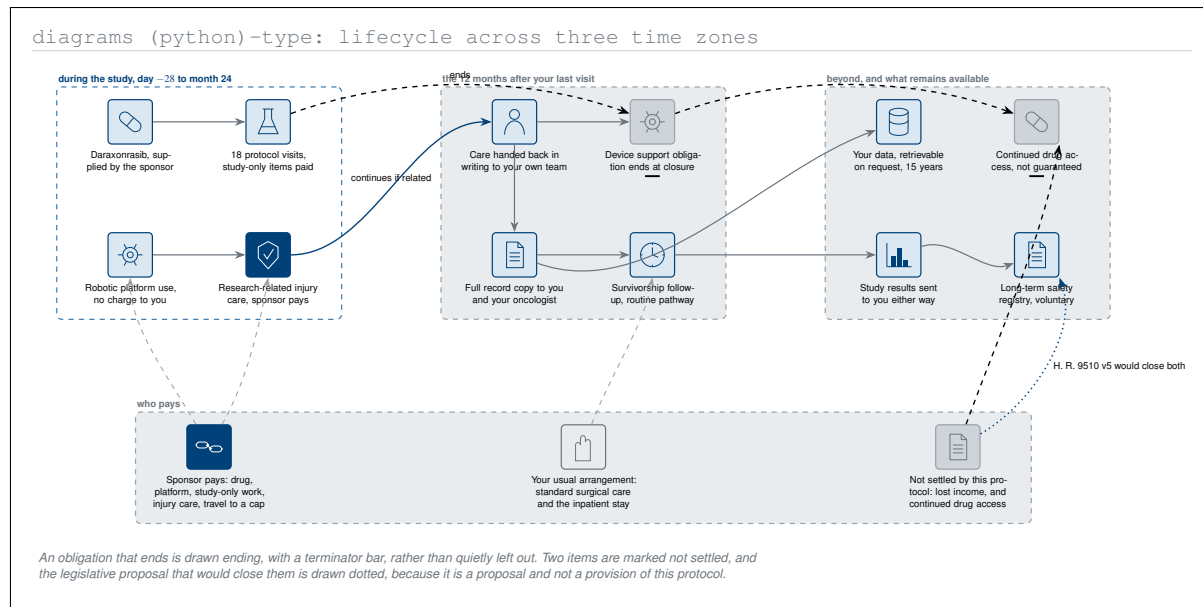


Figure 30. Post-trial continuity across three time zones, during the study, the 12 months after the last visit, and beyond, with the payer attached to every item and the two unsettled items drawn as unsettled rather than being quietly left out of the figure.

Ask which of your concerns is answered by governance alone. There are five, they are listed in §3.8, and a site that claims all twenty-one are answered by hard limits has not read the same protocol this paper read.

Ask for your audit extract to be demonstrated, not described. The extract exists from the evening of the operation. A site that can show the mechanism before you consent is a site that will produce it afterwards.

Those five questions take about ten minutes and they test the five properties the rest of this paper spends forty pages establishing. If the answers are satisfactory, the argument of this paper is that participation is a reasonable choice. If they are not, the argument of this paper is that the participant is entitled to say so and to be believed, and §12.2 is the enumeration of rights that makes saying so safe.

12.6 Closing argument

The argument of this paper is not that this trial is safe because it is new. It is the opposite. Novelty is the reason to demand specifics, and §3 collects twenty-one specific demands that people actually make. What the paper claims is that this protocol answers more of them, more precisely, than the alternative does.

Consider what a participant knows after reading it. They know the force each robotic arm may apply and the force all eight may apply together. They know how long stopping takes, in milliseconds, and who may press stop. They know that no message travels from the model to the arms, and they can see the diagram in which the arrow is absent. They know the read list on every store that holds their data and how many years each is kept. They know the name of the person accountable for each of nine failure classes and which of them to call. They know the two costs the protocol does not cover. And they know that a Phase 1 study of eighteen participants cannot tell them whether this improves survival.

A participant undergoing a standard Whipple outside a trial knows almost none of that, not because anyone is concealing it, but because nobody assembled it. That asymmetry is the argument. It is not an argument that the technology is harmless; it is an argument that a protocol which can be interrogated this closely is a better place to be than one which cannot.

The honest gaps remain where §3.7 put them. Cancer-control effectiveness is answered by a comparator and not by a result. Accountability is answered by a matrix and not by a settled liability doctrine. The human relationship is answered by a notification rule and not by a guarantee. Cost is answered line by line with two lines left open. Sections 10, 11, and 12 say what would have to change for each, and the proposed legislation [2] and the clinician-trust framework [39] are the routes by which some of it might. Until then, this paper states the gaps in the same typeface as the answers, because a proponent who marks their own limits is the only kind worth believing [1].

13 References and Back Matter

13.1 Abbreviations

Term	Meaning	Term	Meaning
AE	Adverse event	IRB	Institutional Review Board
BMI	Body mass index	ISGPS	International Study Group on Pancreatic Surgery
CA 19-9	Carbohydrate antigen 19-9	ISM	Independent safety monitor
CRF	Case report form	LLM	Large language model
CRO	Contract research organisation	OS	Overall survival
DL1 to DL3	Dose levels 1 to 3	PDAC	Pancreatic ductal adenocarcinoma
DLT	Dose-limiting toxicity	PFS	Progression-free survival
DSMB	Data and Safety Monitoring Board	R0	Microscopically clear resection margin
ECOG	Eastern Cooperative Oncology Group	RECIST	Response Evaluation Criteria in Solid Tumors
e-stop	Emergency stop	SAE	Serious adverse event
FDA	Food and Drug Administration	SMA	Superior mesenteric artery
IDE	Investigational Device Exemption	SMV	Superior mesenteric vein
IND	Investigational New Drug	USL	Unified Safety Level

13.2 Figure inventory

Thirty figures, numbered 1 to 30, with no gaps and no duplicates. The counts by type are Mermaid 9, D2 7, PlantUML 5, Graphviz 5, and Diagrams (Python) 4, which follows the purpose each type serves rather than an equal quota.

Fig	Type	§	Subject
1	Mermaid-type	1	The seven commitments made to the participant
2	Graphviz-type	1	Accountability chain rooted at the participant
3	Mermaid-type	2	Journey schema with every decision point marked
4	D2-type	2	Schedule of Activities from the participant's chair
5	D2-type	3	Twenty-one concerns in six disjoint families
6	Graphviz-type	3	Every concern wired to its answering clause
7	Mermaid-type	3	Concern prevalence against answer completeness
8	PlantUML-type	3	Safeguards as capabilities the participant holds
9	Diagrams (Python)	3	Where each concern physically lives

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Fig	Type	§	Subject
10	Mermaid-type	4	Objective, endpoint, and the sentence it authorises
11	D2-type	4	Endpoint registry with a plain-meaning field
12	PlantUML-type	5	Participant status as a guarded state machine
13	Graphviz-type	5	The 3+3 rule as an auditable decision tree
14	Mermaid-type	6	Eligibility as a two-way gate
15	PlantUML-type	6	Direct booking and sponsor response
16	D2-type	6	The five choices the participant keeps
17	Diagrams (Python)	7	Operating room and on-premises stack
18	PlantUML-type	7	Emergency-stop timing budget
19	Mermaid-type	7	Advise, approve, execute, and log
20	D2-type	7	Force caps and vascular no-fly envelope
21	Graphviz-type	7	Hazard fault tree with named barriers
22	PlantUML-type	8	Four withdrawal routes and their data outcomes
23	Mermaid-type	8	Consent lifecycle with forced re-consent
24	Mermaid-type	9	Participant visit calendar with totals
25	Diagrams (Python)	9	Data pipeline with per-store read lists
26	D2-type	9	Ten robot instruction cards, PDAC-scoped
27	Mermaid-type	10	Five-panel quantitative reassurance dashboard
28	Graphviz-type	10	Evidence provenance for every quoted number
29	D2-type	11	Responsibility matrix, one accountable per row
30	Diagrams (Python)	12	Post-trial continuity and the payer per item

13.3 Table inventory

Forty-three tables. Every one is set to the body text width, every fixed column is ragged-right, and every table of more than two rows is breakable with a repeated header, so no table can run off the foot of a page.

§	Table	What it carries
1	The seven commitments	Each commitment against the clause that makes it enforceable
1	Diagram types and counts	Which question each of the five types answers
1	How to check a commitment	The artefact behind each commitment, and who holds it
2	Glossary	Twenty terms in the sense this protocol uses them
2	Schedule of Activities	The conventional matrix, retained for a treating oncologist
3	Concern families	Twenty-one concerns grouped and mapped to sections
3	Answered by governance alone	The five, and why no harder answer is available

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§	Table	What it carries
3	The twenty-one, one row each	Concern, question, answer, and answer class
4	Endpoint table	Every endpoint with its timepoint and plain meaning
4	Endpoint cost	What each endpoint costs the participant to measure
4	Risk classes	Which risks are new to this study and what bounds them
5	Dose levels	What has to be true before each level opens
5	Stopping rules	Seven prespecified triggers, four of them device-specific
6	Two-way gate asymmetry	What differs between the protocol and participant columns
6	Inclusion criteria	Eight criteria, each with the reason it exists
6	Exclusion criteria	Seven criteria, each with the reason it exists
6	Subgroup reporting	What the study can and cannot say at $n \leq 18$
7	Operative limits	Eight limits with cap, worst observed, and what each protects
7	The day, hour by hour	Each step of the operative day, marked study-specific or not
7	Difference from a conventional robotic Whipple	Ten aspects, four constraints added, nothing changed about the operation
8	Withdrawal routes	Four routes and the data consequence of each
8	Two undrawn transitions	Why re-consent and its refusal matter
8	Study-initiated discontinuation	Six triggers and what follows for the participant
9	Visit burden	Visits, inpatient days, clinic days, and totals
9	Store access and retention	Seven stores, retention in years, and the read list
10	Provenance classes	The four class letters and what to do with each
10	Learning-curve comparator	Fistula and mortality by phase, with the protocol response
10	Safety-limit headroom	Six limits with cap, worst observed, and headroom
11	Responsibility matrix	Nine failure classes, accountable party, who to call
11	Oversight bodies	Independence and the strongest action each may take
11	Breach routes	Three likely breaches and the independent route for each
12	Bill provisions	What H. R. 9510 v5 would establish, against current practice
12	Rights during the study	Eleven rights, marked regulatory minimum or beyond
12	Costs	Eleven items, the payer for each, and the two unsettled
13	Abbreviations, figures, tables, version history	This back matter

13.4 Version history

Version	Date	Change
Draft 1.0	July 31, 2026	First release. Thirteen sections, thirty figures across five diagram types, forty-three tables, twenty-one documented concerns answered clause by clause, and an eight-stage build from machine-readable diagram sources through a bracketed draft and a populated full version to a polished final source set.

Data and code availability

Every figure source, every LaTeX source, and every stage narrative is public at github.com/kevinkawchak/robotic-surgeries/tree/main/patient-robot-advocacy. The five diagram-source directories hold the machine-readable originals in Mermaid, PlantUML, D2, Python, and DOT; the three paper directories hold the LaTeX at each stage together with an Overleaf-ready archive. No participant data exists, because no participant exists: this is an independent advocacy paper about a protocol, not a report of a conducted study.

Acknowledgments

The concern evidence in §3 rests on two dated research passes of July 28, 2026, one by Gemini 3.1 Pro and one by ChatGPT 5.6 Thinking Extended, filed in the repository at `research/`. The parent clinical protocol is the author’s own [1]. The LaTeX build across all eight stages was adapted using Claude Code Opus 5.

Ethical disclosures

No human participants, no animals, and no identifiable data. This is an independent research paper and a practical adoption guide. It is not an active IND or IDE, not an approved protocol, and not medical or regulatory advice. It is not endorsed by the FDA, the NIH, HHS, an Institutional Review Board, the International Council for Harmonisation, or any sponsor, contract research organisation, site, regulator, or medical society. The author declares no competing interest other than authorship of the cited works.

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